

PUSH POSSIBLE™



SCALE Solution Design Document

(Multiple sites, Global Design)

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INTRODUCTION

This document is designed to outline the proposed process for the SCALE implementation at Knipper, Inc.'s (Knipper) current distribution centers. This implementation includes existing 2 physical sites to follow the functionality described in this document. Future implementations at other Knipper facilities may result in either changes being made to this document or entirely new functional flows per facility.

SCALE is upgraded from version 2013 to Manhattan Active SCALE ® for this implementation, and Knipper migrates the required functionality into the new version.

The objective of this document is to define the proposed process and scope from a Manhattan Associates' perspective and to identify key extensions to the Manhattan Associates suite of products. This document serves as a reference throughout the process for confirmation of the approach and definition of tasks. It will also serve as a reference for Manhattan Associate's customer support organization after implementation, and potentially a reference point for any future Knipper implementations.

STATISTICS AND LAYOUT

MULTIPLE

Warehouse statistics are listed inline below.



Warehouse Stats.xlsx

Layout

Existing Layouts are used. Layout drawings not available

TECHNOLOGY

- Host ERP-
- Middleware –
- WM Platform: Windows
- Version: Manhattan Active® SCALE
- Warehouse Mobile Vendor: Zebra - procured through Manhattan.
- MHE Vendor: Lightning Pick
- Label Printer: Zebra 203 dpi (Fixed).

KEY DECISIONS / ASSUMPTIONS

- 1) The scope of this document is limited to the standardized flows identified for the following distribution centers. Over time, Knipper may add additional DCs in future phases. One Warehouse will be created in SCALE for each facility list below.
 - Lakewood, NJ
 - Charleston, IN
- 2) Company configuration will be used. Item master in SCALE is maintained with the appropriate company information which represents Knipper's customer. All inventory in SCALE is maintained for these companies. Over time, Knipper may add other companies in future phases through a change management process identified collectively with Manhattan Associates.
- 3) Host assigns company on the interfaced receipts and shipments.
- 4) Item master will be interfaced into SCALE for account which have EDI capabilities. Otherwise, Items will be created manually in SCALE.
- 5) Item master in SCALE includes units of measure, dimensions, and weights for all products for all units of measure Cubiscan will be used to gather the dimension and weight information. Cubiscan is not integrated with SCALE.
- 6) All dimensions are stored using IN (Inches).
- 7) All weights are stored using LB (Pounds).
- 8) Items will not be shared across companies.
- 9) Item master in SCALE maintains Item Cross Reference.
- 10) Knipper has standardized UOMs to have one storage template.
 - EA-PK-CS-PL
 - CS-PL
 - EA-ROLL-CS
 - ROLL-CS-PL
 - PK-CS-PL
- 11) Item Cross Reference is configured in GTIN format and is unique for an item and Unit of Measure combinations. It is of 12 or 14 digits. This information is interfaced from the host for EDI for accounts with EDI, otherwise maintained directly in SCALE.
- 12) Item Cross Reference can be shared across items.
- 13) Knipper owns and sets the quantity um symbol for the quantity unit of measures for the manifesting process.
- 14) Knipper provides the conversion quantity for all units of measure. If the product is never received or shipped in a specific conversion quantity, then that record is not used in SCALE.
- 15) EA, PK, ROLL, and CS units of measure of the storage templates have group during check-in set to 'Yes'.
- 16) All units of measures other than PL have a separate Item Cross Reference. PL UoM does not have a cross reference configured.
- 17) The Quantity unit of measure is a whole number.

Commented [BK1]: Lakewood should be split between KDC and OHW?

Commented [NC2R1]: Per breakout sessions & Vic's final decision is to keep them together.

Commented [NC3R1]: This is resolved

Commented [SM4]: Key Decisions #13: what is this telling us?

Commented [NC5R4]: Knipper owns the UOM

Commented [NC6R4]: This is resolved

Commented [SM7]: Add Pallet "PL"

Commented [NC8R7]: Knipper to take back to MAH for further clarity

Commented [RS9R7]: If we enable grouping for PL UM, SCALE may create one LPN for more than one Pallet. This is not recommended. One LPN should be assigned to one PL.

Commented [NC10R7]: NC12022024: This is resolved

Commented [SM11]: Need more clarity on this.

Commented [NC12R11]: This is resolved

Commented [RS13R11]: Unless PL can have a GTIN like EA, PK and CS, we do not setup item cross reference for PL

Commented [NC14R11]: NC12022024: Pallet has a GTIN for Pallet. The lowest UOM is shipped . This is resolved.

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18) Several items are serial number tracked. These items include tracking as below and are marked on item master. Two different Items can have same serial number, and SCALE should be configured to allow duplicate serial numbers.

- Inbound only
- Inbound and Outbound
- Inbound, Inventory and Outbound
- Outbound only

19) DSCSA workflow is excluded from the scope of this document. There are call outs to inbound and outbound extensions for DSCSA. The integration and deployment strategy for DSCSA with the migration must be reviewed during the conversion planning.

20) None of the items are catch weight enabled.

21) None of the items are immediate needs eligible. Cross-docking with/without immediate needs is out of the scope of this implementation.

22) Lot tracking will be utilized. Lot-tracked items have the lot ID and expiration date available on the product when being received. The lot ID will not be barcoded on all products.

23) For some items, Knipper uses lot ID without the need for expiration date tracking. These items were set up this way during the original implementation and have a dummy expiration date setup. These lots or dates are not used for FEFO.

24) An item on a receipt has only one detail for the same lot, if lot is interfaced.

25) Purchase order and receipt in the host may have 1:M mapping. However, Purchase orders are not sent to SCALE.

26) Treat as Loose Flag is set to Y on the item unit of measure for eligible UM's if the product must be repacked before shipping. The Value will be set to N if it can be shipped in the package it is currently stored in.

27) Dimensions are provided for the shape the product it is shipped in. If for example T-Shirts are shipped as rolled, then the dimension of the Roll will be provided.

28) Location Unit of measure override is not leveraged for this implementation.

29) Item location assignments (Permanent Locations) are used.

30) Item Location Capacity records will be provided for permanent locations.

31) All inventory locations are single item and single lot other than those identified later in the document for exception handling including but not limited to Damages, Held, Destruction, and the Virtual 'See Supervisor' location.

32) All inventory locations are License plate tracked other than forward pick locations.

33) P&D locations are used mostly in all the warehouses in the scope of this implementation.

34) Hazardous materials are shipped using SCALE in the implementation.

35) Inbound systemic QC is not leveraged. A manual SOP using a visual QC is leveraged.

36) Quick receiving will not be leveraged. Quick Receiving is the process in SCALE where the inventory is putaway to final location without any putaway work being generated.

37) Returns are interfaced to SCALE as a Receipt ID Type of RA. For some scenarios with returns due to package not delivered, Knipper would like to utilize custom Receipt from Shipment functionality. Knipper would like to enhance the base functionality to automate extraction of lot and Expiry date from the shipping container data.

38) Inventory Attributes are not leveraged. Inventory attributes are specific attributes to use with inventory for processing reasons.

Commented [SM15]: Need ability to manage a scenario where there are more than one of the same serial number, but for different Items.

Commented [NC16R15]: For non-DSCSA inventory - to allow duplicates. Follow up with MAH

Commented [RS17R15]: SCALE allows same serial number to be captured for two different items. Rephrased the assumption

Commented [NC18R15]: NC12022024: resolved.

Commented [SG19]: Is this correct?

Commented [SM20R19]: If it is not going to be mentioned in this document, where will it be mentioned?

Commented [SM21R19]: EX37B document?

Commented [NC22R19]: MAH will be providing / providing new design documents for the DSCSA Serialization. It will be provided prior to Build. This comment is being resolved

Commented [SM23]: This is what causes issue on 3PL inbound, multi lot issue - PL #11

Commented [NC24R23]: This is resolved

Commented [RS25R23]: Please advise if this should be rephrased

Commented [NC26R23]: NC12022024: resolved.

Commented [RS27]: Reviewing further for unit of measure discrepancies

Commented [RS28R27]: We have an email thread going on with this with Caleb/Vimal. We have already provided recommendation and need final confirmation from Caleb/Vimal to go ahead to resolve

Commented [SM29]: Add to Parking Lot: Need to ensure LP is not assigned to the user who dropped the work off at a drop zone.

Commented [SM30R29]: Dont want to have to scan SN at drop zone if already verified when pulled.

Commented [NC31R29]: MAH to confirm

Commented [RS32R29]: The user should be ...

Commented [NC33R29]: NC12022024: this is ...

Commented [CC34]: please explain

Commented [NC35R34]: MAH will be able to ...

Commented [RS36R34]: The Hazmat documents fo ...

Commented [NC37R34]: NC12022024: Resolved.

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- 39) Picking sequence and/or Putaway sequence are not leveraged. Location template is used in the order-by clause to determine how putaway and picking should occur in the warehouse. The existing setup is used.
- 40) Wave-based dock assignment is leveraged in this implementation.
- 41) Wave-based shipment consolidation is not leveraged in this implementation.
- 42) When a split shipment happens on load confirmation in SCALE, **Middleware or Host system will be able to handle processing of multiple upload files.**
- 43) Transportation Execution is implemented for the supported services of FedEx, UPS, USPS and UPS Mail Innovations.
- 44) Rate shopping is not leveraged and is out of the scope of this implementation.
- 45) Location Check Digit is not leveraged for location verification.
- 46) Host does not send a delete interface for deleting item master. The obsolete items are not marked as inactive either.
- 47) Receiving worksheets are utilized and system-generated LPN labels are used.
- 48) **Country of origin for an item is tracked in SCALE at the Item level.**
- 49) **Item labels (Knipper/Supplier SKU labels) are not required to be printed from SCALE.**
- 50) **Systemic outbound QC is used in this implementation. Knipper would like to enhance QC functionality to be able to perform QC for Pallets with nested containers.**
- 51) Work Orders will be utilized in this implementation.
- 52) **Setting up labor management is not in scope of this implementation.**
- 53) Knipper migrates the existing ODWS to generic configs during the build phase. This is to establish an SOP where reliance on Manhattan cloud services to execute ODWS is alleviated.
- 54) This document uses suggested naming conventions for configurations like locating rule names, zones, work type names, etc. Actual names may change during the configuration phase.
- 55) Vocollect Voice Picking will not be utilized.
- 56) In the warehouse, physically there can be multiple packing locations but systematically in SCALE there is a one pack location per warehouse.
- 57) Parcel international shipments are manifested at shipment level.
- 58) Back-order processing is done at Host. In SCALE we ship 100% of the quantity requested by host. Allocation exceptions and Pick exceptions are handled according to the SOP defined by Knipper.
- 59) Receipt Workbench will be utilized for receiving receipts for most accounts of Knipper (mainly MSM) accounts as most of the products do not have barcode for Item and Lot. For 3PL accounts, Warehouse Mobile receiving will be utilized.

Commented [SM38]: Knipper has discretion to use sequencing as needed.

Commented [NC39R38]: MAH to confirm this can be done for flexibility

Commented [RS40R38]: If Knipper is able to assign Picking/Putaway sequences to the locations, those can be utilized for sequencing while tasks are being created.

Commented [NC41R38]: NC12022024: this is resolved.

Commented [SM42]: Can FedEx be added here?

Commented [NC43R42]: MAH to confirm. Knipper has a current integration with Prologistics. Separate Integration for FedEx

Commented [RS44R42]: I have added FedEx but please note that only carriers which were mentioned in the SOW were UPS, USPS and UPS Mail Innovations for parcel carrier.

Commented [NC45R42]: NC12022024: What is the impact to the SOW

Commented [RS46R42]: I do see that FedEx parcel service is being used in SCALE 2013 currently. In that case, we should be able to migrate in SCALE Active as well

Commented [RS47R42]: Checked the SOW again and FedEx is included. We can resolve this.

Commented [SM48]: Knipper would like to use ...

Commented [NC49R48]: MAH to confirm if any ...

Commented [RS50R48]: The configurations will need ...

Commented [NC51R48]: NC12022024: This is ...

Commented [SM52]: Blind LPNs are used for Blir ...

Commented [NC53R52]: MAH to add. Current ...

Commented [RS54R52]: The blind LPNs can be use ...

Commented [NC55R52]: NC12022024: this is ...

Commented [CC56]: What do the Item labels look ...

Commented [RS57R56]: Currently this is not ...

Commented [NC58R56]: NC12022024: this is ...

Commented [SM59]: Can we get a demo? KMW ...

Commented [NC60R59]: Configuration based. This ...

Commented [SM61]: Can we get a demo? KMW ...

Commented [NC62R61]: Hardware integration - this ...

TERMINOLOGY

Client Terminology	MA Terminology	Definition
WMS	SCALE	Supply Chain Architected for Logistics Execution warehouse management.
EDI	HOST / ERP	Knipper' ERP system integrating with Scale
ITEM	ITEM	Item Identifier for regular inventory. For Knipper, this is item code, style, color, and size.
GTIN	ITEM CROSS REFERENCE	Global Trade Item Number. Definition of an SKU in a specific Unit of Measurement (UoM).
PURCHASE ORDER / RECEIPTS	RECEIPTS	Goods purchased by Knipper. Purchase orders and receipts have a 1:M mapping. SCALE maintains these as receipts with PO interfaced on receipt header.
TRAILER ID (IN BOUND)	TRAILER ID	A container or truck being received into the DC. One inbound trailer has one receipt.
TRACK & TRACE ITEM	ITEM	Items needing tracking and traceability in SCALE for the source and valid paperwork.
ORDER	SHIPMENT	Goods to be shipped to stores, wholesale customers, or retail supermarkets. the order in ERP and shipment in scale will be a 1:1 mapping.
WAVE	WAVE	A wave represents the different steps that the system uses to retrieve orders from the pool, and process them into the outbound portion of SCALE
RUN WAVE	RUN WAVE	a group of orders when processed resulting in inventory allocation
POOL	POOL	Any orders pending processing to be shipped immediately after interface to SCALE
PICK TASK	WORK INSTRUCTION	Transaction tracked by SCALE to coordinate the movement of inventory for a variety of warehouse processes. A group of work instruction is work unit

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CARTON / CASE	SHIPPING CONTAINER	An object that can be used to hold or transport inventory for a shipment.
CARTON / CASE (INBOUND)	RECEIVING CONTAINER	An object that can be used to hold or transport inventory for a receipt.
LPN	LOGISTICS UNIT	An object that can be used to hold or transport inventory. When nested, the tree unit is referred to as the parent logistics unit.
	P&D	Pick up and Drop location
ODWS	ODWS	Override Data Wave Step. SCALE provides a wave step that allows performing crud operation to data during the wave process.
EXIT POINT	EXIT POINT	An external process that performs logic in line with the base process. Used to update data or perform additional logic.

UOM DEFINITION:

These are examples of a unit of measure definition displaying four UOMs and their respective representation to the other UOMs. This is not the baseline unit of measures.

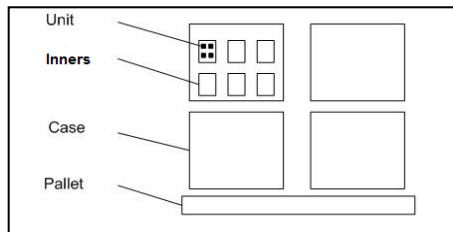


Figure – Unit of Measure

I. INTERFACES

Host system downloads information into SCALE using API-based methods of interfacing. Upload from SCALE to Host will be handled using XML file-based method of interfacing. This allows the Host system to create records with key information for downloads, and SCALE reads this key information to download the information into the SCALE production tables. XML upload method also allows the Host system to read key information from SCALE's upload XML records.

Each interface touch point below can be run manually as well as through scheduled jobs as defined by Knipper. The specific schedule can depend on the Host system, warehouse processing times and SCALE interface execution times.

Warehouse alerts can be configured to notify Knipper employees when a download or upload interface has failed for any reason.

1.0 DOWNLOAD FROM HOST TO SCALE



Figure: Download Touchpoints

Commented [RS63]: This picture will be updated with the details of host/middleware used by Knipper

Commented [RS64R63]: Please advise the details on the host and middleware system so that this can be updated.

Commented [NC65R63]: NC12022024: Vimal to provide

1.1 Item Master

For majority of Knipper's accounts (Companies), Item master is managed manually in SCALE as there is no EDI capabilities currently. The Item download interface is automated only for EDI accounts. For EDI accounts, the Host system downloads a record into the SCALE Item Master when an item is modified or created. If an item already exists in the Item Master, then the record is flagged as a change and SCALE updates the existing item with the latest information bridged from the host. Information such as item number and item description are downloaded from the host system into SCALE. Item unit of measure and item cross reference is also downloaded as part of item master interface.

1.2 Receipts

For majority of Knipper's accounts, Receipt creation is managed manually in SCALE as there is no EDI capabilities currently. The Receipts download interface is automated only for EDI accounts. For EDI accounts, Receipts will be interfaced to SCALE. This data is sent immediately upon creation of ASN record in Host and could sit in SCALE potentially across several shipments or deliveries until the entire receipt has been received (or manually closed). The information downloaded to SCALE from host will always contain both header and detail level information. This is the typical format for Knipper for ASNs and Returns. This data is illustrated in the table below:

SCALE Table	Description
Receipt Order Header	Header Level Receipt Data
Receipt Order Detail	Line Item Detail Receipt Data

In Future, Knipper may explore capability to send container level information. These true ASN records are periodically processed within SCALE to create Receipt records. The information downloaded to SCALE will always contain header, detail, and container level information. This data is illustrated in the table below:

SCALE Table	Description
Receipt Order Header	Header Level Receipt Data
Receipt Order Detail	Line Item Detail Receipt Data
Receipt Container	Box level Receipt Data

1.3 Shipments

For majority of Knipper's accounts, Shipment creation is managed manually in SCALE as there is no EDI capabilities currently. The Shipment download interface is automated only for EDI accounts. For EDI accounts, Shipments are created in Host and Host reformats the data into SCALE format and sends with a warehouse-specific shipment

header, one or more shipment details, and optional shipment comments in SCALE. The shipment header contains information such as customer, customer address, ship to address, carrier, and scheduled ship date information. The shipment detail is the item level detail for the shipment containing information such as item and ordered quantity. Comments can be linked to the shipment header or details to validate additional processing requirements within the warehouse. Until the shipment has been waved, SCALE can process updates and deletions from the Host against the shipment.

2.0 UPLOAD FROM WM TO HOST

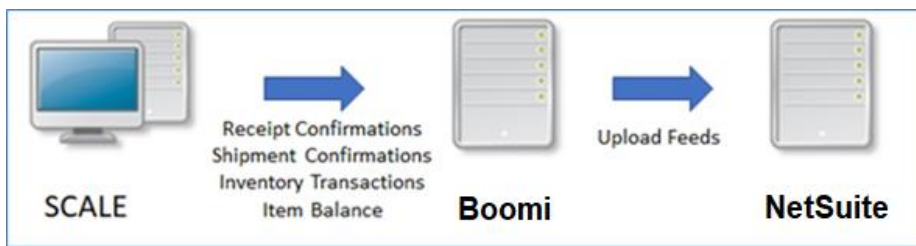


Figure: Upload Touchpoints

Commented [RS66]: This picture will be updated with the details of host/middleware used by Knipper

Commented [RS67R66]: Please advise the details on the host and middleware system so that this can be updated.

Commented [NC68R66]: NC12022024: Vimal to provide

2.1 Receipt Confirmation

The Interface Data option creates the receipt upload files from SCALE for all receipt containers that have reached the status 'Closed'. The receipt upload files are the output of the receiving processes within SCALE. The upload includes receipt header, receipt detail, and receipt container.

For Knipper, the upload will be performed at container level when the status of the receipt container reached 'Closed'. This is a global setting and will apply for all kind of receipt types.

2.2 Shipment Confirmation

The Interface Data option creates the shipment upload files from SCALE for all shipments that have reached status 'Load Confirm Pending' (shipment has been loaded to the truck). The shipment upload files are the output of the outbound process within SCALE. These files are generated after the execution of the load confirmation. The upload includes shipment header, shipment detail, shipment comment, and shipping container information.

2.3 Inventory Transactions

SCALE maintains 4-wall inventory at a detailed level and communicates any changes in inventory levels to Host through the Inventory Transactions Interface.

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All status changes and quantity changes in inventory are eligible to be uploaded to host system. During the cycle count process if the initial count produces discrepancy, then that transaction (suspense quantity) is also uploaded as inventory transaction upload to host.

If a receipt is manually closed in SCALE, the receipt close transaction is also uploaded as part of inventory transaction upload.

2.4 Item Balance

The total on-hand inventory for a given item and lot combination and inventory status can be uploaded to the host system on-demand through the Interface Data option in SCALE, or on a scheduled basis as configured in the Scheduled Job option. Any time the total on hand quantities are needed to compare against the host system, this option can be run and sent to the host system for reporting & comparison purposes.

1. For Knipper, include 0 inventory items in item balance upload will be set to No.
2. Item balance will not include inventory from receiving dock locations.
3. Item balance will include inventory from shipping dock locations.
4. Update and Delete for the Download Touchpoints

As part of download interface SCALE allows Host to modify or delete the records that were interfaced to SCALE. There are certain validations that SCALE performs before allowing these changes. The Validations are:

1. Shipment can be Updated / Deleted when the Leading and Trailing status of the shipment are in status of In Pool. Based upon the Action Code sent as part of the interface, SCALE determines what action (Insert, Update, Delete) needs to be carried out.
2. Receipts can be updated / deleted when Leading and Trailing status is in "Check In Pending".

Commented [RS69]: This is the recommendation so that Receipts in the Host system can be closed automatically in case of short receiving scenario. Need to confirm with Knipper if it is possible to implement this on the host system..

Commented [NC70R69]: NC12022024: We close when we receive the remaining inventory for receipt

Commented [RS71R69]: This recommendation is more on SCALE generating an electronic confirmation that Receipt has been still closed if it cannot be completely received. If this is not a valid use case, we can resolve this.

II. INBOUND

3.0 PROCESS OVERVIEW

Knipper performs Item Level Receiving for all receipt types. Once the ASN is created, host system will send receipt download records to SCALE. From this, the Receiving Worksheet

can be printed to assist in the receiving process. Each item and quantity are then received. SCALE will then perform locating to identify a suitable storage location based on item characteristics. Lastly, the system directs the receipt containers to a storage location for putaway.

Each step mentioned in the overview are described in detail in the sections below.

Note: Knipper currently uses a customized Receiving worksheet and same will be ported over to Active SCALE.

Receipt Num:TB_ERMA_01		RECEIVING WORKSHEET		ERP Order #: 8	
Receipt ID Type:ERMA				Receiving Date: 2/25/2021	
Warehouse TB_UKDC 375 Kings Mill Rd GA 30253 US		Source Info SL901 Total Quantity Rcvd: 3			
Line #	Item	Item Description	Total Qty	Open Qty	
1	5JMW01-000-OS	LOVE RELENTLESSLY EAU DE PARFUM SPRAY- 3.4 OZ / 100 ML	3	3	
Tare:			L	W	H
					LBS

Figure: Receiving Worksheet sample

4.0 PRE-RECEIVING

4.1 Receipt Creation

4.1.1 Interface

Host will produce receipt download records that are interfaced to SCALE for the ASNs and Returns. Host sends these records as download messages to SCALE. These records are then processed and validated through the SCALE interface to ensure the data format is correct.

4.1.2 Receipt ID Types

To group receipts from SCALE screens and to drive processing rules, it is helpful for SCALE to store different *Receipt ID Types* and *Receipt Types*.

The following list of Receipt ID Types will be configured in the system:

- ASN
- Packing List

- Returns

Receipt type is a free format field on a receipt record. It is commonly used to identify what type of supplier sent the receipt, such as manufacturer, vendor, etc. The system does not validate this value.

1.1.3 Create Receipt from Purchase Order Insight

Knipper does not create receipts manually from Purchase Order.

1.1.4 Create Receipt using Receipt Insight

Knipper does create receipts manually using **Receipt Insight** screen. For majority of Knipper's accounts, Receipts are not interfaced automatically. The Receipts are created manually based out of the packing List document received from the vendor.

1.1.5 Create Receipt from Shipment

Knipper does create receipts from shipments to receive the returns. The Receipt from Shipment is used to create the Returns receipt for the undelivered shipments which are returned with all the original packaging.

Note: Currently Knipper uses this functionality to create Receipt from Shipment. However, the receipt lines are created without the lot information populated. Knipper would like to enhance the functionality to populate lot information based on lot shipped on the shipments **[EX40 – Custom Receipt from Shipment]**.

1.1.6 Blind Receipts

Knipper does leverage blind receiving in some scenarios when there is no ASN receipt exists.

4.2 Viewing Receipts

All receipts that are downloaded or created in SCALE can be viewed from the **Receiving Insight** screens. The lines can be viewed using **Receipt Line Insight**. If ASNs are downloaded or created, they can be viewed using **Receipt Container Insight**.

Commented [SM72]: > 1.1.5 does not provide much detail to how the new system creates receipts from shipments.

>Need to be able to modify receipt data.

> Once LPs are created, need the ability to locate/un-locate at the line level.

Commented [NC73R72]: This can be resolved. MAH to provide EX 40

Commented [SM74]: Add Lot and Expiration Date to this.

Commented [RS75R74]: We will have a separate design document which will have all the details.

Commented [NC76R74]: NC12022024: this is resolved.

Commented [SM77]: Want the ability to Blind Receive in the future.

Commented [NC78R77]: Knipper leverage blind receipts. MAH to update

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The screenshot shows the 'Receipt insight' application. At the top, there are four summary cards: Receipts (93), Lines (215), LINKS (123.4k), and Volume (18.3m). Below these is a 'select columns' bar. The main table lists receipt lines with columns: Receipt id, Receipt id type, Company, Trailer id, Trailer location, Receipt date, Closed date time, Leading Status, and Trailing Status. The table shows several rows of data, including receipt IDs like 20200407073427991 and ASNTES1001. On the right side, there are two buttons labeled '3 Lines' and '3 Containers'.

Receipt id	Receipt id type	Company	Trailer id	Trailer location	Receipt date	Closed date time	Leading Status	Trailing Status
20200407073427991	Purchase Order				4/27/2022		Check In Pending	Check In Pending
ASNTES1001	Purchase Order				1/20/2022		Closed	Check In Pending
hghghgh	Purchase Order				1/17/2022		Check In Pending	Check In Pending
PO0002-02	Purchase Order				12/7/2021		Locate Pending	Check In Pending
PO28MARICH0001-01	Purchase Order				3/28/2022		Closed	Check In Pending
POS436434	Purchase Order				4/27/2022		Check In Pending	Check In Pending
PO67823	Purchase Order				2/26/2021		Closed	Check In Pending

Figure – Receipt Insight

The screenshot shows the 'Receipt line' application for receipt ID 3270011053.2. It features a sidebar with navigation options: Item info, Quantity info, Item dimensions, Item characteristics, Processing values, Reference info, Categories, and User defined. The main area displays detailed information for the selected item, including item ID (3270011053.2), company (HARTZ), description (551 STA CHATNA 1402), lot controlled status, and quantity information (total quantity 20500, open quantity 20500, original net quantity 20500). It also shows item dimensions and other processing values.

Figure – Receipt Line Insight

Receipt container insight

Account ▼

Basic criteria

License plate

Receipt id

Order receipt number

Company

Warehouse

Order receipt number

Company

Order receipt number

Order receipt number

Order receipt number

Advanced criteria

Receipts1

License plates33

Quantity6.5k

Weight27.4k

Volume147.3m

select columns

Item	License plate	Group number	Status	Item	Description	Company	Quantity	Quantity unit	Receipt id	Status failed
000001	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000002	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000003	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000004	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000005	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000006	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000007	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000008	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000009	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000010	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000011	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000012	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000013	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000014	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000015	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000016	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	
000017	0		Closed	000000000.0	PET CENTRAL CDS SECT	HARTZ	200	EA	3410-130000	

Figure – Receipt Container Insight

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4.3 Generate Receiving Documents

Once a receipt has been created and validated, documents can be printed against it. Knipper prints the receiving Work sheet (**DOC01**) for all receiving flows. This document contains various data from the receipt that can be printed from SCALE by selecting a receipt from the **Receipt Insight** and choosing the Print Selected Documents option from **Actions**. The Receiving Worksheet is used to aid in the systematic receiving process and serves as a receiving journal after the receiving process.

Commented [SM79]: Update for 3PL to use DOC01 and DOC02; OP-101D: Receiving Worksheet & 3PL-QA-001A: QA Inspection of Incoming Material

Commented [NC80R79]: MAH to update to the 2 documents

Commented [RS81R79]: Please confirm if the 2nd document is OP-101D or QA-001A. Also, I assume these are currently being used and will be ported over.

Commented [NC82R79]: NC12022024: 1st document OP-101D and 2nd document QA-001A. Yes we are porting. This is resolved

The screenshot shows the 'Receipt Insight' interface. On the left is a sidebar with 'Basic criteria' and 'Advanced criteria' sections. The main area displays a table of receipts with columns: Receipt id, Company, Receipt id type, Trailer id, Trailer location, and Receipt date. A summary bar at the top shows 103 Receipts, 1.2k Lines, and 861.9k Units. An 'Actions' menu is on the right.

Receipt id	Company	Receipt id type	Trailer id	Trailer location	Receipt date
✓ 1800102095	62	Advanced Shipping Notification	5338		3/21/2018
☐ 180015139	764	Advanced Shipping Notification			9/21/2018
☐ 180017662	1234	Advanced Shipping Notification			9/24/2018
☐ 180019006	764	Advanced Shipping Notification			9/24/2018
☐ 180200996	764	Advanced Shipping Notification			9/25/2018
☐ 180200966	764	Advanced Shipping Notification			9/26/2018
☐ 180200928	764	Advanced Shipping Notification			9/26/2018
☐ 180200968	764	Advanced Shipping Notification			9/28/2018
☐ 180201055	764	Advanced Shipping Notification			9/28/2018
☐ 180201064	764	Advanced Shipping Notification			9/28/2018
☐ 180200962	764	Advanced Shipping Notification			10/2/2018
☐ 180201057	764	Advanced Shipping Notification			10/3/2018
☐ 180201032	764	Advanced Shipping Notification			10/3/2018
☐ 180201052	764	Advanced Shipping Notification			10/7/2018

Figure – Printing Receiving worksheet

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RECEIVING WORKSHEET			
Receipt Num: NODIMS-01		ERP Order #:	
Receipt ID Type: ASN		Receiving Date: 4/27/2020	
Warehouse		Source Info	
TN 3819 Logistics Way TN 37013 US		Total Quantity Rcvd: 20	
Line #	Item	Item Description	Total Qty
1	3000	Bleach	1
TVHT:		0 L	0 W
		0 H	0 LBS
Line #	Item	Item Description	Total Qty
2	34000032129	HsyNgt DkCh/AI 10.56oz	20
TVHT:		10 L	0 W
		7 H	8 LBS

Figure – Receiving work sheet sample

5.0 APPOINTMENT SCHEDULING

5.1 Operations

Knipper handles inbound appointment scheduling currently outside of SCALE. With this implementation, Knipper continues to handle the appointment scheduling outside of SCALE.

A typical workflow for using systemic appointment scheduling is listed here for reference if Knipper wants to leverage it in the future.

To better facilitate the receiving, Knipper personnel may assign an appointment for the given receipt using the **New Appointment** function in the **Receipt Insight** screen option. This is documented for reference purposes in case appointment visibility is needed inside SCALE. This needs a receipt ID visibility in SCALE in advance. Users can then go back and view what receipts are scheduled for a given day by viewing the Appointments folder in the **Receipt Insight**.

Commented [SM83]: Knipper will have discretion to schedule in or out of scale.

Commented [NC84R83]: MAH to confirm no impact for Knipper using discretion

Commented [RS85R83]: Please note that appointments can only be scheduled for Inbound shipments (Receipts). There is no functionality to manage appointments for shipping loads.

Commented [NC86R83]: NC12022024: This is resolved.

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Figure – Appointment Schedule Window

This allows for enhanced management of dock door locations, as well as the personnel required to unload the trailers at the dock. Users may enter the following information when scheduling an inbound appointment:

- Trailer ID
- Dock Door
- Start Date/Time
- End Date/Time

Note: Appointments can only be scheduled for open receipts in SCALE. Appointments can't be created without associating a receipt with it.

A graphical calendar screen for managing inbound receipt appointments is also available. The screen shows activity for the day per dock door and allows the user to easily see and schedule open spots on the calendar. From this screen, the user can preview appointment details, schedule new appointments, change existing appointments, or delete appointments.

Figure – Receiving Appointment Schedule Window

6.0 UNLOADING

6.1 Unloading Trailer

Upon the arrival of the truck and verification of the appointment schedule at the guard check, delivery documentation is obtained from the driver and matched to the corresponding receipts in SCALE. The delivery documentation includes a packing slip, BOL, Seal ID, and other information. Once it is verified that the receipt exists in SCALE, the truck is docked at the assigned door and items are unloaded from the truck to that pre-receiving dock area. For Knipper, this area is the receiving dock.

Note: Seal id and Truck id is available as paperwork for the receipts. This information is not available in the interface. Knipper manually updates this information on the receipt header.

Some of the floor-loaded trailers may not have decent stacking/layering of the product. Multiple people may unload such trailers. The same item on the receipt may be present in the nose, mid, or tail of the truck and that leads to some single-item pallets taking more time to build before systemic check-in can begin.

At this point, a visual QC check is conducted outside of SCALE if required. Knipper may perform a supplier compliance check outside of SCALE and leverage user-defined fields on the receipt header to capture compliance codes. This supplier compliance process is out of the scope of this implementation is not defined here.

Once all items are unloaded and verified, the truck leaves the warehouse. If mixed item pallets are unloaded, they are broken into single item pallets. For lot tracked items, always, single item, single lot pallets are created.

Tribal knowledge is leveraged at the DCs, for receipts by vendor, to decide using putaway groups or not. Since moving heavy items across pallets is labor intensive, the receiving clerks makes this decision during trailer unload. Manhattan recommends flagging heavy items on the receiving worksheet to provide visual aid to the clerk and users before unloading begins.

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7.0 QUALITY AUDIT

7.1 Inbound Quality Control

Knipper performs a visual QC as an SOP on the inbound products and receives with a default status of 'QC Hold'. The inventory status will be set to 'QC Hold' for all the Inbound LPNs even though physical inspection is performed on few selected LPNs. The quality inspection for the selected LPNs will be performed though all LPNs will be putaway to the reserve location before that.

The inventory status of the LPNs will be changed based on the outcome of the quality inspection. The status will be updated as below

1. QC Pass – User will change Inventory status from QC Hold to Available from Insight screen or Warehouse mobile. Note: The user will search LPNs by one lot at time to update the status as one lot is received on one receipt.
2. QC Fail – The disposition of the inventory will be updated based on the decision from the customer. The LPNs will be transferred to a designated location depending on the final disposition. If the final disposition is destruction, the LPNs will be transferred to the designated location.

Commented [SG87]: Please mention that User will search by Lot and 1 Lot is 1 Receipt

Commented [NC88R87]: MAH to add

Commented [RS89R87]: Added

Commented [SG90]: Will they then transfer to another location?

Commented [NC91R90]: MAH to respond

Commented [RS92R90]: Added

8.0 RECEIVING / PALLETIZATION

After unloading the truck and physically building the pallets if needed, and executing manual quality audit SOP, systematic receiving is then initiated via a check-in process. Checking In product is the process that creates the inventory inside of SCALE based on Receiving Preference of the user performing the action.

8.1 Item Receiving (Pallets) – 3PL/Track and Trace Receiving

Item level receiving is used for receiving single item, single lot pallets. The receiving clerk uses Receiving Worksheet (**DOC01**) to decide and use this receiving preference. These pallets may either be manually built pallets or pre-built pallets.

The receiving preference is configured in SCALE as below:

- Create Putaway Work: Yes
- Nest During Check In: No
- Disposition Code Required: No
- License Plate Assignment: System
- RF Workflow: Header – Item
- Execution Method: Check in and Locate (immediate)
- Container Locating Method: Parent
- Default Inventory Status – QC Hold/Client Hold

Figure - Receiving Preference Configurations

The user selects *Receiving* option from Warehouse Mobile Main Menu. User selects the Item Level Receiving option.

Once selected, the user scans:

- Receipt ID (from Receiving work sheet **DOC01**)
- Item code on the product
- Quantity (Key in) in the applicable UoM. The lowest UoM will be always defaulted, and the user selects the UoM to receive.

[EX39 – GS1 Label Scanning] – To extract GTIN, Lot and Expiration Date from GS1 label. If no label, the details will be entered.

After assigning a system-generated license plate number, the user either scans the Lot or keys in (if no barcode).

If the lot and expiration date are interfaced on the receipt detail, the information is prepopulated.

If the Item is Serial Number Tracked for Inbound, the user will scan the serial number from the barcode on the product or keys in (if no barcode).

At this time, SCALE locates the LPN and putaway work is created and receipt Container Label (**LBL01**) is printed. The label is applied at the front bottom case of the pallet. SCALE determines a putaway location based on the locating rule assigned during the Check-In process.

Commented [SM93]: Need to locate at line level

Commented [NC94R93]: MAH to confirm if this can be done.

Commented [RS95R93]: The locate is always at the LPN level even though same rule can be assigned to a line.

Commented [NC96R93]: NC12022024: this is resolved.

The system-generated label (**LBL01**) has flowing key information on it and is the same label currently used.

- License Plate #
- SKU (Item)
- Locating location
- Lot
- Expiration Date

Note: Knipper uses a customized LPN label and same will be ported over to SCALE Active.

The user continues to receive the next item for the same receipt or exit the current receipt and start over with a new receipt ID. Exiting a receipt on the Warehouse Mobile does NOT close that receipt. Users can go back to that receipt at any point in time.

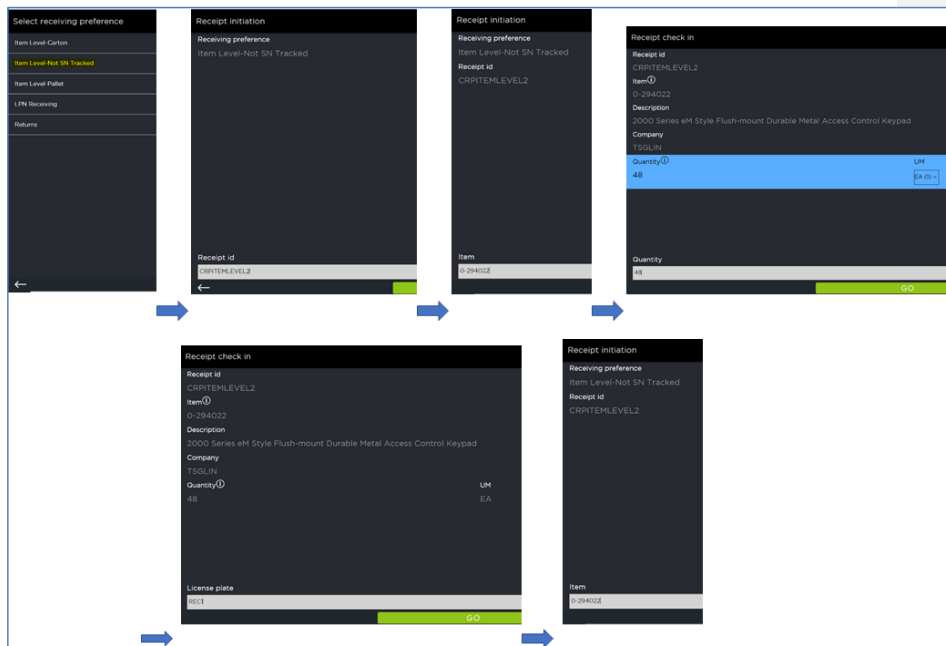


Figure: Item Level Receiving flow

Note: [EX37 – DSCSA Inbound Processing] will be developed to enhance the Item Level Receiving to receive DSCSA tracked Items. The screen will be customized, and

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details will be documented in the extension specification document. The DSCSA Inbound Processing module will only be developed for Warehouse Mobile (RF).

Receipt workbench

KMW103906

Parent container id

Item

Partial line Full line All lines Cancel all

Lines

Line	Item	Description
1	59148-008-95	Abilify 10mg 1unitx1 sleeve/7 tabs per sleeve 48 units per case

1 - 1 of 1 records

Containers

Receipt info

Check in partial quantity

Item: 59148-008-95

Company: 62

Description: Abilify 10mg 1unitx1 sleeve/7 tabs per sleeve 48 units per case

Open quantity: 43,680 UOM: Each

Check in quantity: 43,680 UOM: Each

Locating rule: KMW Abbvie REFG RSV

Location:

Check in Cancel

Commented [SM97]: Need to re-verify EX37B is still accurate.

Commented [SM98R97]: Need to be able to scan SSCC and verify quantity.

Commented [NC99R97]: This is resolved. New documents coming for MAH

Figure – Receipt Workbench Receiving

Check in partial quantity

Item: 5513-001

Company: 727

Description: XEN STERILE INJECTOR US_R3D_PLANT1619

Open quantity: 10 UOM: Each

Check in quantity: 10 UOM: Each

Locating rule: KMW Allergan RX RSV Rule

Location:

Check in Cancel

Lot entry

Item: 5507-001R2

Company: 727

Description: XEN Sterile Injector - (Trade)_U_GER

Lot: 12062024 Expiration date: 1/31/2025

Quantity: 10 UOM: Each

Save Cancel

Figure – Receipt Workbench Receiving – Lot not Prepopulated

Note: Receipt Workbench will also be utilized for receiving Receipts in OHW (NJ) and KMW (IN) warehouses.

8.2 Damage Receiving

Damage receiving will be used for receipts of receipt type ASN and Packing List when the inventory is damaged, or inventory can't be sold. Receipt will be at line level. The inventory will be received on a single item pallet.

Commented [SM100]: Desktop Check In - need to be able to verify the Lot and Expiration Date to be checked in.

Commented [NC101R100]: MAH to confirm and add

Commented [RS102R100]: If the lot/expiration date is already assigned at the line level, then no additional scan is needed. If not, user is asked to scan. Please let me know if additional screenshots are needed to illustrate all the steps.

Commented [NC103R100]: NC12022024: Please provide additional screenshot

Commented [RS104R100]: Added a screenshot with an example where lot is not pre-populated

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Receiving Preferences - Edit Existing

File Actions Help

Preference Name: **KDC DAMAGE RECEIV**

Description: **KDC DAMAGE RECEIVING**

General RF Workbench Assigned Users User Defined Data

Container Locating Method

☐ Parent

☒ Child

Advanced Options

☒ Disposition Code Required

☐ Reason Code Required

☐ QC Inspection Active

☒ Verify Item Dimensions

☐ Verify Item Unit of Measure

☐ Inactive

OK Cancel

Figure - Receiving Preference Configurations

Receipt workbench

10890528

Parent container id

Item

Partial line Full line All lines Cancel all

Lines

Line	Item	Description
25099376	USA1SHDM00916	novoeight HCP Pillar Purity (1+1 Piece, SW Pack Of 10)
25099387	USA1SHDM04061	HCP Leave Behind (1+1 Piece, SW Pack Of 10)
25099398	USA1SHDM04980	tretten PI (1+1 Piece, SW Pack Of 20)
25099399	USA1SHDM05031	novoeight PI (1+1 Piece, SW Pack Of 20)
25099400	USA17BIO00011	RBD Brochure Update (English) (1+1 Piece, SW Pack Of 10)
25099401	USA17BIO00208	tretten HCP Brochure (1+1 Piece, SW Pack Of 10)
25099402	USA17BIO02140	AH Toolkit (1+1 Piece, Pack Of 10)
25099403	USA17BIO01999	rebisyn PI
25099404	USA17BIO02170	rebisyn HCP Core Visual Aid (1+1 Piece, SW Pack Of 10)
25099405	USA17BIO03979	Formulary Plan FWH Product Comparison Plan

Check in partial quantity

Item: USA1SHDM00916

Company: 61

Description: novoeight HCP Pillar Purity (1+1 Piece, SW Pack Of 10)

Open quantity: 30 UM: Each

Check in quantity: 30 UM: Each

Locating rule: Returns_Loc Rule Location:

Reason code: Damages Disposition code: Rework

Check in Cancel

Figure – Damage Receiving

Damage Receiving will be performed from the Receipt Workbench. The user selects the Receipt Workbench option from the Receiving menu. The user selects Damage Receiving Preference (OHW and KMW). Once complete, the user scans or enters the Receipt ID from the Receiving Worksheet (DOC01). Next, the user selects the item and enters the quantity to be received for the item. If the item doesn't have unit of measure information or dimension information, the system notifies the user of this. The user selects the Disposition Code. A Disposition Code indicates the status of the inventory within an LPN.

Commented [SM105]: Upon receipt, LPN placed on "Received Damaged" hold.

Commented [NC106R105]: This can be resolved

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For example, the item(s) could be in good condition, damaged, broken, etc. These codes are linked to a specific locating rule that the system can locate product differently based on its disposition code. At this time, SCALE performs locating. The work unit of the putaway work will be the LPN value.

The LPNs will be putaway to the designated damaged area with appropriate Damaged status awaiting the communication on further action. The inventory will be shipped for destruction upon approval from Knipper's customer.

8.3 Returns

Customer returns are received and stocked into the inventory based on the disposition. Returns will be stored in a designated area.

Return receiving is used for receipts of receipt type RA/RMA that are downloaded to SCALE from Host. For some returns, receipts are created using Shipments.

Returns use their receiving preference, and the default inventory status is 'Available/Awaiting Client' and located to a returns location. Knipper will maintain designated Returns location for each warehouse.

Commented [SM107]: Default location for Damages: KMW-QA-Damage

Commented [NC108R107]: This can be resolved

Commented [SM109]: Does not go on QC hold, should go onto either available or awaiting client disposition depending on the nature of the product returned.

Commented [NC110R109]: MAH to update

Commented [RS111R109]: updated

Figure - Receiving Preference Configurations

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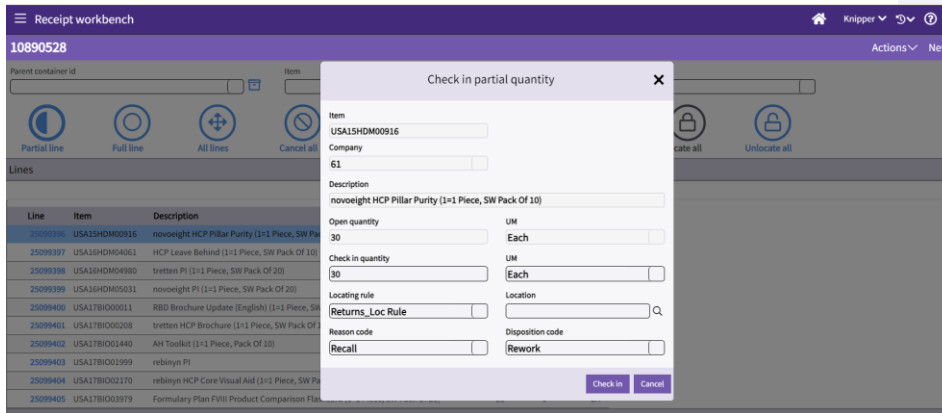


Figure – Returns Receiving

Returns Receiving will be performed from the Receipt Workbench. The user selects the Receipt Workbench option from the Receiving menu. The user selects Returns Receiving Preference (OHW/KMW). Once complete, the user scans or enters the Receipt ID from the Receiving Worksheet (DOC01). Next, the user selects the item and enters the quantity to be received for the item. If the item does not have unit of measure information or dimension information, the system notifies the user of this. The user selects the Disposition Code. A Disposition Code indicates the status of the inventory within an LPN. The user selects a Reason code indicating the reason for Returns. For example, the item(s) could be in good condition, damaged, broken, etc. These codes are linked to a specific locating rule that the system can locate product differently based on its disposition code. At this time, SCALE performs locating. The work unit of the putaway work will be the LPN value.

The LPNs will be putaway to the designated Returns area with appropriate inventory status awaiting the communication on further action. The inventory will be shipped for destruction upon approval from Knipper's customer.

8.4 Blind Receiving

Blind receiving is used for receiving when there is no ASN receipt exists. The operator will create a receipt using the RF on the fly and add line items to the receipt.

The receiving preference is configured in SCALE as below:

- Create Putaway Work: Yes
- Nest During Check In: No
- Disposition Code Required: No
- License Plate Assignment: System

Commented [SM112]: PL #20 - Good for Rep Returns using integrated RMA to receive against.

Need a process to address undeliverable DTP (Direct to practitioner). Want to be able to scan the OB container number from original shipment, pull the data of the shipment to receive back in.

Commented [NC113R112]: MAH to confirm

Commented [RS114R112]: This section describes the process of receiving into the warehouse. However, the receipt creation process you are asking will be documented in a separate design document as it will be a custom process.

Commented [NC115R112]: NC12022024: this is resolved.

Commented [SM116]: Do the users not verify lot and expiration date?

Commented [NC117R116]: MAH to add/confirm

Commented [RS118R116]: Knipper would like to pre-assign lot/expiration into the lines prior to receiving. Hence, the user will not scan lot/expiration during receiving

Commented [NC119R116]: NC12022024: this is resolved.

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- RF Workflow: Blind
- Execution Method: Check in and Locate (immediate)
- Container Locating Method: Parent
- Default Inventory Status – QC Hold/Client Hold

The user selects *Receiving* option from Warehouse Mobile Main Menu. User selects the Blind Receiving option.

Once selected, the user enters:

- Receipt ID
- Item code on the product
- Quantity (Key in) in the applicable UoM. The lowest UoM will be always defaulted, and the user selects the UoM to receive.

After assigning a system-generated license plate number, the user either scans the Lot or keys in (if no barcode). If the Item is Serial Number Tracked for Inbound, the user will scan the serial number from the barcode on the product or keys in (if no barcode).

At this time, SCALE locates the LPN and putaway work is created and receipt Container Label (**LBL01**) is printed. The label is applied at the front bottom case of the pallet. SCALE determines a putaway location based on the locating rule assigned during the Check-In process.

The system-generated label (**LBL01**) has flowing key information on it and is the same label currently used.

- License Plate #
- SKU (Item)
- Locating location
- Lot
- Expiration Date

Note: Knipper uses a customized LPN label and same will be ported over to SCALE Active.

The user continues to receive the next item for the same receipt or exit the current receipt and start over with a new blind receipt ID. Exiting a receipt on the Warehouse Mobile does NOT close that receipt. Users can go back to that receipt at any point in time.

Commented [SM120]: Need to locate at line level

Commented [NC121R120]: MAH to confirm if this can be done.

Commented [RS122R120]: The locate is always at the LPN level even though same rule can be assigned to a line.

Commented [NC123R120]: NC12022024: this is resolved

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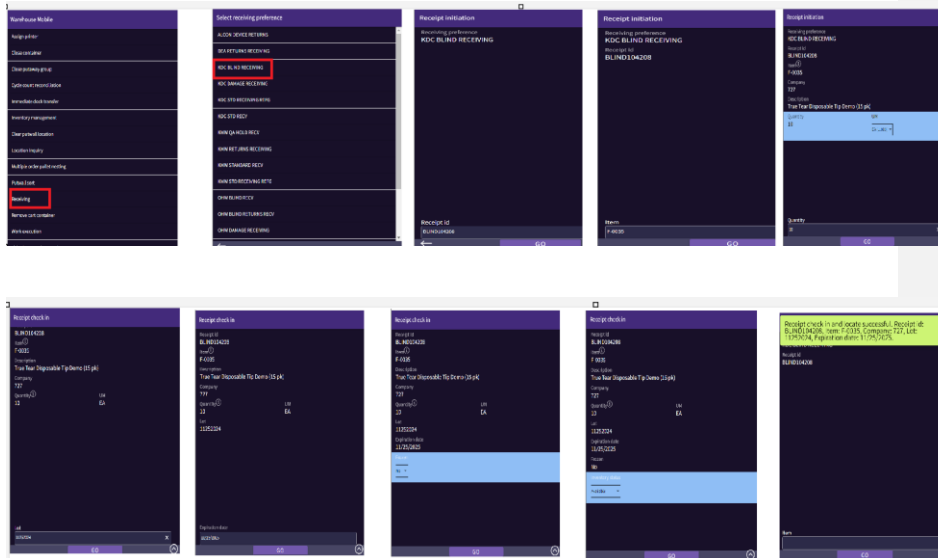


Figure: Blind Receiving flow

8.5 Disposition Codes

Knipper uses disposition codes for damages and returns primarily. There are multiple overlapping disposition codes setup by site. Knipper will revisit them during the build phase for cleanup.

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Receipt check in

CRP-RETURN
Item ⓘ

009851
Description

LACTATED RINGERS INJ 12X1000ML LF
Company

WA
Quantity ⓘ

1
Lot

D4567
Expiration date

12/26/2024

Disposition code

06 - Quality Control
06 - Quality Control Vault
40 - Quality Control
50 - Quality Control
60 - Quality Control

CANCEL OK

Figure – Disposition codes

8.6 Verify Receipts

For receipts where all receipt details are received complete, SCALE automatically closes the receipt upon the putaway of the last LPN. If all detail lines were not received completely (such as with a short ship), the receipt must be closed manually. To close a receipt manually, a user selects the receipt that is completed and uses the Close Receipt action from the **Receipt Insight**. Upon confirmation of the close, SCALE updates the receipt status to Closed and does not allow additional products to be received.

Note: A closed receipt can be re-opened in SCALE to allow additional receiving

The screenshot shows the 'Receipt Insight' application. On the left is a sidebar with 'Basic criteria' and 'Advanced criteria' sections. The main area displays a table of receipts with columns: Receipt id, Company, Receipt id type, Trailer id, Trailer location, and Receipt date. A table summary at the top shows 103 Receipts, 1.2k Lines, and 861.9k Units. On the right, an 'Actions' menu is visible with options like New, Edit, Delete, New container, New line, Assign trailer id, New appointment, Edit appointment, Delete appointment, Cancel close, Check in, Create pre-check in containers, Close, Print preview, Immediate needs, Print default docs, Print selected docs, and Yard check in & out. The 'Close' option in the actions menu is highlighted with a red box.

Receipt id	Company	Receipt id type	Trailer id	Trailer location	Receipt date
1800113305	62	Advanced Shipping Notification	5338		3/21/2018
1800191399	764	Advanced Shipping Notification			9/21/2018
180017862	1234	Advanced Shipping Notification			9/24/2018
180019006	764	Advanced Shipping Notification			9/24/2018
180200998	764	Advanced Shipping Notification			9/25/2018
180200956	764	Advanced Shipping Notification			9/26/2018
180200928	764	Advanced Shipping Notification			9/26/2018
180200968	764	Advanced Shipping Notification			9/28/2018
180201055	764	Advanced Shipping Notification			9/28/2018
180201064	764	Advanced Shipping Notification			9/28/2018
180200602	764	Advanced Shipping Notification			10/2/2018
180201057	764	Advanced Shipping Notification			10/3/2018
182201032	764	Advanced Shipping Notification			10/3/2018
180201052	764	Advanced Shipping Notification			10/7/2018

Figure – Receipt Insight – close receipt

9.0 EXCEPTIONS

9.1 Items missing weight or dimensions (New Items)

Knipper captures the dimensions for the new items using the Cubiscan which is updated manually in SCALE by the operations team.

9.2 Overages

Knipper does not over receive. If during unloading the truck, the user determines the inventory unloaded is more than the Receipt line quantity, then the receiving user reaches out to receiving supervisor. A new PO/Receipt is generated by the Host-based on the request from operations to receive the additional items into SCALE. The receiving team manually moves this product away from the receiving dock to avoid any obstruction until the new Receipt is available.

Config Note: Receiving Preferences will have 'Allow over receiving' disabled.

9.3 Shortages

When a receipt is not received in full, the user must manually close it, if the rest of the balance is no longer expected.

Commented [SM124]: PL#26

Commented [NC125R124]: MAH to confirm

Commented [RS126R124]: Please elaborate on PL#26

Commented [RS127R124]: This is currently being reviewed for HLE. A detailed design document will be created if Knipper would like to go ahead with the Integration.

Commented [NC128R124]: NC12022024: this is resolved.

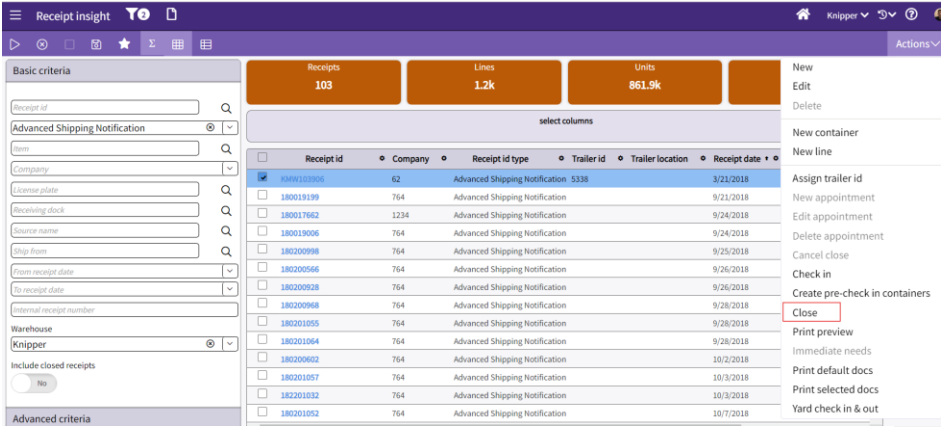


Figure – Closing receipt Shortages.

9.4 Receipt not in SCALE

In this case, the product is delivered by a carrier for a receipt that is not in SCALE. The warehouse cannot Check-In the product without the receipt information in SCALE. The Operations team will work with the appropriate team members (IT/Client account Representative) to create the Receipts.

9.5 Unknown Product

When an item is available on the trailer but exists neither on the receipt nor the item master, it is called an Unknown Product. The check in clerk informs the supervisor, who coordinates with the supplier team. Knipper manually moves this product away from the receiving dock to avoid any obstruction. The supervisor coordinates with the procurement team and/or vendor, an item master is interfaced with this item if it must be received. A new receipt is then created in the host for this known (unknown before item master download) product and interfaced into SCALE.

9.6 Damages

Damaged goods are received under the Damage Preference. The inventory received is put in a Held Inventory Status (Damage Hold) based on the disposition code entered by user and assigned a Locating Rule to direct the product to a damaged area.

Note: Knipper configures a virtual multi-item and license plate tracked location called Supervisor/Error Location to handle locating exceptions. This is used to troubleshoot these exceptions.

9.7 Manually Closing the receipt

If a receipt is manually closed in Host, in receiving interface updates the receipt header with the closed date time field. No changes will be done at the receipt detail level.

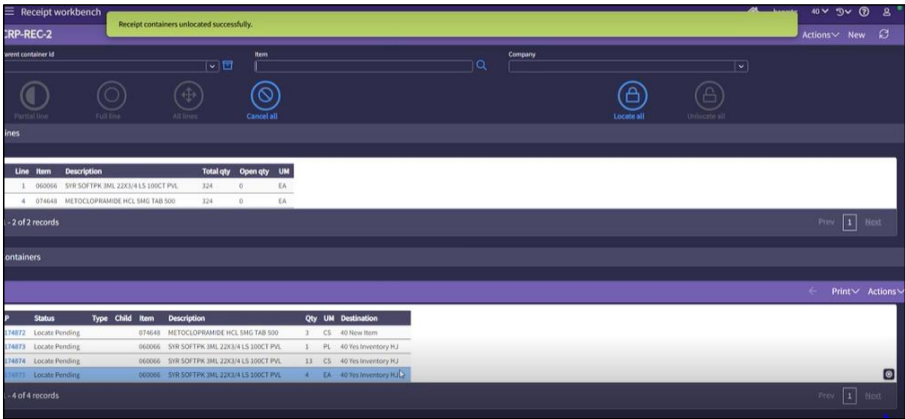


Figure – Closing receipt Shortages

9.8 Using Receipt Workbench for troubleshooting

Receiving supervisors may use **Receipt Workbench** for troubleshooting exceptions like locating failure as well receiving in some exception scenarios.

- Commented [SM129]: Receiving via workbench is in scope for exception process, such as multi lot issue.
- Commented [NC130R129]: MAH to remove out of scope and is in scope because there will be exceptions
- Commented [RS131R129]: Just removed the section to avoid the confusion
- Commented [NC132R129]: NC12022024: this is resolved.

10.0 PUTAWAY

Put away involves executing the task to move the inventory to the system-directed location from the receiving dock.

Locating takes each receipt container from the Check-In process and tries to find a place in the warehouse to store the product. Locating Rules are based on a series of sequences that use a combination of strategy and Location selection to try and find the right place in inventory for the item. If the system locates the receipt container to a location, work is created to move the product from the Receiving Dock to the destination location in the warehouse for all receipts.

Knipper uses a manual SOP to troubleshoot if a container is located to the 'Supervisor Location'. Transaction history and process history can be leveraged to review the locating rule sequences to determine the root cause. Knipper cancels the check-in (if not uploaded) for this container and check in after making the needed corrections.

10.1 Locating Rules

Once inventory has been checked in to SCALE, a putaway location is found using locating rules. Knipper would like to use item's categories to determine the locating rule.

To allow for easy addition of new items and locating rules, the locating rule is assigned on the Receipt Detail using the Locating Rule Assignment configuration. Knipper can still manually set the locating rule after the Receipt Detail is downloaded by opening the receipt line and selecting a different locating rule.

To allow for easy addition of new items and locating rules, the locating rule is assigned on the Receipt Detail using the Locating Rule Assignment configuration.

Configuration Note: The Receiving System Value Locating Rule Assignment During is set to Receipt Check-In.

	Locating Zone	Description
1	L-DEA Reserve	- Single Item - LPN Tracked - Max Lot = 1
3	L-DEA>Returns	- Multi Item - LPN Tracked - Max Lot = 0
4	L-Refrigerated	- Single Item - LPN Tracked - Max Lot = 1
5	L-OHW-Reserve	Single Item

Commented [SM133]: What is the Supervisor Location?

Commented [NC134R133]: MAH to confirm

Commented [RS135R133]: It is the ERROR location currently used by Knipper. If for some reason, SCALE cannot find a location for an LPN, it is the last resort to avoid the process to fail.

Commented [NC136R133]: NC12022024: This is resolved

Commented [SM137]: Why would you cancel check in, if putaway had the issue? That is downstream of check in, correct?

Commented [NC138R137]: MAH to confirm

Commented [RS139R137]: This is not talking about issues with Putaway rather correcting the destination location before the Putaway is being executed.

Commented [NC140R137]: NC12022024: this is resolved.

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		• LPN Tracked • Max Lot = 1
6	L-KMW-Reserve	• Single Item • LPN Tracked • Max Lot = 1
7	L-Freezer	• Single Item • LPN Tracked • Max Lot = 1
9	L-3PL-Reserve	• Single Item • LPN Tracked • Max Lot = 1
8	L-Damaged	• Multi Item • LPN Tracked • Max Lot = 0
10	L-KMW>Returns	• Multi Item • LPN Tracked • Max Lot = 0
11	L-OHW>Returns	• Multi Item • LPN Tracked • Max Lot = 0
12	L-See-Supervisor	• Multi Item • LPN Tracked • Max Lot = 0

Note: The locating zone listed above is a sample. Knipper will be configuring/porting many more additional locating zones created per building and some customer specific.

Locating Sequence for Item with item category2 = Refrigerated:

Sequence	Strategy	Location Selection	Split Quantity
10	Consolidate - Exclude differing lots	L-Refrigerated	No
20	Empty Location	L-Refrigerated	No
30	Use specific (multi-item) location regardless of status	L-See-Supervisor	No

Locating Sequence for Item with item category2 = DEA Controlled:

Sequence	Strategy	Location Selection	Split Quantity
10	Consolidate - Exclude differing lots	L-DEA Reserve	No
20	Empty Location	L-DEA-Reserve	No

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30	Use specific (multi-item) location regardless of status	L-See-Supervisor	No
----	---	------------------	----

Locating Sequence for Item with item category2 = General:

Sequence	Strategy	Location Selection	Split Quantity
10	Consolidate - Exclude differing lots	L-KMW-Reserve/ L-OHW-Reserve	No
20	Empty Location	L-KMW-Reserve/ L-OHW-Reserve	No
30	Use specific (multi-item) location regardless of status	L-See-Supervisor	No

Locating Sequence for Item with item category2 = Freezer:

Sequence	Strategy	Location Selection	Split Quantity
10	Consolidate - Exclude differing lots	L-Freezer	No
20	Empty Location	L-Freezer	No
30	Use specific (multi-item) location regardless of status	L-See-Supervisor	No

Locating Sequence for Item with item category2 = General and Item category8 = Literature:

Sequence	Strategy	Location Selection	Split Quantity
10	Consolidate to location that already contains the item	L-KMW-Reserve/ L-OHW-Reserve	No
20	Empty Location	L-KMW-Reserve/ L-OHW-Reserve	No
30	Use specific (multi-item) location regardless of status	L-See-Supervisor	No

Locating Sequence for Item with item category2 = General and Item category8 = Medical Devices:

Sequence	Strategy	Location Selection	Split Quantity
10	Consolidate to location that already contains the item	L-KMW-Reserve/ L-OHW-Reserve	No

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20	Empty Location	L-KMW-Reserve/ L-OHW-Reserve	No
30	Use specific (multi-item) location regardless of status	L-See-Supervisor	No

Locating Sequence for Damaged Item

Sequence	Strategy	Location Selection	Split Quantity
10	Use specific (multi-item) location regardless of status	L-Damaged	No

Locating Sequence for Returns

Sequence	Strategy	Location Selection	Split Quantity
10	Use specific (multi-item) location regardless of status	L-KMW>Returns/L-OHW>Returns	No

Note: The Locating sequence listed above are few samples. Actual rules configured will be more than what has been listed here.

Note: Knipper would like to mass adjust out the inventory from one Item to another to capture the Title Transfer. This requirement is handled as an extension in SCALE.

10.2 Putaway Work Creation

Upon receiving and palletization of the items, if necessary, SCALE generates system work records for each located Pallet / LPN. The work unit for the putaway work equals the Pallet / LPN ID. This allows the user to scan the Pallet / LPN ID in the Warehouse Mobile Work option to initiate the putaway in a user-directed mode. Knipper creates work for Receipt Putaway and uses User directed putaway.

10.3 Putaway Work Execution

To initiate putaway work, Knipper utilizes the option *Work* on warehouse mobile. A user enters the number associated with the *Putaway Work Profile* selection or selects it from the menu options. Upon selection, SCALE prompts the user to enter a work unit, which in this scenario is the LPN (Pallet having cases on it). The work profile is set to be *User Directed*.

Upon scanning the LPN barcode on the pallet, SCALE assigns the work unit to the user. SCALE then displays the current receiving dock location and item information and displays the number of units associated with the work unit. Upon confirmation, SCALE updates the LPN(s) to status In Putaway and displays the putaway location for the first item. To

complete the step of putaway, the user is prompted to confirm the putaway location. The user scans the location for validation upon putaway.

The user may skip the putaway instruction and proceed to the next item on the work unit if multiple work instruction lines exist. This is done by using the *Skip* button. If the user skips the putaway, the system continues directing the user through the putaway locations in the location sequence before looping back to put away the skipped items.

Upon putaway to the final inventory location, SCALE updates the LPN to status Closed and updates the on-hand quantity at the final putaway location. The closed LPN is then eligible to be uploaded to Host as part of the receiving upload interface.

The figure illustrates the putaway work execution process through three sequential mobile application screens:

- Pick confirmation:** Displays work unit 459131, location D-RECV, license plate 459131, item 08460K, and description Kimchi Hot Chicken Ramen. It also shows lot PO546543434, expiration date 04/04/2024, quantity 480, and unit of measure EA.
- Putaway confirmation:** Displays work unit 459131, location A-01-01-I-B, item 08460K, and description Kimchi Hot Chicken Ramen. It also shows lot PO546543434, expiration date 04/04/2024, quantity 480, and unit of measure EA.
- Putaway confirmation:** Displays work unit 459131, location A-01-01-I-B, item 08460K, and description Kimchi Hot Chicken Ramen. It also shows lot 654654745654545, expiration date 04/04/2024, quantity 480, and unit of measure EA.

Figure– Putaway work execution.

For DSCSA receiving, the status of the serial number is updated in the custom DSCSA serial number table and record Receive event in rTS system through an API. For details, refer **EX37 - DSCSA inbound processing** specification document.

Note: The Inventory status of the LPNs will remain as QA Hold (any status assigned during check in process) after the putaway to the inventory location. Once inspection process is

completed or based on the results, the status of each LPN will be changed to the new status (Available/ HOLD FOR DEST AUTH).

10.4 Location Override

Location override will be used when required. Over time, Knipper will evaluate this, and modify as needed.

Certain users can be granted security to override the system's suggested location for the inventory. If this needs to happen, the user picks the Parent LPN from the receiving dock, just as with the other work; however, when they are prompted for the Putaway Screen the user selects the Location Override option on the screen.

This redirects the user to a screen where they can scan the location name from the barcode in which they want to put the product away. The system validates that there is nothing else directed to that location and that the location is valid before accepting the user's override. After validation passes, the system updates work, the LPN to note the new location, and writes Transaction History noting the change in the putaway location. Finally, the user is presented with the Putaway Confirmation screen where they complete the Putaway the same as with other work units.

When a user presses the Location Override button on the screen, after a user is redirected to the override screen, the user also has the option to click on the Locate button. When the Locate button is pressed the user is presented with an option to select a locating rule. The system will then decide on a suitable putaway location based on the locating rule.

If a user selects the option to override, SCALE has the option to create an activity-based cycle count at the original putaway location.



Figure– Putaway location override

III. INVENTORY CONTROL

11.0 INVENTORY MANAGEMENT

11.1 Adjustments

Inventory adjustments either increase or decrease the on-hand quantity of an item in a location. The adjustment types are configurable and are created to indicate reason codes such as Damaged, Scrap, etc. Each adjustment type can be set up to have minimum and maximum adjustment quantities. Security is maintained to control which users have access to which adjustment types. Adjustment types can be configured to not upload to the Host.

All adjustments are entered in the **Inventory Management** option. This option can be initiated blindly from the main menu or **Warehouse Mobile Inventory Management**. It can also be initiated by selecting a location/item combination in the *Inventory Insight*. The user proceeds by entering the quantity to adjust, where the quantity is specified as a negative value for negative adjustments. After confirming, SCALE adjusts the inventory in the location and creates a history log of the inventory transaction.

The screenshot shows a mobile application interface for 'Inventory adjustment'. At the top, there's a header bar with a menu icon, the title 'Inventory adjustment', and user information 'DEF02'. Below the header, there are three buttons: 'Actions', 'Create work', and 'Adjust'. The main form contains several input fields: 'Adjustment type' (set to 'Positive Adjustment'), 'License plate' (LP0411 Q1), 'Location' (26006), 'Item' (NOR0001), 'Company' (empty), 'Description' (Crayola Crayons 64 ct. with Built in Sharpener), 'List' (empty), 'Expiration date' (empty), 'Quantity' (empty), 'UM' (EA (1)), 'Status' (Available), and 'User defined fields' (empty). There are also search icons next to the License plate, Location, Item, and Company fields.

Figure - Inventory Adjustment Screen

Commented [SM141]: Can we find out the current and potential max characters for UDF in the adjustment.

Commented [NC142R141]: Adjustment User defined fields (Transaction history) -
UDF1 to UDF6 = 50 chars
UDF7 and 8 = Numeric

Inventory adjustment

Adjustment type
ER

Location
CA034D

Item
068926

Company
WA

Description
PROPARACAIN HCL 0.5% OPTH SOL 15ML

Quantity
UM
EA (1.00)

Quantity
GO

Figure - Inventory Adjustment Screen

Note: Inventory cannot be adjusted in the receiving dock location. Only located items can be adjusted using Inventory Management.

Typical adjustment types are listed below.

Adjustment Types

- + Adjustment
- – Adjustment
- Cycle Count

11.2 Transfers

Inventory transfers move the on-hand quantity of an item from one location to another within the four walls of the warehouse. The transfer types are configurable and are created to indicate reason codes such as Consolidation, Back to Stock, etc. Each transfer type can be set up to have minimum and maximum transfer quantities. Security is maintained to control which users have access to which transfer types.

The Inventory Transfer option is initiated blindly from the main menu or **Warehouse Mobile Inventory Management**. This can also be initiated by selecting a location/item combination in the *Inventory Insight*, in which case the “from location” and item

Commented [HJ143]: If we remove access from profiles to do cycle count adjustments? Will this prevent pending cycle counts from being reconciled?

Commented [NC144R143]: No action - this can be resolved

Commented [HJ145]: There are occasions when transfers are created from building to building. Examples are Archives and inventory from the printshop in OHW to KMW.

Commented [NC146R145]: This can be resolved

automatically default. In the scenario where Inventory Management is blindly initiated, the user is forced to specify the "from location", item and quantity being transferred.

As part of inventory transfer, users can also create work. This way a supervisor can decide what inventory needs to be moved and then a picker on the floor will get the work to physically move the product. Inventory transfer with work must be created on the insight screen. The work execution can happen on the RF device.

Inventory transfer

Actions: Create work **Transfer**

Adjustment type: Transfer

License plate: ASDASDADU

From location: 2A002

To location:

Item: KEL-2M-00H-L

Description: Keller 2M Item 011T

Lot: 2019T50

Quantity: 0

Status: Available

Company: KFI 1 - John's Manville

Expiration date: 11/30/2019

UM: EA (1)

User defined fields

Figure – Inventory Transfer with work

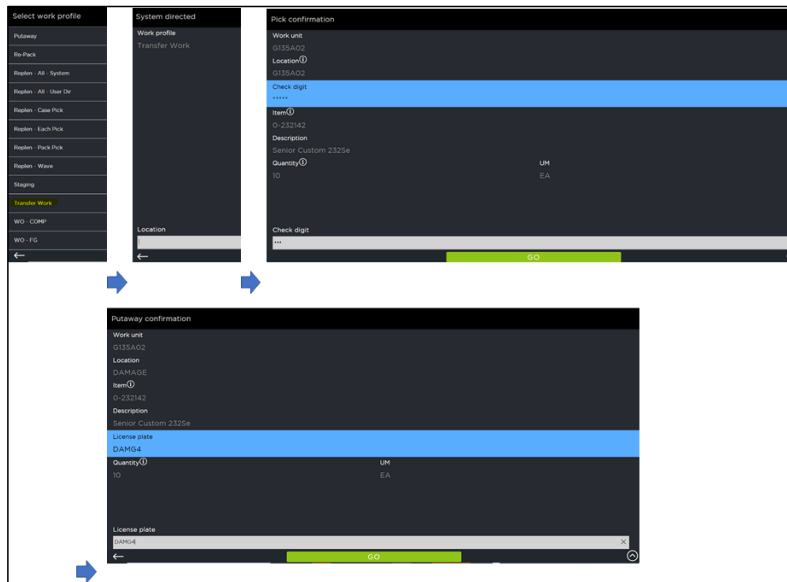


Figure - Inventory transfer work execution

Note: Inventory transfer work execution will be system directed for majority of the operations. Knipper would use User directed work execution as well and would like to print work units document in bulk. [EX40 - Printing multiple work unit document from work insight] will be developed if approved by Knipper

11.3 Status Change

Inventory is received into SCALE with a default status of Available, and HQ for items flagged as inbound QC required. However, Knipper may choose to have different inventory statuses to represent contrasting conditions of inventory such as Hold, damaged, rejected etc. The Warehouse personnel can use an Inventory Status Change to update inventory status. These status change types are configurable, and security is maintained to control which users have access to which status change types.

The **Inventory Management** option can be initiated blindly from the main menu wherein the user specifies the location and item. It can also be initiated by selecting a location/item or a license plate (per configuration) combination in the **Inventory Insight**, in which case the location and item automatically default. The user then specifies the new inventory status. At confirmation, SCALE updates the inventory status for the item and location combination. SCALE also creates a history log of the inventory transaction.

Commented [RS147]: We will create an extension if it is really needed for Knipper

Commented [NC148R147]: NC 11/13: Current state can only print 1 at a time. Ops wants to be able to multi-select. BOL printing functionality - MAH to confirm if an extension is needed

Commented [RS149R147]: Added the details on the extension needed. Please note it is not BOL printing but the work unit document

Commented [NC150R147]: NC12022024: this is resolved

Commented [HJ151]: Can we have the option to change the default status during the receiving process and adjustment in process without having to do the additional steps after these processes to change the status? We have client's that require inventory to be placed on a specific status upon receiving and the inventory departments has to do adjustments in on items that are not always available status.

Commented [NC152R151]: NC 11/13: It is all receiving preferences for items. Following SOPs and client specific. This is resolved

Note: Single location cannot hold the same item with a quantity in status Available and another quantity in status Damaged if the location is not License plate tracked

Knipper does update inventory status of multiple lots at the same time by selecting multiple lots when managing items for destruction.

11.4 Location Inquiry

The warehouse mobile location inquiry enables Knipper to search and view the location inventory records on warehouse mobile. This helps to validate the items actual quantity with that of the quantity physically present in a location. Also, users can access the adjust or transfer inventory actions from this screen.

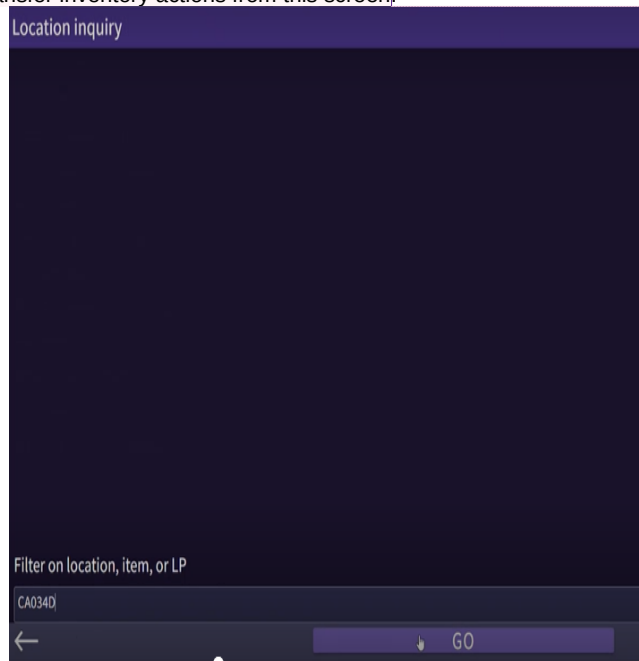


Figure – Location Inquiry

Commented [SM153]: On inbound the system has located a receipt locate into a location with different status - see consolidation rules.

Commented [NC154R153]: NC 11/13: MAH to review

Commented [RS155R153]: Yes, it will allow even for this scenario as long as the location is license plate tracked

Commented [NC156R153]: NC12022024: this is resolved.

Commented [MA157]: Want to make sure this is not for every profile. If it is user based that is fine. This cannot be a default setting for everyone with RF Access.

Commented [MA158R157]: Adjustment access should be at the user profile level.

Commented [NC159R157]: NC 11/13: This is based on user profile. This can be resolved

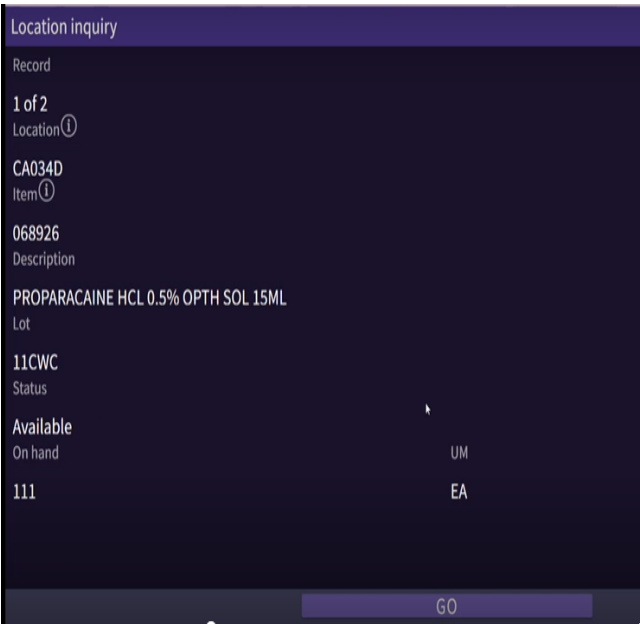


Figure – Location Inquiry results

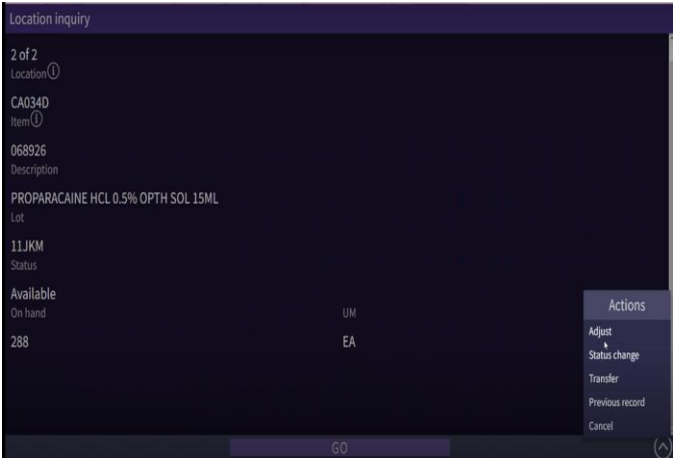


Figure – Location inquiry actions

Commented [SM160]: There is more information now in Locn Quick Find than this Locn Inq. (LPN for example)

Commented [NC161R160]: NC 11/13: No lot, no expiration date, conversion rate - we need more information. MAH to confirm this can be added

Commented [RS162R160]: Please note that this screen is of RF module and not the full screen. The Inventory Insight is the screen which is replacing the location quick find and that screen will have more fields than this.

Commented [NC163R160]: NC12022024: this is resolved.

12.0 CYCLE COUNT

12.1 Generating Cycle Counts

12.1.1 Plan Based Cycle Counting

SCALE utilizes *Cycle Count* Plans for everyday cycle counting. Knipper personnel define the Cycle Count Plans in the **Cycle Count Plan Insight**. A Cycle Count Plan is used to define a range of items and/or locations for which to generate cycle count work. For an example of location criteria, Knipper can exclude a location that was counted within the last 30 days. For an example of item criteria, Knipper can use the Item categories (Client Code, etc.). Once the plan has been created, SCALE creates work to count a specified number of locations within the criteria of the Cycle Count Plan. Each location determined for cycle count is generated as a separate work unit in SCALE.

Some Plan based counts that Knipper uses are:

- Daily Count for PM locations
- Daily Literature Items Count
- Daily Count KMW Reserve
- Daily Count OHW Reserve Bausch
- Weekly Refrigerated Locations Count

Note: The below are just for reference and Knipper creates Weekly/Daily/Monthly cycle count for different accounts as per the requirements.

Commented [SM164]: Currently to set up a schedule, users need to set up a plan via CC Location Criteria, CC Item Criteria, and CC master; do these steps get consolidated with scale v'24?

Commented [NC165R164]: NC 11/13: MAH to confirm

Commented [RS166R164]: This process will remain same in the new version

Commented [NC167R164]: NC12022024: this is resolved.

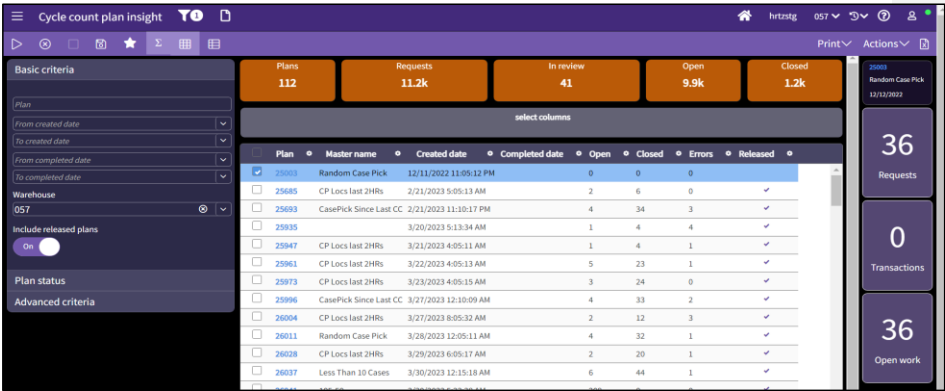


Figure - Cycle Count Plan Insight

12.1.2 Activity Based Cycle Counting

Activity-based cycle counting is the concept of triggering the generation of a cycle count request after a warehouse activity (i.e., Short Picking). The triggering of these requests is tied to the location being processed. After the transaction is executed, the location is reviewed to see if cycle count work should be generated.

For Knipper, SCALE is configured to create activity-based cycle count work for a location when short picked.

Note: Threshold Count is utilized. Knipper would like to enable threshold count and perform threshold counts immediately for some work zones.

Knipper would like to perform counts for PTL (Pick to Light) Mod locations after completing picking of every wave and would like to create the count automatically after work unit is completed. **[EX19 – Pick to Light Integration]** will be enhanced to create the count automatically.

Commented [SM168]: Knipper does not short pick.

Commented [NC169R168]: NC 11/13: This can be resolved

Commented [SM170]: Parking Lot Item #17

Commented [HJ171]: If there is a discrepancy in the pick mod, we cannot reconcile the location before the full inventory investigation is completed due to the reconciliation making an inventory adjustment in SCALE. If the count stays open, this will keep the wave open, and operations cannot process anymore orders from that location until the cycle count is reconciled. If the cycle count is used as an adjustment after an investigation, how is it tied back to the wave?

Commented [NC172R171]: NC 11/13: MAH is creating an announcement - this will be taken care of

12.2 Executing Cycle Counts

12.2.1 Work Execution

For Knipper the cycle count execution is set to *Standard* count for active and reserve locations. To confirm the cycle count, users sign onto a warehouse mobile device, chooses the *Work* option, and then specify the *Cycle Count Work Profile*. The user is then prompted to scan a location for SCALE to assign a Work Unit in closest proximity. Work can also be user-directed, if the user knows exactly which locations need to be counted. If User directed, then the user needs to scan the location that they intend to count.

Once done, SCALE assigns the work to the user and displays the location to be counted on the work execution screen and the item present in that location. User then must specify the quantity present in the location.

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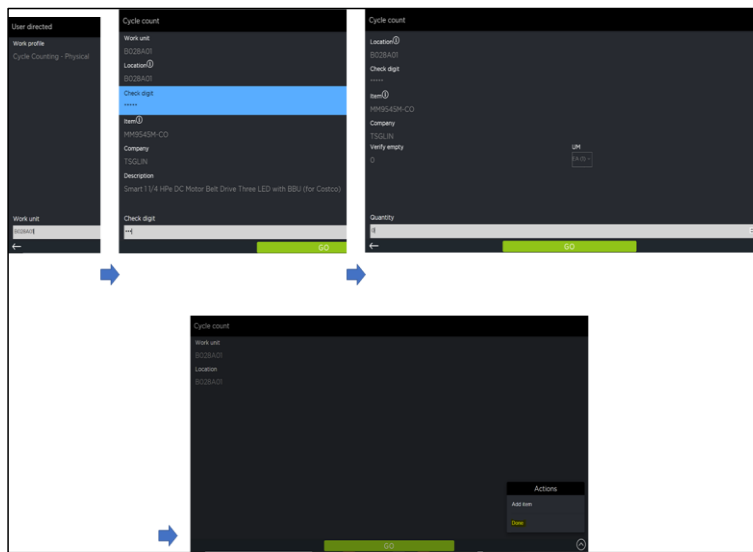


Figure - Cycle Count Work execution

Note: Cycle Count Preferences are set up to "Verify Bad Count".

If the quantity entered by the user is the same as the system quantity, then the system accepts that quantity and displays the cycle count screen where the user can count other items in the location (multi-item location). If the location is a single item, then the user needs to click on the Done to indicate the cycle count work execution is complete. If no discrepancy in count the cycle count request is updated to close.

If the quantity entered by the user for the first time doesn't match the system quantity, SCALE displays verify count screen and ask the user to confirm the quantity again. If the user enters verify count, they need to enter two consecutive counts of the same value to complete the count.

Upon entering the information, if there is a discrepancy in the count then SCALE determines whether to post the inventory adjustment or update the cycle count request to status Pending Review. For Knipper the Cycle Count Tolerances will be set to 0 (all discrepancies must be reconciled) by default. So, SCALE updates the cycle count request to pending review, and reconciliation needs to be completed for the cycle count request to be updated to closed status.

Commented [SM173]: Cycle count screen needs to display the standard Locn, LPN, Item, Batch, Exp Date

Commented [NC174R173]: NC 11/13: MAH to add the above

Commented [RS175R173]: The cycle count execution screen would display all these fields.

Commented [NC176R173]: NC12022024: this is resolved

Commented [SM177]: Option to Cycle Count the Location or the LPNs

Commented [NC178R177]: NC 11/13: MAH to confirm we can add this functionality

Commented [RS179R177]: There is an option to count total license plates of an item in a location. There is a configuration which needs to be turned on for this to work. Please note that the count will also have to be for a location/item whether a count is done by License plate or not.

Commented [NC180R177]: NC12022024: 2 options need to be available cycle count the location and the LPNs

Commented [RS181R177]: The configuration can be utilized if the user need to scan the LPN during the count. If this is fine, please go ahead and resolve this.

Last Modified: 12/10/2024 10:30:00 PM

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12.2.2 Cycle Count Reconciliations

For cycle count transactions that fall outside of a user's tolerance, SCALE updates the status of the Cycle Count work to Pending Review.

To reconcile a cycle count, a supervisor uses *Reconcile* option from the **Cycle Count Request Insight** screen.

From the **Cycle Count Request Insight**, the supervisor selects the appropriate cycle count in the status Pending Review. On each of the counts requiring review, the supervisor uses the Reconcile action to complete the cycle count adjustment. Once in the reconcile screen, the supervisor enters the correct On-Hand quantity for the specific item in the location.

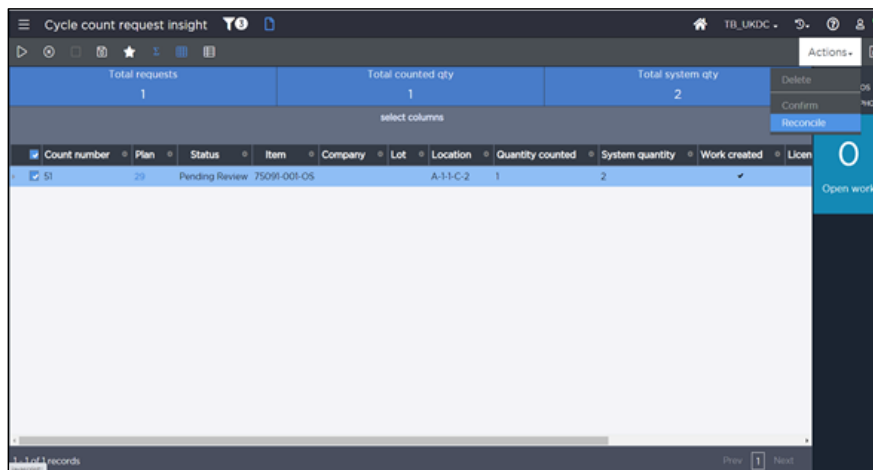


Figure - Cycle Count Reconcile using Insight screen

Commented [SM182]: Currently we confirm cc is accurate, and then go back and do a manual adjustment so the INCA # can be logged (for drug product).

Inventory Check and Adjustment Form (INCA)

Commented [NC183R182]: NC 11/13: this can be resolved

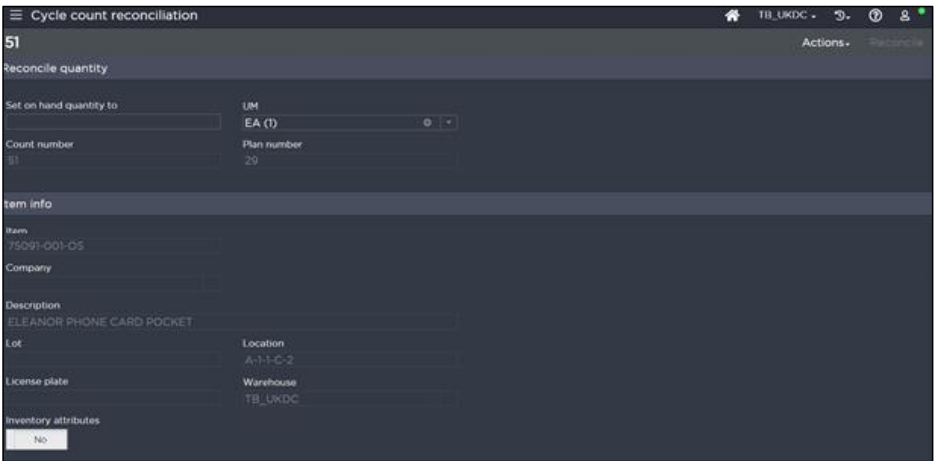


Figure - Cycle Count Reconciliation insight screen

The supervisor confirms the on-hand quantity, and then SCALE updates the inventory in the location, records an inventory transaction, and closes the cycle count request.

Cycle Count reconciliation can also be performed using the warehouse mobile option.

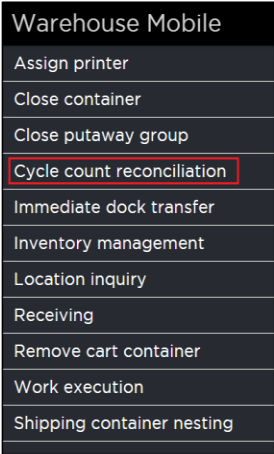


Figure - Cycle Count Reconciliation using warehouse mobile

Cycle count reconciliation

Location ①

HRO1PDO

Item ①

TOOLBOX SET

Description

TOOLBOX SET

Company

Apex

Counted ①

0.00000

On hand quantity

UM

EA

UM

EA (100000)

On hand quantity

Figure - Cycle Count Reconciliation using warehouse mobile

13.0 REPLENISHMENT

Replenishment is setup using Replenishment Master records. A replenishment master record helps define how a replenishment request is generated. It specifies the parameters for replenishing product based on either location need or demand generated for a wave/order pool. The sequence records for a replenishment rule can be edited. Each sequence rule identifies how the system will perform replenishment (strategy).

Setting up the replenishment is a manual process and leverages using the high-volume item demand report and the setup of item location assignments and capacities as defined earlier in the wave selection section.

Commented [SM184]: 3 means of replenishing: Wave Generated, Capacity based on Perm Locns, manual xfer.

Commented [NC185R184]: NC 11/13: no action required

Replenishment master - Edit existing

File Actions View Help

Replenishment name: Heavy Case

Description: Heavy Case - (A,B,C) - LS

General Demand ums Replenishment criteria Replenishment m

Replenishment type

- ☐ Item location capacity
- ☐ Top-off item location ca
- ☐ Demand from pool
- ☒ Demand from wave
- ☐ Demand from RF outbou

Create work method

- ☒ Automatic
- ☐ Manual/scheduled job

Priority: 5

Allocation rule: LS Case Replen

Replenishment i...: Pallet

☐ Allocate all ums to clear reserve location

☐ Create multiple requests for excess demand

☐ Consolidate replenishment requests

☐ Inactive

OK Cancel

Figure – Replenishment Master

13.1 Demand Replenishment

Each demand from the wave is evaluated against flow rack inventory to determine if there is enough available. If not, SCALE will request the product in Case increment (round up) to be replenished to each pick primary bins.

If the demand is greater than the Capacity of flow rack location, then SCALE will direct the additional inventory into empty dynamic active locations.

Case demand from the wave is evaluated against the pallet rack location inventory to determine if there is enough available. If not, SCALE will request the product in Pallet increment (round up) to be replenished to pallet rack locations.

If the demand is greater than the Capacity of pallet rack location, then SCALE will direct the additional inventory into empty dynamic active locations.

Note: Knipper utilizes both dynamic and permanent locations at Pick to Light and other primary locations at multiple areas (MSM, 3PL, DEA, Refrigerated and Freezer) of the warehouse. There will be multiple Replenishment masters created for the replenishment of all the areas of the warehouse.

Commented [SM186]: Parking Lot # 13 needs to be addressed.

Commented [NC187R186]: NC 11/13: MAH to look into it

Commented [RS188R186]: This will be discussed during the call on 11/25

Commented [NC189R186]: NC12042024: This can be resolved

Commented [SM190]: KMW : wave replen is normally a pick to zero. Not necessarily case level replen.

Commented [NC191R190]: NC 11/13: this can be resolved

13.2 Capacity Replenishment

Capacity based manual replenishment is utilized at Knipper (Lean time replenishment) daily. Manual replenishment is kicked off when requested by the user from the **Inventory Insight and through Scheduled Jobs**. SCALE requests the product in EA increment (Fill location) for primary bin locations based on the selected replenishment master(s). Inbound teams execute this capacity based replenishment.

Note: Knipper will utilize scheduled jobs for manual replenishment

Commented [SM192]: KMW - replenishes at case UOM not always EA. Would this be driven at the item config?

Commented [NC193R192]: NC 11/13: This can be resolved - this is done at the replenished level

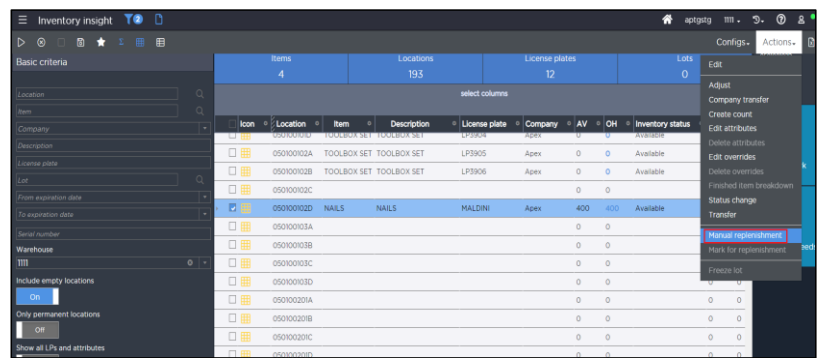


Figure - Manual Replenishment

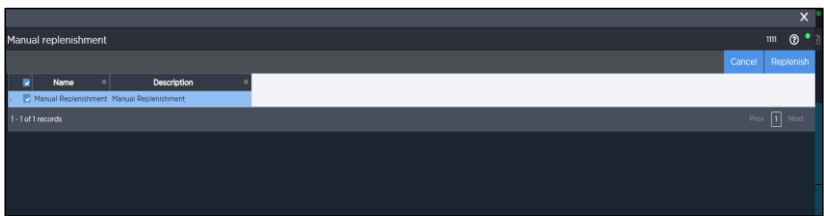


Figure - Manual Replenishment selection

When manual replenishment runs, it evaluates all permanent active locations and attempts to replenish any location that is below its configured minimum percent. Replenishment Allocation Rules, Item Criteria, and Allocation Strategies are configured to direct replenishment to the appropriate primary bin locations.

Note: To utilize Capacity based replenishment, Item location assignment and Item location capacity configurations should be present in SCALE.

13.3 Real Time Replenishment

Threshold based real time replenishment is not leveraged for the go live. Over time, Knipper will evaluate this and may use it.

The need to replenish is evaluated against the primary picking locations to determine if there is enough available. If the quantity in the forward pick floor locations is below the minimum replenishment threshold percentage configured, SCALE will request the product in configured increments of UM.

Figure – Item Location Capacity with Replenishment percentage

Since demand replenishment runs throughout the day in advance to the waves, adding real time replenishment at the same time may bring more than required inventory to the forward locations causing operational challenges. When setup, Knipper leverages real time replenishment for fast-moving items on a scheduled basis, and leverages alerts/reporting to monitor open capacity-based replenishment and delete those open replenishments request.

Commented [SM194]: How is this different from Demand Replenishments? Or is this a hybrid of demand and capacity for non-perm locns?

Commented [NC195R194]: NC 11/13: No action is needed
Manual
You must execute a replenishment attempt on the Location Explorer Window. When the user chooses the Replenishment option from the Location Explorer, this replenishment master will appear in the window for selection to be run. The user will have to check this replenishment master for it to be run.

Real Time

The system will run replenishment (via a scheduled job) throughout the day, for locations that would typically need frequent replenishment (you can make a location eligible for this on the Location Window). The replenishment process will run quicker than usual because the system will check for replenishment need only the locations that actually are currently in the need of replenishment, and ignore the ones that do not currently need replenishing.

This type of replenishment is helpful because you can have replenishment masters running throughout the day, and keep your forward picking locations full for order processing.

Wave

The system will automatically execute a replenishment attempt during the wave cycle. Whenever any launch is run that has Replenishment as a step in its launch flow, then this replenishment master will be executed.

Commented [SM196]: Wouldnt the waves trigger the demand replen?

Commented [NC197R196]: NC 11/13: This is resolved

13.4 Replenishment Allocation

Knipper' allocation for replenishment is to attempt to allocate using First Expiration First Out for lot tracked items, and available most available first for non-lot tracked items, that fills the entire forward location from the rack location. The rules are ported over from the existing setup.

Zone		Description
1	PM Locations (KMW, KDC and OHW) – EA Pick	• Single Item • Permanently Assigned • No License Plate Tracking • Single Lot • Allocate In Transit - Y
2	PM Locations (KMW, KDC and OHW) – CS Pick	• Single Item • Permanently Assigned • No License Plate Tracking • Single Lot • Allocate In Transit - Y
3	PTL Locations (KMW, KDC and OHW) – EA Pick	• Single Item • Permanently Assigned • No License Plate Tracking • Single Lot • Allocate In Transit - Y
4	PTL Locations (KMW, KDC and OHW) – CS Pick	• Single Item • Permanently Assigned • No License Plate Tracking • Single Lot • Allocate In Transit - Y
5	3PL PM Locations – EA and CS	• Single Item • Permanently Assigned • No License Plate Tracking • Single Lot • Allocate In Transit - Y
6	3PL Case Pick Locations	• Single Item • Permanently Assigned • No License Plate Tracking • Single Lot • Allocate In Transit - Y
7	DEA PM Locations (KMW, KDC and OHW)	• Single Item • Dynamically Assigned • License Plate Tracking • Single Lot • Allocate in Transit - Y

Commented [SG198]: What is KDC and ODW?

Commented [NC199R198]: NC 11/13: updated from ODW to OHW

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8	DTP Refrigerated PM (KMW, KDC and OHW)	• Single Item • Dynamically Assigned • License Plate Tracking • Single Lot • Allocate in Transit - Y
9	DTP Freezer PM (KMW, KDC and OHW)	• Single Item • Dynamically Assigned • License Plate Tracking • Single Lot • Allocate in Transit - Y

Replenishment Allocation:

Lot tracked Items - CS

Sequence	Strategy	Location Selection	Eligible UMs
10	First Expiration, First Out	Reserve (General, DEA, Refrigerated, Freezer and 3PL)	CS

Commented [MA200]: Are we going to use the term "general" here over "controlled room temp?"

Lot tracked Items - PL

Sequence	Strategy	Location Selection	Eligible UMs
10	First Expiration, First Out	Reserve (General, DEA, Refrigerated, Freezer and 3PL)	PL

Commented [NC201R200]: NC 11/13: We can update with the correct verbiage and remove General

Commented [202R200]: We do not want to remove "general" due to legacy products.

Non lot tracked items - CS

Sequence	Strategy	Location Selection	Eligible UMs
10	First In, First Out	Reserve (General, DEA, Refrigerated, Freezer and 3PL)	CS

Non lot tracked items - PL

Sequence	Strategy	Location Selection	Eligible UMs
10	First In, First Out	Reserve (General, DEA, Refrigerated, Freezer and 3PL)	PL

Note: The existing replenishment rules are ported over for Knipper and during the build phase Knipper evaluates all replenishment masters and allocation rule for other item categories as well.

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13.5 Replenishment Work Creation

The replenishment work creation process is like the work creation process performed during the locating portion of the receiving process. After performing replenishment, SCALE creates a *Work Unit* to pick the inventory from its reserve location and transport it to Primary bins or dynamic active locations. Based on the configuration in the Work Group, Work Type, Work Criteria, and Work Creation Master, the system analyzes, sorts, and bundles the replenishment requests to create a *Work Unit*. This *Work Unit* consists of only one items from one storage or reserve locations.

Most DCs create replenishment work based on Hi or Low zones. However, this is not a standardized setup and DCs may change this on their discretion. Manhattan recommends to have standardized replenishment masters for layout and/or operating profiles.

Commented [MA203]: Is this separate from the web-based scale? How are we able to make these changes and have them update into the web-based scale?

Commented [NC204R203]: NC11182024: This is resolved.

Replenishment work criteria - Edit existing

File Actions Help

Record type: REPLENISH

Filter name: 40 - Capacity Replen Hi

Description: 40 - Capacity Replen Hi

Table name: Replenishment request

Filter data Order by

Filter criteria

☒ A Attribute: Operand: Value:

Replenishment_Request FROM_WHS = N'40'

AND (Replenishment_Request FROM_WORK_ZONE = N'W - Hi Re

OR Replenishment_Request FROM_WORK_ZONE = N'W - Cooler F

AND Replenishment_Request.REPLENISHMENT_MODE = N'Manu

Up Do

Add rule

Delete selected r

Delete last rule

Insert left parenth

Insert right paren

Delete parentheses

☐ Inactive

☐ System created

OK Cancel

Figure – Replenishment work criteria

Replenishment work criteria - Edit existing

File Actions Help

Record type: REPLENISH

Filter name: 40 - Capacity Replen Hi

Description: 40 - Capacity Replen Hi

Table name: Replenishment request

Filter data Order by

Order by criteria

☒ Ascending ☐ Attribute: ☐ Create work unit

Attribute	Ascending/desce...	Create	
Replenishment_Request.FROM_TEMP...	Ascending	Yes	Add sort criteria
Replenishment_Request.FROM_TEMP...	Ascending	Yes	Deleted selecte
Replenishment_Request.FROM_LOC	Ascending	No	Delete last sort i

☐ Inactive ☐ System created OK Cancel

Figure – Existing replenishment work criteria – Hi Zone

Replenishment work criteria - Edit existing

File Actions Help

Record type: REPLENISH

Filter name: 40 - Capacity Replen Lo

Description: 40 - Capacity Replen Lo

Table name: Replenishment request

Filter data Order by

Filter criteria

☒ A Attribute: Operand: Value:

Replenishment_Request.FROM_WHS = N'40'
AND (Replenishment_Request.FROM_WORK_ZONE = N'W - Lo Re
OR Replenishment_Request.FROM_WORK_ZONE = N'W - Cooler F
OR Replenishment_Request.FROM_WORK_ZONE LIKE N'%Static
OR Replenishment_Request.FROM_WORK_ZONE LIKE N'%Active
AND Replenishment_Request.REPLENISHMENT_MODE = N'Manu

Add rule
Delete selected r
Delete last rule
Insert left parenth
Insert right paren
Delete parentheses

☐ Inactive ☐ System created OK Cancel

Figure – Replenishment work criteria

Replenishment work created in the wave will have a higher priority than the replenishment work created by manually capacity based replenishments. The user then receive this replenishment work first, as replenishment is system directed work in most work profiles configured currently.

13.6 Replenishment Work Execution

Replenishment work is performed within SCALE as *System Directed* work. Users sign onto an RF warehouse mobile device and choose the **RF Work** option. Users then choose the *Demand Replenishment* or *Capacity Replenishment* work profile. The user is then prompted to scan a location for SCALE to assign a Work Unit in closest proximity. The user is assigned a Work Unit and SCALE then presents the user with the first pick which displays the location, item, and quantity to be picked. Users are required to verify the pick by scanning the location check digit and then hitting OK. Once all the picks have been completed (all picks can be any number of items – this is defined by Knipper), SCALE displays a putaway screen where the user confirms the putaway to forward pick location(s) that the inventory should be stored in. These forward pick locations can also force validation from the user (this is configurable – location or check digit validation). After this, the user can be assigned the next Work Unit.

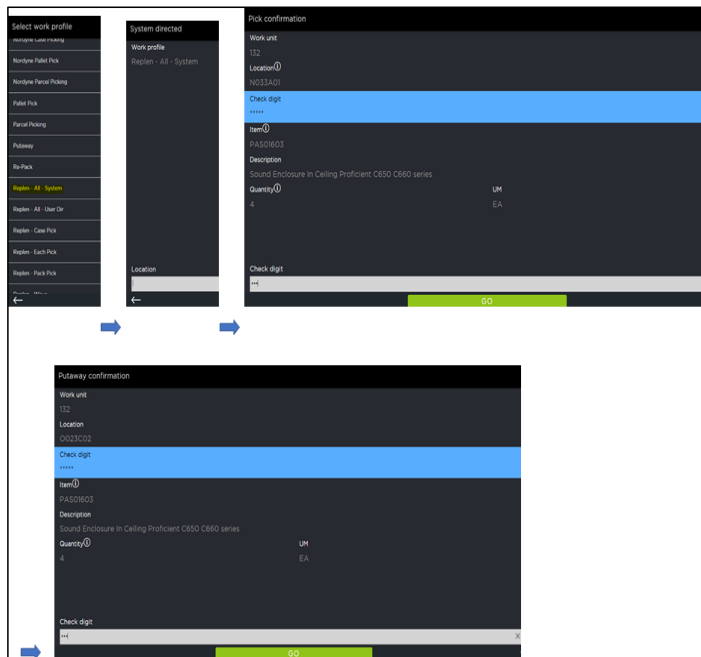


Figure - Replenishment Work Execution

Once the Replenishment Work is completed the inventory will be available at the pick face location. The inventory on the pick location is then available for shipment picking.

14.0 WORK ORDERS

Knipper use work orders in the manufacturing facilities to build finished items.

15.0 BILL OF MATERIALS (BOM)

BOM will be used as part of this implementation. Knipper will own the configuration of BOM as required. The BOM is used when work order is manually created in SCALE. The component items are sent on the work order when they are downloaded to SCALE.

16.0 WORK ORDER CREATION

Work orders for manufacturing finished goods will be interfaced from Host to SCALE. The work order will include the finished item to be built and the quantity to be built. Work order will not have the build location information. User needs to update the build location before they allocate the work order.

17.0 COMPONENT PULLING

Once the work order exists in SCALE, it can be allocated whenever Knipper is ready to work on it. To allocate a work order, users will go to the Work Order Insight, select the work order, and choose the Allocate all option. Allocating the work order initiates the allocation and work creation that is required to bring forward all required components to the designated build area. Once all the allocations and work created are verified, users will release the work order to commence the next steps in building the finished items.

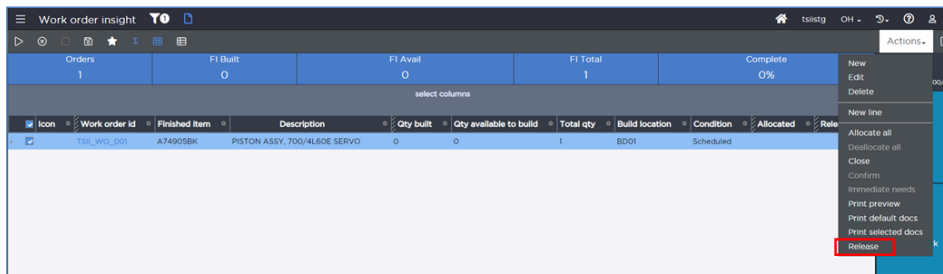


Figure: Work Order insight – Release Option

The allocation and work creation work the exact same as the standard shipment allocation within SCALE. Each component will have an allocation rule, which defines how component inventory should be pulled from the warehouse.

Component Allocation:

Sequence	Strategy	Location Selection	Eligible UMs
10	First in First Out	A-General Reserve – Non-Lot	

Commented [SM205]: BOM is in scope.

KMW: PO is dropped into scale. - Abbvie is integrated.

Commented [NC206R205]: NC11182024: BOM is in scope - MAH to confirm the comment BOM is not in scope

Commented [RS207R205]: Updated the description

Commented [NC208R205]: NC12042024: This is resolved.

Commented [CC209]: OHW manually enters work orders for production kitting and labeling work. Work orders are also Loaded into SCALE from our internal OMS not sure what this is referring to

Commented [NC210R209]: NC11182024: resolved.

Commented [CC211]: please explain we create the BOM docs in OHW

Commented [SM212R211]: SOP 105 A

Commented [NC213R211]: NC11182024: resolved.

Commented [SM214]: KMW: When adding components to master BOM, asks for allocation rule. Allocate all in BOM fails because the allocation rule is on the line level. Must be removed.

Commented [NC215R214]: NC11182024: MAH to confirm if this can be automated and not manual. Pull allocation from reserved location.

Commented [RS216R214]: SCALE can be configured to automatically allocate upon Release. As long as the allocation rules are correct, this should work as expected.

Commented [NC217R214]: NC12042024: This is resolved.

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20	First Expiration First Out	A-General Reserve - Lot	
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Work is created and executed for these allocations in the same fashion as other work creation within SCALE. Below is a sample Work Creation Master for component pulls.

Work Creation Master - Edit Existing

File Actions Help

Work Creation Name: WO Component Pulls

Description: WO Component Pull

General Maximums User Defined Data

Work Type: WO Component Pulls Process: Work Order Component W

Work Unit Field: Internal Instruction Num Work Criteria: All Components

Creation Method: Pre-build

Priority: 1

Auto Print Documents

Figure: WO Component Pulls work creation master

Knipper will be creating work units by following rule:

For Component picks, one work unit is created per work order.

Component pulls will be executed via user directed work. This work can be done separately via a Component Pulling work profile.

System directed Work profile WO Component Pick

Pick confirmation Work unit TSL WO 001 Location 022012 Item 022012 Description TC ACDE/4R70W 93-09 Quantity 1 UM EA

Pick confirmation Work unit TSL WO 001 Location 022012 Item 022012 Description TC AX45/NL L94-UP Quantity 1 UM EA

Putaway confirmation Work unit TSL WO 001 Location 022012 Item 022012 Description TC AX45/NL L94-UP Quantity 1 UM EA

Figure: Component Pull work execution

Commented [SM218]: Need the current RF details displayed. (Lot number, expiration date, qty conversion, etc.)

Commented [NC219R218]: NC11182024: MAH to add

Commented [RS220R218]: The lot will be displayed on the screen. The Expiration date and conversion are not.

Commented [NC221R218]: NC12042024: This is resolved

17.1 Component allocation failures

If for some reason the entire component allocation fails, SCALE displays the allocation failure message and moves the work order into In process folder. Here the user will review the reason for allocation failure and once the inventory is fixed, user can navigate to the component section of the work order and select Allocate All option to re-allocate the components.

If for some reason, only few component allocations fail then SCALE moves the work order into In Process folder and create work for components that allocated successfully. User can review the reason for component failure and fix the inventory. User can navigate to component section and

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select the component that failed allocation and select Allocate option to re-allocate the component.

SCALE would use the '#' symbol to differentiate duplicate work units.

Since Component pull work is user initiated as soon as the work is created it is available for users to execute. Even with some component failing allocation, component pull work will be available to user if any components succeed allocation. Hence it is a good practice to use an SCI report/Inventory Insight to verify component inventory before releasing the work order.

18.0 FINISHED GOODS CREATION

As component materials arrive at the build location, finished goods can be built and processed in SCALE. To process the production, users will process as follows:

- Go to the Work Order insight screen
- Key/scan the Work Order ID
- Press the Confirm button
- Enter quantity built
- Scan pre-printed license plate value for each putaway unit built and affix label to pallet

The above process relieves the component inventory from SCALE and creates new finished goods inventory to be put away in stock.

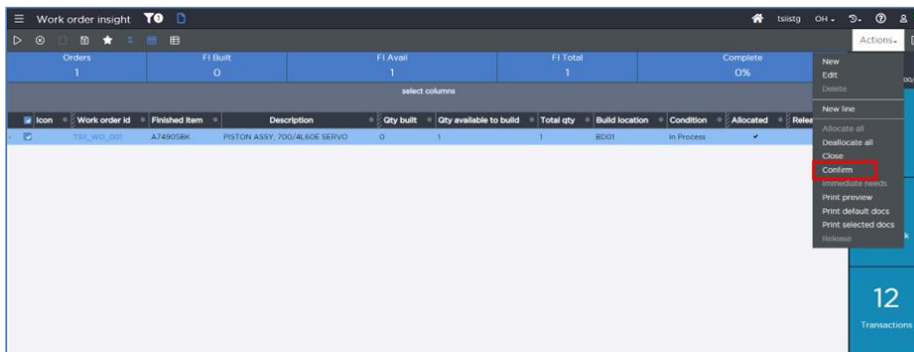


Figure: Work Order Insight Screen – Confirm Option

Commented [SM222]: OHW: Once the work order is released then the users can start the work. Need the release step included.

Commented [SM223R222]: OHW and KMW process is different. KMW does transfer before allocation.

Commented [NC224R222]: NC11182024: MAH to confirm comment

Commented [RS225R222]: The work order release is a must process to start the work. This is already mentioned in 17.0 section

Commented [NC226R222]: NC12042024: We need the ability release on the line item level. Just because we have an issue on 1 item it shouldn't hold up the whole order

Commented [RS227R222]: The Release is at the Header level and it will not hold up other line items if there is any issue with one of the line items. If this is fine, please resolve.

Commented [SM228]: SCI?

Commented [NC229R228]: NC11182024: MAH to confirm what does SCI mean.

Commented [RS230R228]: SCI means Supply Chain Intelligence. It is a reporting tool. Alternatively, Inventory Insight can be used to verify the inventory before the release.

Commented [NC231R228]: NC12042024: This is resolved.

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Work order confirmation

TSII_WO_001

Quantity1UMEach

Quantity available to confirm1UMEach

Work order idTSII_WO_001Putaway locationQM123

Item info

ItemA74905BK

Company

DescriptionPISTON ASSY, 700/AL60E SERVO

LotExpiration date

License plateF1001

Figure: Work Order Confirmation

19.0 FINISHED GOODS PUTAWAY

Once the work order is complete, SCALE will locate the inventory and create putaway work. This locating and work creation works the exact same as the standard receipt locating within SCALE (described above). All finished goods will be located to inventory location using the locating rules configured in system, SCALE creates putaway work.

Work is created for these put away tasks in the same fashion as other work creation within SCALE. Finished goods putaway will be executed via user directed work. This work can be done separately via a Work order finished work profile.

Select work profile

User directed

Pick confirmation

Putaway confirmation

Figure: Finished Item Putaway work execution

VI. OUTBOUND

Outbound is the process in SCALE whereby inventory is shipped out of the warehouse via Shipments. From a high-level perspective, the following processes are required. Rectangles represent processes normally done by the system and trapezoids represent processes normally requiring user interaction.

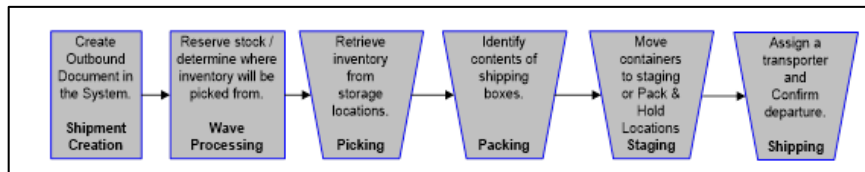


Figure 52 – High-Level typical SCALE Outbound Process

Shipment Creation

This is the process by which a shipment is entered into the system. This is normally a system process that is initiated on a schedule to interface records from an existing host system.

Wave Processing

Assigning shipments to a wave and running the wave is normally a system process, whereby shipments are grouped logically based on user-defined criteria to be processed together during the outbound process. Example groupings include: 'Priority Orders', 'High Volume Shipments with only one Item' Once assigned to a Wave, the wave can be run either automatically or manually. Running the wave initiates, the processing of Wave Steps may include the following: Allocation, Work Creation, Carrier Assignment, Load Building, etc.

Picking

This is typically a user process whereby the system directs users to picking locations in a logical sequence to retrieve inventory and bring it to packing stations, consolidation areas, or staging areas.

Packing

This is typically a user process where inventory is placed into boxes and the boxes are identified in the system. There are three main options for performing packing (1) During the Wave where the system uses dimensions and configured Container Types to calculate the most efficient way to pack the items, (2) During the picking process where users identify what boxes the items are being picked into while they are being picked, (3) After picking where items are dropped off at discreet packing stations and a separate team packs items into boxes

Commented [CC234]: Can the user override the container chosen by the system and enter a different one

Commented [NC235R234]: NC11182024: MAH to confirm if this can be done

Commented [RS236R234]: If containers created during the wave, it can be changed after picking before closing the container. When we say during picking, it means pick into container option (no containers created during wave) where user enters a container id and chooses container type.

Commented [NC237R234]: NC12052024: This is resolved.

Staging

This is a user process where containers are taken from a packing location to a staging area. Here they await the arrival of the method of transport. At this point, multiple shipping containers can be consolidated onto parent pallets for easier transport. This can also be a holding area where shipments are waiting for future shipping.

Shipping

This is a user process where shipments can be assigned to specific Shipping Loads and subsequently Load Confirmed. A Shipping Load can represent a lorry, a sea container, a van, etc.

The assignment of a Carrier to a shipment can be done systematically during the wave or manually after the wave process.

Loading is the action in which a user takes the shipping containers \ parent pallets and confirms them onto the truck via a dock door.

Load Confirmation is the process that represents the truck, container, etc. has already been loaded and has left the warehouse. The shipping process encompasses the dock management functionality of SCALE. Load Confirmation is the final status in the order flow in SCALE and represents the removal of stock from within the four walls of the warehouse

Shipment Trailing & Leading Statuses

These values are defined on the shipment header record. The trailing status is the least advanced status associated with the record; the leading status is the most advanced status associated with the record. The system obtains this information from the containers created for the header. You can use this information to research container activity and perform troubleshooting tasks.

Below are all the Outbound Execution default statuses available in SCALE:

- (90) In Pool Pending: The shipment's status before it is processed via the interface. The purpose of this status is to be a temporary status used by the interface download process to ensure that shipments do not get picked up for processing while the interface process is running. Also, shipments in this status cannot be added to a wave.
- (100) In Pool: The shipment has not been processed in a wave but has been created in the pool or downloaded via the interface. No work can be performed against this shipment.
- (200) Wave Pending: The wave is being built. It has not been run.
- (201) In Wave: A wave run was initiated, and the wave has not been released.

- (300) Picking Pending: A wave was released, but none of the work has been initiated.
- (301) In Picking: At least one work instruction associated with this shipment has been assigned to an employee.
- (400) Packing Pending: At least one work instruction associated with this shipment has been pick confirmed. It can now be packed.
- (401) In Packing: At least one work instruction associated with this shipment has been packed.
- (600) Staging Pending: All items of a shipment have been picked, packed, and/or consolidated.
- (650) Loading Pending: All items of a shipment have been picked, packed, or consolidated, and/or staged.
- (700) Ship Confirm Pending: At least one container associated with this shipment has been closed.
- (800) Load Confirm Pending: All shipments on the load have been ship confirmed and the load is ready to be confirmed.
- (900) Closed: The shipping load has been confirmed.
- (998) Delete Rejected: A quantity on a shipment detail line was rejected. The quantity will be deleted from the system.
- (999) Rejected: During allocation, if any quantity on the line is rejected, this status will indicate that rejected quantity

20.0 WAVE PROCESSING

A wave is a method for retrieving shipments from the pool and processing them through a wave flow. All shipments are processed through the system in a wave (a wave is simply a collection of shipments that generate an amount of work that can be handled by the operation in a single session). All wave activity is defined on the wave master record.

Outbound shipments are created in Host and sent to SCALE. On a scheduled basis, these shipments are interfaced into SCALE. Any orders that fail validation are logged for further review and will be reviewed manually by looking at the **Interface Error Insight** screen for corrective actions. Once corrected, the shipment(s) can be reprocessed during the next scheduled interface download (or the interface can be manually invoked).

Once successfully downloaded, the shipments are viewable as SCALE shipments in the Pool in the **Planned Shipment Insight**. The shipments can be viewed by Order Type and

Commented [CC238]: Are corrective actions easy fixes for the person waving to correct or does this require Inventory involvement?

Commented [NC239R238]: NC11182024: this is resolved

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Scheduled Ship Date (Knipper can create Planned shipment filter criteria records to view shipments in pool by other criteria of their choice).

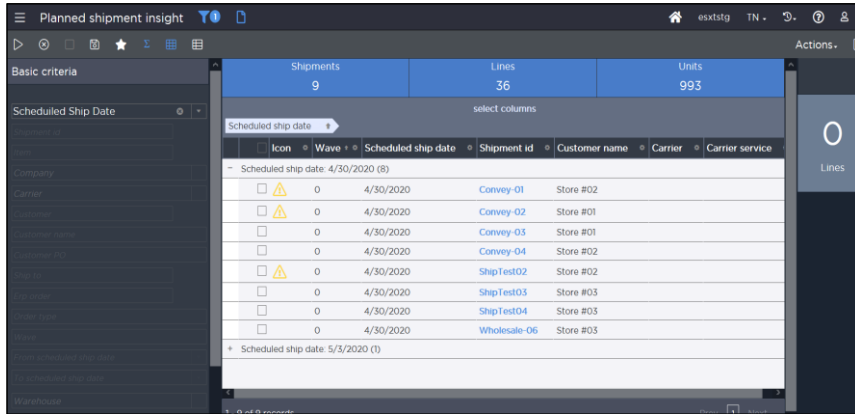


Figure - Planned Shipment Insight

The following order profiles are used by Knipper Host.

Order Profile	Description	Interfaced	Frequency
Direct to Physicians (DTP)	Orders for samples to physicians	Y/Manual	
Direct to Rep (DTR)	Orders for samples to Medical Reps	Y/Manual	
3PL	Orders for 3PL accounts	Y/Manual	
Destruction	Shipping inventory for destruction	Y/Manual	
Direct to Consumer (Ecom)	Direct to consumer orders	Y/Manual	

20.1 Waving Strategy

Knipper' waving strategy is to wave by priority or cut-off times for the orders in the pool. For LTL/TL orders, Knipper does not wave for a truck – aka, do not wave by cubing for a truck, hence, one wave may result in multiple load numbers.

Shipments interfaced into SCALE are processed differently depending on the shipment order type. Every shipment that is interfaced has a unique item per line with the requested quantity in the lowest unit of measure.

The waving supervisor monitors the **Planned Shipment Insight** with preconfigured filters called *Planned Shipment filters*. Knipper runs several waves throughout the day.

Planned Shipment Filters display planned shipments based on the criteria that you select. You could, for example, define a rule in SCALE configuration for a particular customer, carrier, and order type. You could then display shipment records that meet those criteria in the Planned Shipment Insight. You can define pool views using any combination of shipment or shipment line values. If you specify a shipment line in your filter criteria, the system will use it to determine if shipments should be included in a pool view. If a shipment line matches a rule, then the system will select the entire shipment for the pool view.

Knipper uses the existing *planned shipment filter* criteria are created aiding in the waving strategy. Knipper may create additional filters as needed.

Commented [CC240]: Will all our current pool views be imported into new SCALE or will they require reconfiguring. When will this happen? I would like to clean up old Pool views no longer in use.

Commented [NC241R240]: NC11192024: Knipper to clean up pool views before going into new SCALE

20.2 Wave Flow

A wave flow is a grouping of wave steps. These flows determine which wave steps that a shipment will be processed through, and the order in which the wave steps will be processed. Once a wave flow is created, it can be associated with the appropriate wave on the Wave Master Window. (Defined later in the document)

Knipper uses existing wave flows and wave masters.

Sequence	Description
10	Start Wave
20	Override Data: Set Default Status Flow
30	Override Data: Set Allocate Complete Flag
32	Override Data: Set Packing Class
34	RTS Address Verification
40	Allocation
42	DT Location Assignment
43	DTP Wave Splitting
50	Container Creation

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60	Routing
70	Pallet Building
80	VAS Assignment
90	QC Assignment
100	Dock Assignment
110	Load Building
120	Work Creation Shipping Container
130	Paperwork - Labels
999	Override Data: Check for No Work
1000	Complete Wave

Standard wave flow

20.3 Build Wave – Future Use

Knipper wants SCALE to build wave instead of wave planner selecting what shipments needs to be waved. Knipper will configure wave criteria which will define rules as what shipments are eligible to be picked up to be waved. During the build wave process, wave criteria records determine which shipment(s) will be included in a wave. Knipper can associate these criteria records with wave master. Also, user can define wave criteria records using any combination of shipment or shipment line values. If user specify a shipment line in filter criteria, the system will use this to determine if shipments should be included in a wave. If a shipment line matches a criteria rule, then the system will select the entire shipment for the wave. Note that all the shipment lines do not have to match for a shipment to be selected. These Build waves are run on a scheduled job so that they can pick up shipments from pool over a period.

20.4 Run Wave during build wave process

Wave masters configured for Knipper will have mode set to Automatic. Mode Automatic means as soon at a wave is built, the system will run it.

Commented [SG242]: Only 1 waveflow? What about ODWS?

Commented [NC243R242]: NC11182024: what does this mean?

Commented [RS244R242]: The question is if Knipper would use only one wave flow which is not true. Knipper would migrate all the wave flows from the current version which will still be based on this standard wave flow template.

Commented [NC245R242]: NC12052024: This is resolved.

Commented [CC246]: I'm looking for some clarity on 4.3. Is this a way have a wave be created automatically? It will build a wave and run the wave based on specific criteria?

Commented [NC247R246]: NC11182024: MAH to confirm

Commented [RS248R246]: Yes, the Build wave is a scheduled job which can generate waves based on criteria. We can have multiple criteria defined which can create waves with different wave flows.

Commented [NC249R246]: NC12052024: This is resolved.

Commented [CC250]: As shipments are added to a wave, I want to manually release the wave unless the wave is set up as in 4.3 if I understand that correctly

Commented [NC251R250]: NC11182024: this is resolved

Figure: Wave Master – Mode config

Upon confirmation of the run wave request, the system executes the steps detailed in the Wave Flow associated with the wave. Upon completion of running the wave, the system moves the wave into the Completed status.

20.5 Add to Wave – If Build Wave is not used

To initiate the wave process, wave planner selects a shipment (or multi-selects more than one shipment) from the **Planned Shipment Insight** and selects “*Add to Wave*” from the Actions menu. Next, the wave planner is prompted to add the shipment(s) selected to an existing open wave, or they may select *New Wave* to manually assign the shipment(s) to a new wave. When executing the “*New Wave*” action, the system prompts the user to select a “*Wave Master*” to act as a template that manages the movement of shipments through the wave cycle. The Wave Master defines the Wave Flow, Replenishment Master(s), and Paperwork Master for the wave. The wave flow defines the sequence and specific steps SCALE performs when running the wave. Examples of Wave Flow steps include allocation, work creation, printing documents, etc. The Wave Flows are defined via the Wave Flow windows in the fixed station Configuration option. The Replenishment Master defines what replenishment masters are eligible to evaluate demand within the wave. The paperwork master defines the documentation that is printed with the wave.

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For Knipper the following wave masters will be configured in the system:

- Project (REPS)
- EX17 Re Carton Cooler Wave Domestic LTL wave master
- DTP – PTL
- PTL – Case Pick
- Destruction Wave Master
- 3PL
- PM Manifest
- DEA
- 3PL - PM LTL FT
- 3PL - PM UPS
- 3PL - PM ADRS LTLFT DSCSA

Note: Some wave masters would be replicated with facility/Business unit/client with some additional wave steps to add some specific processing step through an override data wave step.

After confirming the “Wave Master” for the wave, the system assigns a wave number (via a next up counter) to group the selected shipments. The shipments are removed from their previous pool view. The wave is created in the Planned Shipment sections of the **Wave Insight**. The wave is now ready to be run.

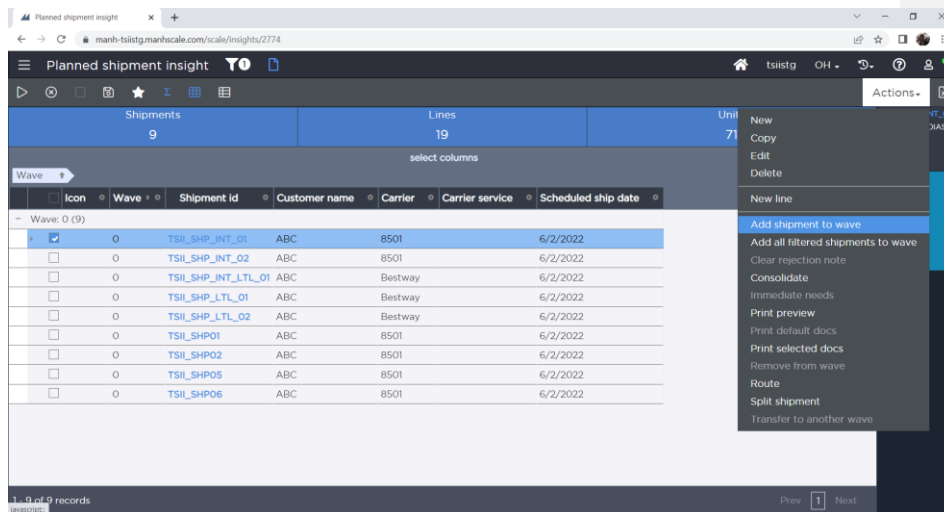


Figure: Planned Shipment insight – Add shipment to wave option

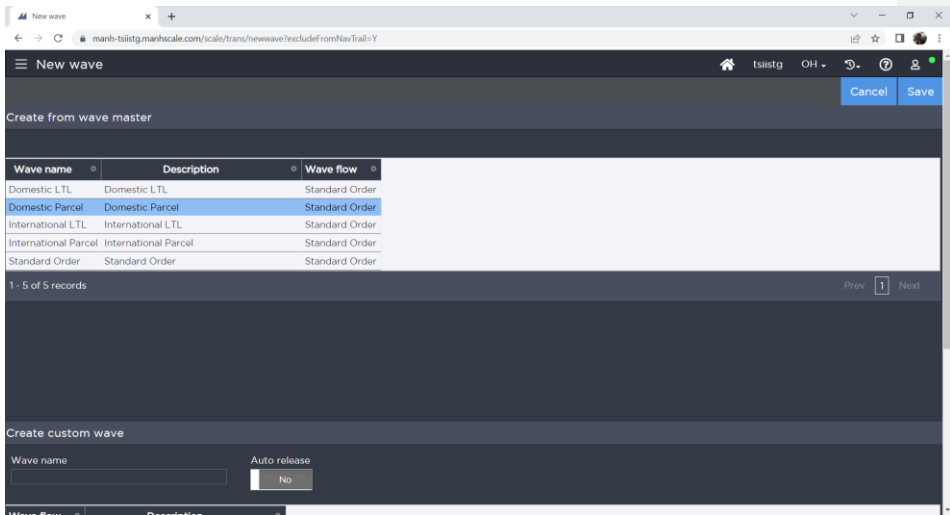


Figure: Selection of Wave Master

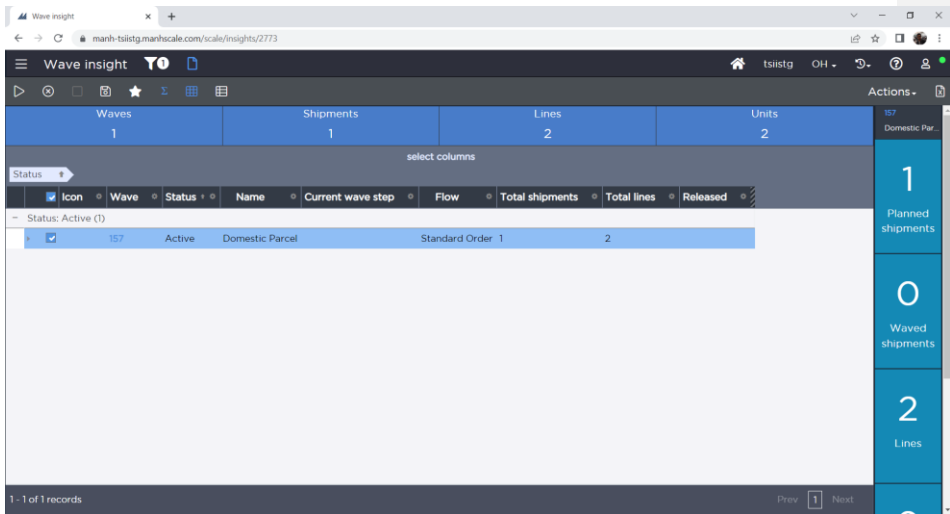


Figure: Wave Insight screen

20.6 Run Wave

After reviewing a wave and making any necessary changes, wave planner uses the Run Wave action on the desired wave in the *Active Wave* view of the **Wave Insight**. Upon confirmation of the run wave request, the system executes the steps detailed in the Wave Flow associated with the wave. Upon completion of running the wave, the system moves the wave into the *Completed* status. Knipper would like Wave Insight to auto refresh while waves are running [EX48 – Auto Refresh Wave Insight Screen].

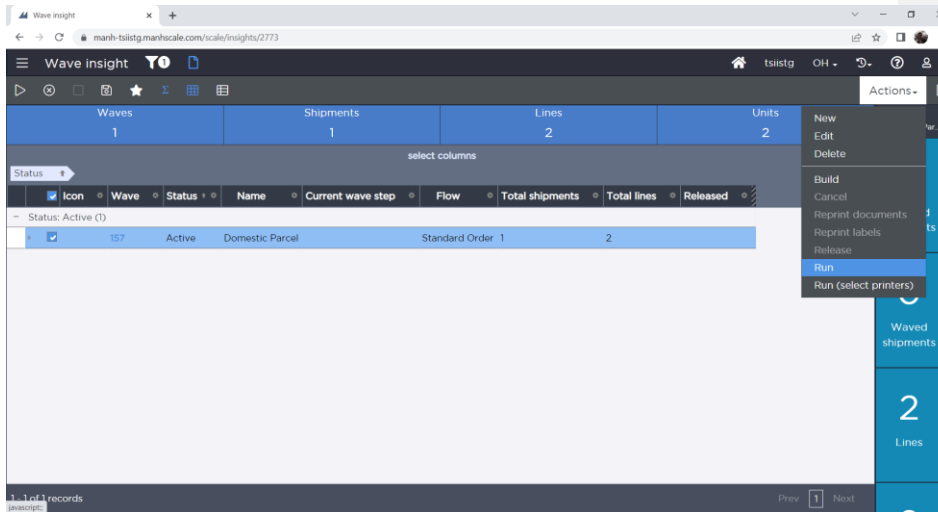


Figure: Wave Insight – Run wave option

[EX13 – Pick to Light Wave Splitting] will allow all Direct to Physician orders to be run through a single wave that groups and sequences the orders into new waves to be fulfilled across the same group of pick to light locations.

20.7 Wave Steps

The following sections explain the logic the system uses when executing the various key wave steps that are included in standard wave flow. Please note that there may be additional wave steps that may be added during the build phase.

20.7.1 Start Wave

Start Wave is required for all wave flows and marks each shipment with a status of *In Wave*.

20.7.2 Override Data: Set Default Status Flow

This Override Data Wave Step updates the Status Flow field of the Shipment Details to a value based on the below table:

Various Custom Status flows that will be configured for Knipper are below:

Note: Knipper currently has configured site specific status flow names which are redundant. The status flow configurations can be consolidated with generic names.

- Status Flow - Parcel

Parcel	
Status	Status Name
100	In Pool
200	Wave Pending
201	In Wave
300	Picking Pending
301	In Picking
401	In Packing
700	Ship Confirm Pending
800	Load Confirm Pending
900	Closed

- Status Flow – Non-Parcel/LTL

Non Parcel/ LTL	
Status	Status Name
100	In Pool
200	Wave Pending
201	In Wave
300	Picking Pending
301	In Picking
401	In Packing
650	Loading Pending
700	Ship Confirm Pending
800	Load Confirm Pending
900	Closed

20.7.3 Replenishment

The each/case demand from the wave is evaluated against the primary picking locations to determine if there is enough available. If not, SCALE requests the product in case (or configured UM) increments by rounding up. These replenishments create "In Transit" inventory to the forward picking locations which are then allocated by shipments during the Allocation wave step. This allocation is done leveraging FEFO strategy and is explained in more detail in the Replenishment section of the document.

20.7.4 Override Data: Set Packing Class

As part of this wave step, SCALE will assign a Packing class value on the shipment detail record which will indicate what items can be combined in a single shipping container and the type of shipping container to be used.

Note: Knipper will utilize override data wave step to assign packing class at the shipment detail level as required to handle some special processing. Knipper will need to review the override data SQL fetch the packing class values from generic configuration based setup instead of hardcoded values.

20.7.5 RTS Address Verification

As part of this custom wave step, SCALE will verify the ship to addresses with RTS system. This custom wave step is part of **[EX38 – Serialization Integration with Rfxcel for Outbound]**. If the ship to address is not successfully verified by RYS system, the shipments will be cancelled back to the pool.

20.7.6 Allocation

SCALE allocates inventory towards a shipment by using the allocation rule defined on the shipment detail. Allocation rules define what locations and units of measure are eligible for allocation and how SCALE should allocate that inventory. For example, an allocation rule can be configured to allocate inventory only from mid-level bins and to allocate the inventory with the FIFO. Allocation rules can have multiple sequences. SCALE attempts to allocate inventory using the first sequence and if inventory could not be 100% allocated, SCALE continues to the next allocation rule sequence to attempt to allocate the remaining inventory.

Allocation rules are set on the shipment detail in one of three ways. The allocation rule can be set at the item level (in the Item Master configuration) which in turn automatically defaults on the shipment detail. The interface can also set the

allocation rule on the shipment detail. In addition, the allocation rule can be determined in the wave just prior to allocation occurring. This is done using the Allocation Rule Assignment functionality. Allocation Rule Assignment sets the allocation rule on the shipment detail based on user-defined criteria. For example, based on the customer and order type, the allocation rule can be set to one that allocated in only full cases. If no allocation rule is set on the shipment detail, the *Default allocation rule is used. To allow for maximum flexibility and easy addition of new items and allocation rules, the allocation rule is assigned on the Shipment Detail using the Allocation Rule Assignment functionality in the wave. Knipper can still manually set the allocation rule in the interface or via the Shipment Detail screen as needed.

The below table covers majority of locations to allocate inventory for Shipments. The zone names in the below table are generic and multiple warehouses uses the same zone name with warehouse name appended in the name.

Note: Knipper will utilize override data wave step to assign allocation rules at the shipment detail level as required to handle some special processing. Knipper will need to review the override data SQL fetch the allocation rule values from generic configuration based setup instead of hardcoded values.

	Allocating Zone	Description
1	A-DTP-PM A-DTP A-DTP REF A-DTP Cart	• Single Item • Dynamically Assigned and Permanent • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available
2	A-DEA A-DEA-CAGE A-DEA-PM	• Single Item • Dynamically Assigned and Permanent • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available
3	A-DEA-FLOOR	• Multiple Item • Dynamically Assigned • License Plate Tracking • Allocate In Transit • Inventory Status = Available
4	A-3PL-CP A-3PL-PM A-3PL-LANE A-3PL-RESV	• Single Item • Dynamically Assigned • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available

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5	A-Shelves (KMW and OHW)	• Single Item • Dynamically Assigned and Permanent • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available
6	A-FREEZER (KMW and OHW)	• Single Item • Dynamically Assigned and Permanent • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available
7	A-REFG (KMW and OHW)	• Single Item • Dynamically Assigned and Permanent • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available
8	A-PNP	• Single Item • Dynamically Assigned and Permanent • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available
9	A-RESV (KMW and OHW)	• Single Item • Dynamically Assigned • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available
10	A-CP (KMW and OHW)	• Single Item • Dynamically Assigned • Not License Plate Tracking • Allocate In Transit • Inventory Status = Available

Allocation Sequence for non-Lot controlled:

Seq	Strategy	Location Selection	Eligible UMs	Inventory Status	Clear Loc?	Lot
10	First In First out	A-RESV	PL	Available		
20	First In First Out	A-CP	CS	Available		
30	First In First Out		PK, EA	Available		

Allocation Sequence for Lot controlled:

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Seq	Strategy	Location Selection	Eligible UMs	Inventory Status	Clear Loc?	Lot
10	First Expiration First out	A-RESV	PL	Available		
20	First Expiration First out	A-CP	CS	Available		
30	First Expiration First out		PK, EA	Available		

20.7.6.1 Allocate Complete

By setting the 'Allocate Complete' flag on the shipment header, SCALE does not perform any allocation for this shipment unless the shipment can be 100% allocated. For Knipper, allocate complete functionality is used for defined order profiles.

20.7.6.2 Allocation Rejections

If the shipment line is not completely allocated, then the entire Shipment is sent back to the pool on a back order.

Note: Default status when a shipment is rejected is set to 'In Pool'

20.7.7 DTP Location Assignment

This custom wave step is part of **[EX13 – Pick to Light Wave Splitting]**. This wave step evaluates the orders allocated at Virtual Locations and assigns them to a series of Pick to Light Locations in groups across one or more sub waves. As the wave step cycles through the Pick to Light locations of each sub wave, orders are assigned to Pick to light locations based on "Most common items" that have the "fewest number of lines."

20.7.8 DTP Wave Splitting

This custom wave step is part of **[EX13 – Pick to Light Wave Splitting]**. Once Pick to Light Assignment has been completed, a separate Wave step is executed to move the shipments, their allocations, and replenishments into a new wave for the Sub Wave Master. After this information has been transferred to the new wave, the Replenishment and Allocation Requests are moved from their Virtual Location to the Assigned Pick to Light Location in the sub wave.

20.7.9 Container Creation

Commented [CC252]: We reject the whole shipment

Commented [NC253R252]: NC11182024: This does not happen today. MAH to update

Commented [RS254R252]: Updated

Commented [NC255R252]: NC12052024: This is resolved.

Container creation is the process in which SCALE determines the number of containers and each container's contents for every shipment on the wave. A unique container number, commonly referred to as a UCC 128, is assigned to every container as it is created.

SCALE first takes all items whose allocated unit of measure is set up as a 'shippable unit' and creates full containers. Shippable units are defined in the item unit of measure configuration. Full containers are containers that are shippable without repacking into another container (box). Pallets, gaylords, and certain cases are generally configured as shippable units.

SCALE then takes the remaining 'loose' items whose allocated unit of measure is set up as 'not shippable' and groups them together by Packing Class. A Packing Class is defined on the Item Master and then defaulted on the Shipment Detail. Each Packing Class is associated with a container group. A container group is a listing of container types (box sizes) listed from largest container to smallest container. SCALE attempts to cube items into the least number of containers possible. The total weight and volume are calculated for all the items on the shipment with the same packing group. SCALE first tries to cube into the first container of the container group. If capacity still exists, the system tries cubing into the next container type and continues until either capacity in the next priority is reached or there are no additional priorities.

If there is not enough capacity in the largest container, then SCALE cubes as much as possible into the largest container, calculates the remaining weight and volume, and starts over. SCALE considers the critical dimensions of each item it creates containers. An item is not cubed into a container if any one of its critical dimensions is greater than the corresponding container dimension.

If an item doesn't have unit of measure record defined, then SCALE treats the item to have dimension of 0*0*0 and of weight 0 LB. Knipper can build oSCI report that will provide the list of all items that have missing dimension in pool and use that to handle the exception. Knipper can also create personal alerts-based Notification to send all the items that have missing dimensions.

Knipper ships cooler and Freezer items. These require icepacks and dry ice to be present inside the shipping container. **[EX17 – Re-Cartonization of Containers based on Season]** will be used to handle special Cartonization requirements for some customers.

20.7.10 Override Data: Assign Accessorial

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Knipper uses third party / customer account for billing some of their parcel shipments while the others are billed to Knipper. For the third-party billing, Knipper assigns accessorials to the required cartons for the parcel orders.

Third party billing information is applied via an override data wave step (ODWS) in wave. Knipper provides the account number, billing address and other information that must be billed and interface to shipment header through shipment download interface. This ODWS reads the information and applies to the cartons when SCALE communicates to FedEx. During the build phase, the scope of this ODWS will be evaluated.

Knipper will be using the below accessorial.

- 3rd Pty Billing
- Alternate Address
- Auto POD
- COD
- Collect
- Commercial Invoice Method
- Consignee 3rd Pty Billing
- Delivery Notification
- Department
- Dlv Confirm
- Dry Ice
- Duty Tax Payment Type
- DVL
- EEI
- FedEx Ref
- Flats
- HAZMAT
- HLD
- Home Delivery Type
- Insurance
- Pickup / Delivery
- POD
- Premier Service
- Print Alt Return Add Flag
- Proactive Recovery
- Proactive Response
- Quantum View Dlv
- Quantum View Exception
- Reference
- References
- Registered Mail

Commented [SG256]: Looks like lot of custom accessories

Commented [NC257R256]: NC11182024: ok

- Return Delivery
- Return Shipment Indicator
- Sat Dlv
- SED
- Sign Rlse
- Signature Required
- Signature Required - Adlt
- Signature Required - Dirc
- VATEIN number

20.7.11 QC Assignment

The QC Assignment wave step assigns QC activities to shipping containers for any matching criteria configured within SCALE's QC Assignment Criteria. For Knipper, every shipping container created in wave will be assigned to QC. For scenarios where pallet is created using the pallet building wave step, QC will be assigned at the pallet level.

Note: For QC to be assigned, containers must be created as part of wave. Also, for Knipper we need to enable QC execution at pallet level. This will be handled as an extension (TBD) in SCALE.

20.7.12 Pallet Building

Knipper uses wave-based pallet building for LTL/TL order waves.

Pallet Building consists of a set of requirements, strategy and the Container Type for the pallet being built. The Container Type drives the dimensions, weight and height of the pallet so that these constraints are known so the build pallet meets the specifications defined in the configurations.

The Pallet Building Requirements help define the types of containers that should be included, but also sort these containers so that the pallets are built in the most efficient but cleanest way to keep boxes from crushing other boxes if stacked incorrectly. Finally, the criterion helps define if any attribute should automatically break to a new pallet regardless of if the height or weight is met. For customers that require that a pallet can contain only a single item, this break field is used to make sure that all items are on their own pallet.

Pallet Building Strategies take the Requirements and Container Type and build out the pallet. The standard strategy takes the containers and builds out layers until the height or weight of the pallet meets the limits on the Container Type. This

strategy also needs to incorporate the grouping of items, so that SCALE does not attempt to ship two mixed pallets with the same item on both pallets. This strategy looks out for any items that are not grouped together and consolidates these items onto the same pallet if they can be physically fit based on the quantity being shipped.

Full cases picked from the selective reserve or case area may be palletized. Knipper may use the Maximum Height and Weight Strategy for pallet building and uses the pallet building criteria to exclude full allocated pallets. Over time, Knipper may setup additional criteria and strategies based on the product category. Pallet building criteria can be ordered by descending shipping container weight.

Knipper uses 'Build Pallets using Height-Weight' strategy.

20.7.13 Load Building

With this wave step, SCALE attempts to assign all shipments on the wave to a shipping load (using shipment criteria) for a carrier. The system executes this process as follows:

- The system orders all of wave's shipments by carrier, ignoring shipments that do not have a carrier assigned.
- The system takes each shipment and looks for any open loads for the carrier, scheduled ship date, and route. If the system finds one, the shipment is added to that load, unless the load is not flagged to stop additional shipments to it.
- If there are no loads in the system for the carrier, scheduled ship date, and route, SCALE creates a new load for these parameters and places the shipment on this new load.

20.7.14 Dock Assignment

Dock Assignment determines where product is moved to, after it is picked. This could be any one of the following: manifest station, staging location, or dock door. Knipper defines a custom status flow for each of the different process flows in the system (defined above). Based on the status flows, the Dock Assignment wave step assigns the correct packing, staging or dock door location for an order. This is controlled by three configurations: *Dock Area Carrier Assignment*, *Dock Area Anchor Criteria* and *Dock Management Flow*.

A dock management flow record includes information that determines how the system will assign a dock location destination to a shipment line/container and if the line/container is eligible for assignment. Each flow record is made up of a series

of detail records that identify what selection and assignment strategies that the system should use when a quantity hits a certain status in your status flow. You can indicate what type of dock location you want the system to assign to this entity. Also, both the flow header and detail(s) have a default location that the system will assign as a "fallback" option in case no eligible dock area/positions are found.

For Knipper, the dock assignment is leveraged for TL/LTL as well as parcel shipments. SCALE assigns single staging locations to a shipment based on customer and carrier. Multiple dock management flow is used based on each of these parameters and dock door assignment happens manually later to create work.

If a new shipment on a different wave needs to be added to a load that is staged, by base the new shipment (new wave) gets default staging lane that can be different than what is on the original. Knipper uses manual SOP to assign the status flow for staging lane with the existing load.

20.7.15 Work Creation

The work creation process performed during the wave is like the work creation process performed during the locating portion of the receiving process. After performing allocation and container creation, SCALE creates a work unit to pick the inventory from a location (bins or racks) and transport the inventory to the shipping area based on the configuration in the Work Group, Work Type, Work Criteria, and Work Creation Master. Each location can be assigned a numeric value called Picking Sequence that can be used to sort the picks in an order other than alphabetical by location.

20.7.16 Work Creation Replenishment

The work creation process for replenishment during the wave is only for any demand-based replenishments that were created. The types of work that are generated from this wave step can be viewed in the Replenishment section of this document. There is no change to existing work creation setup.

20.7.17 Work Creation - Shipping Container

Following key Work Types are created:

Work Type: Group (Cart) Pick

One work unit is created per shipping container. User will group a set of containers onto a Cart (this will enable group picking). If a cart has more than one item, the work instruction (picks) is sequenced by pick sequence.

Commented [SG258]: Under wrong heading

Commented [NC259R258]: NC11182024: if this is under the wrong heading - please update

Commented [RS260R258]: The header has already been corrected.

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Note: This type for picking will be utilized by both the business units (3PL and MSM) in all the facilities.

Work Type: 3PL Pick

One work unit is created per pallet built using the pallet building wave step. The work unit will include Repack boxes and Full Cases nested onto a pallet. The work is ordered based on the pick sequence. This work type includes Full cases and Repack boxes that are picked from ambient zone.

Work Type: Cooler Pick

One work unit is created per shipping container. Within the work unit the work instruction (picks) is sequenced by picking sequence. This work type includes Items that are picked from Cooler zone.

Note: This type for picking will be utilized by both the business units (3PL and MSM) in all the facilities.

Work Type: 3PL Pick DSCSA

One work unit is created per pallet built using the pallet building wave step. The work unit will include Repack boxes and Full Cases nested onto a pallet. The work is ordered based on the pick sequence. This work type includes picking for DSCSA tracking enabled Items.

[EX38 – Serialization Integration with Rfxcel for Outbound] enables DSCSA tracking to validate GTINs and Serial numbers during picking for different UMS. The extension design document describes the entire flow in detail.

Work Type: REPS Pick

One work unit is created per Wave. The work is ordered based on the pick sequence. This work type includes Full cases and Repack boxes that are picked for medical representative orders.

Work Type: DEA Pick

One work unit is created per shipping container. Within the work unit the work instruction (picks) is sequenced by picking sequence. This work type includes Items that are picked from DEA zone.

Note: This type for picking will be utilized by both the business units (3PL and MSM) in all the facilities.

Commented [MA261]: KMW will need DSCSA Work Types as well.

Commented [NC262R261]: NC11182024: All Facilities NJ (OHV, KDC and KMW) have the ability to do DSCSA - MAH please update this work type

Commented [RS263R261]: We can always create multiple work types by warehouse. However, it is recommended to keep one which can be used by all the warehouses.

Commented [NC264R261]: NC12052024: This is resolved

Work Type: PM Pick (non MHE)

One work unit is created per shipping container. Within the work unit the work instruction (picks) is sequenced by picking sequence. This work type includes Items that are picked from Pick Module zone.

Note: This type for picking will be utilized by both the business units (3PL and MSM) in all the facilities

Work Type: PTL Pick (Pick to Light MHE)

One work unit is created per shipping container. This work type includes orders that are picked from PTL (Pick to Light) controlled zone.

Note: This type for picking will be utilized by both the business units (3PL and MSM) in all the facilities for different types of orders (e.g. DTP). **[EX19 Pict to Light Integration]** will auto execute the work units upon receiving pick confirmation messages from PTL.

Work Type: PTL Case Pick (Pick to Light MHE)

One work unit is created per shipping container. This work type includes orders that are picked from PTL (Pick to Light) controlled zone for full case picks.

Note: This type for picking will be utilized by both the business units (3PL and MSM) in all the facilities for different types of orders (e.g. DTP).

[EX19 Pict to Light Integration] will auto execute the work units upon receiving pick confirmation messages from PTL.

20.7.18 Paperwork - Labels

Knipper uses existing paperwork setup.

Container content labels (LBL03) and Shipping labels (LBL04) will be printed per container for repack boxes for Parcel carrier shipments.

Container content labels (LBL03) and Shipping labels (LBL04) will be printed for Full cases for parcel carrier shipments.

Container content labels (LBL03) and Vendor label (LBL02) will be printed for LTL/TL carrier shipments.

These Labels are printed at the time of wave release.

20.7.19 Paperwork – Documents

Knipper prints Shipment Pack List during the wave and will use multiple different templates for customer requirements.

Note: Knipper utilizes an external printing process (VVIP) to print special formats of shipment pack list for specific customers. VVIP requires SCALE to generate an input file (XML format) per wave. An extension (TBD) will be developed to accomplish this requirement.

20.7.20 Override Data: Check for No Work

This Override Data Wave Step ensures that the Work Creation configurations are correct, and every Shipment has work created successfully, otherwise marks the wave for failure.

20.7.21 Multiple ODWS placeholder

Knipper may use multiple other override data wave steps. This is a placeholder for it. There are other steps in current system included in other wave flows that mostly have programmatic checks ensuring that ineligible order types are not waved in those wave flows.

20.7.22 Complete Wave

Complete Wave updates each shipment with a status of *Picking Pending*. Shipments with this status are now eligible for release.

There may be additional wave steps to the above that Knipper can leverage for various wave flows though the ones mentioned above are the key. Examples may include updating user-defined fields for any reporting needs.

21.0 WAVE MANAGEMENT

After building and running the wave, the wave supervisor reviews the results of the wave using the full-screen **Transaction History Insight** and **Work Insight** options. Depending on the results, the user may perform one of two options. These options are identified in the following sections.

21.1 Cancel Wave

If the results of the entire wave are not satisfactory users may cancel the wave using the Cancel action in the Completed Wave window of the **Wave Insight**.

Commented [BK265]: 5.1 Cancel Wave - when a wave is released but work has started, it can no longer be cancelled at the user level and IT staff needs to be asked to step in. When systemic driven work has been scanned and the wave has "started" this presents a problem. We would prefer a wave be able to be interrupted, cancelled after being started.

Commented [NC266R265]: NC11222024: MAH to provide options for this

Commented [RS267R265]: This is the standard behavior. We cannot cancel a wave once it is already released. Also, we cannot cancel/delete work units which are already in Process. The process would be to complete the work and cancel the shipment.

Commented [NC268R265]: NC12052024: This is resolved.

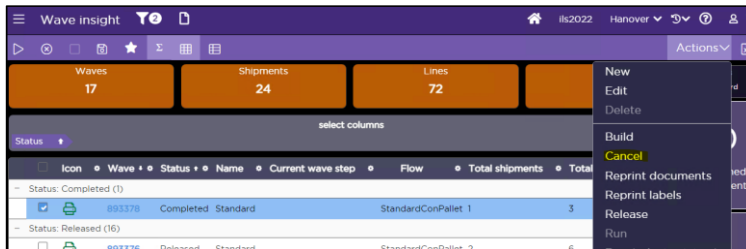


Figure – Cancel a wave

The cancel process backs out allocation and deletes work instructions and shipping containers. As allocations are backed out, the corresponding inventory is now available for order fulfillment again. At the point of Cancel, Knipper personnel can move the canceled shipments onto another wave or move them completely back to the Pool.



Figure – Return canceled shipments to the pool or another wave.

21.2 Release Wave

If the results of the wave are satisfactory, Knipper personnel release the wave. The Release action performs multiple operations:

- Releases the generated work to the warehouse floor (if applicable) by removing the Hold Code from the work units – this allows the work to now be eligible for picking
- Prints Wave Documents and labels as applicable if configured.

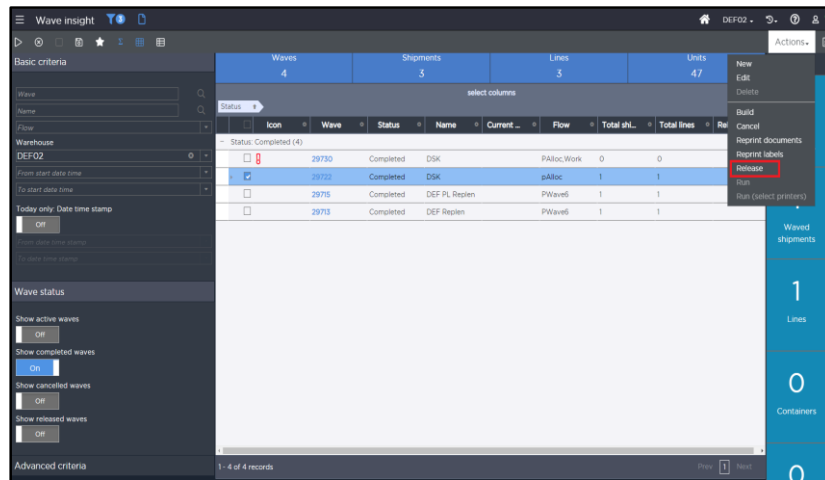


Figure – Release a wave

21.3 Hold Codes

Hold Codes allow work to be temporarily placed on hold so that no further processing can be done against it. Work created through a wave initially has a Hold Code of **Wave Not Released**. When a wave is released, the Hold Code is removed.

21.4 Print Wave Documents or Labels

Releasing a wave prints the Wave Documents. This is the normal process Knipper uses to print these documents. If for some reason the Wave Documents ever need to be reprinted at a later point in time, this is always possible by choosing the Reprint Wave Documents option from the **Wave Insight**.

Commented [BK269]: 5.3 Hold Codes: It would be helpful to have the wave status of released, held (with hold code) or cancelled visible in work quick find view to assist supervisor in releasing warehouse "printed" work. Also recording the time of release, hold, release from hold, or cancel be recorded and visible in transaction and process history

Commented [NC270R269]: NC11222024: MAH to confirm if 1. held (with hold code 2. cancelled visible

Commented [RS271R269]: The wave insight shows whether the wave has been Releases/cancelled. The hold code would be visible in Work Insight screen. It will indicate if is held due Wave not released

Commented [NC272R269]: NC12052024: This is resolved.

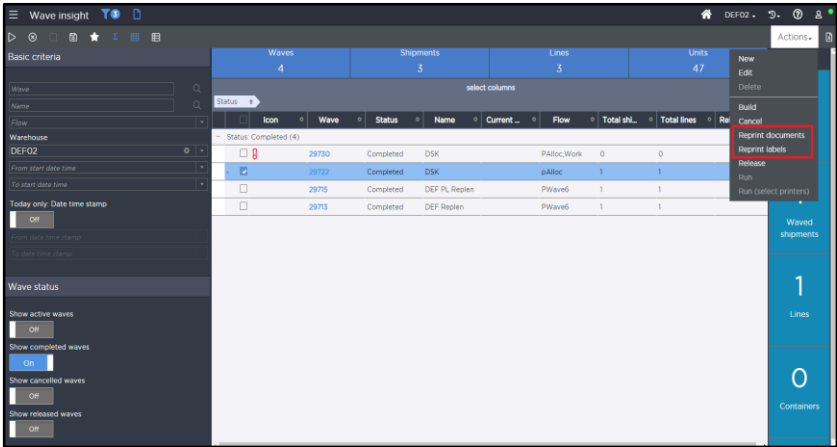


Figure – Reprint wave labels

21.5 Post Wave Shipment Changes

If a change is required for an order that has been waved and released (but not picked), the shipment can be canceled by choosing the Cancel option from the **Shipment Insight**. Canceling a shipment de-allocates the inventory for the order, deletes the created shipping containers and work, and moves the shipment back into the pool. Once an order is partially or completely picked, it can be canceled by the same previous process, but any items that were picked would need to be transferred back to an inventory location using the **Inventory Management** option.

Note: If an order is canceled it cancels all the shipping work tied to the order. Replenishment work will not be canceled (Replenishment work can be manually canceled). Also, while we cancel an order, if any work related to the order is being actively executed by picker(s), then order cancellation fails. Canceling of the shipment after the wave is released can only be performed by a warehouse user. After the wave has been released, the Host will not be able to cancel the order (The only time the Host system can make a change is when a shipment is in 'In Pool' status).

Commented [CC273]: This is not the current case, today we can cancel an order if it is being picked the order cancellation does not fail

Commented [NC274R273]: NC11182024: this is resolved

22.0 WORK MANAGEMENT

22.1 Work Viewing

Using the fixed station **Work Insight** or **Work Monitoring**, personnel can monitor the progress of work. The **Work Insight** contains views grouping work by condition: Open, In Progress, and Closed. Additionally, this option enables personnel to view the types of work that are open and in progress to determine if additional users are needed to help with receipt putaway, replenishment, or picking work. Personnel can inquire about the specifics of the work unit, such as the 'from' and 'to' location, item(s) and quantity being moved, receipt/shipment id, the user performing the work, etc.

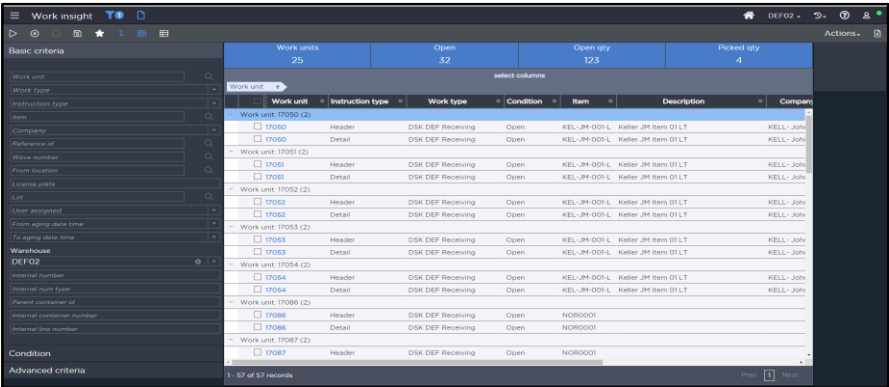


Figure – Work Insight

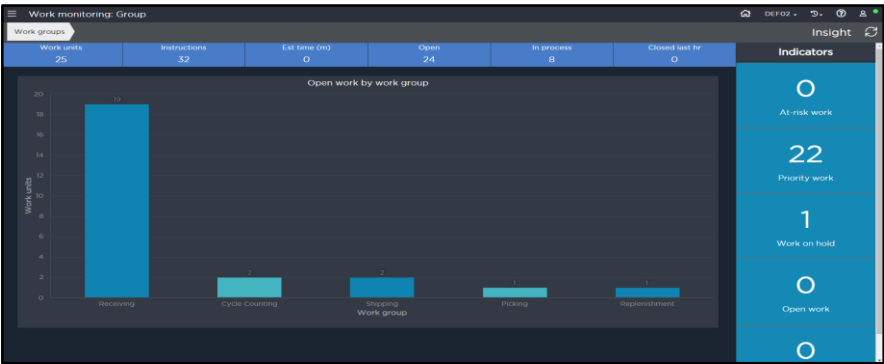


Figure – Work Monitoring Group

22.2 Work Priority

SCALE allows system directed tasks to be assigned in one of three ways:

- Priority/Location/FIFO
- Location/Priority/FIFO
- FIFO

The above defines the sort order that system-directed tasks will be assigned to users within the warehouse. Within this structure, the "priority" portion from above can be critical. Most tasks will need to be created with the same priority so that they are truly processed by location proximity in ascending order. However, over time Knipper may want to bump up the priority of certain tasks. This is possible via the Work Priority Escalation Criteria configuration. From this configuration, Knipper can define what types of work should be bumped up. This can then be used in conjunction with an interval that is defined by a scheduled job for the work priority.

23.0 PICKING

23.1 Group (Cart) Pick

Assumptions:

- Containers are created in wave.
- One Work unit is created per shipping container.
- At Wave release container content labels are printed.
- User will physically pick the inventory into final boxes or to a Tote. Determination is made by the picker.
- User will physically build the cart before starting the work execution.
- User will build the cart based on the zone where the items are picked.

The user starts picking by signing into the **Warehouse Mobile on RF** and selecting a Work Profile of *Group (Cart) Pick*. This is a user directed work assignment, which means user needs to tell SCALE which work unit they want to execute. The user will start building the cart by scanning the ContainerID from the container content label. SCALE will assign a spot systematically and display the Spot and the container type. If the picker is picking to the final box then they need to place the right container type into the spot displayed. If picking to Tote, then place the tote into the displayed spot. Place the container content label inside the Tote or the final box. Continue scanning the ContainerID and clicking Go button until the cart is Full. When cart has been built, the user will select the Begin picks option to initiate the pick. System will assign the task to the user and SCALE then presents the user with the pick which displays the location, item, lot, and quantity to be picked **[EX24 - Display Item Alias in RF Pick confirmation screen]** displays the Item Alias if exists for an item. The picker confirms the pick by scanning the Location and item **[EX39 - Work Confirmation Item validation]** for validation.

Commented [BK275]: 6.2 Work Priority: I am unaware if MHE truck type can be brought up here or at another point in this document. Meaning I would like to prioritize work or assign work by machine type - reach or order picker for example. Our guys are constantly jumping from reach, to order picker and then back again. If an entire pallet is being pulled then maybe a pick sub-type "reach" could be assigned along with PM Replen (for example). Then management can assign sub type to a single user.

Commented [NC276R275]: NC11222024: MAH to confirm if this can be done

Commented [RS277R275]: I am not sure if I understood this completely. Please let me know if we need a quick call with Kenneth?

Commented [BK278R275]: In other words, some replenishments are for 1 case or 2, from a reserve location which would enable the driver to complete multiple replenishments with an order picker in 1 trip. Some replens are for a pallet, requiring a reach truck and the ability to only make that 1 pick. If the driver is picking less than total quantity, he would know to use an order picker, but if picking the location complete he uses another machine. Feel free to call me any time on teams

Commented [RS279R275]: If it helps, these can be created as separate work types (based on the qty UM replenished) in SCALE and can be assigned to two different users. Please let us know if this is fine and we can resolve.

If item being picked is part of only one container on the cart, the slot number and Container ID will be displayed on the pick confirmation screen itself. If the item is serial number tracked, then user needs to scan the serial number at the unit level.

If the item being picked is part of multiple containers on the same cart, then once user confirm the pick, SCALE will display the slot and quantity where they need to put the inventory. The user can scan the ContainerID Barcode from the container content label in that box to confirm they are putting inventory into correct box. If the item is serial number tracked, then user needs to scan the serial number at the unit level. Once the user confirms the slot it will display another slot where the same item needs to be putaway. Once the picked item is putaway completely, SCALE will display the next pick based on the pick sequence.

While the user is picking, if there is not enough inventory in the location that the user is prompted to pick from, the user presses the '*Skip*' button to continue with the next instruction on that work unit. If after all the picks are complete for the work unit and they are returned to the screen with the short item, if the picker has permission to short, picker will adjust the quantity and then click on the short button. SCALE will ask for a short reason and user must select the reason from the drop down. If picker doesn't have permission to short, the picker presses the '*Pass*' button to back out of the work unit (to leave it suspended) and moves the cart to hospital area. The supervisor intervenes and determines how to handle the out-of-stock situation.

Note: Short option is controlled using security and Knipper can decide which pickers will have access to short option. If picker doesn't have short option, then once they click on the pass button, they need to take the cart to a designated hospital area and leave the cart there for supervisor to research on the issue.

Once all the picks are completed user will confirm putaway to PACK location and will take the cart to packing station and unload the boxes.

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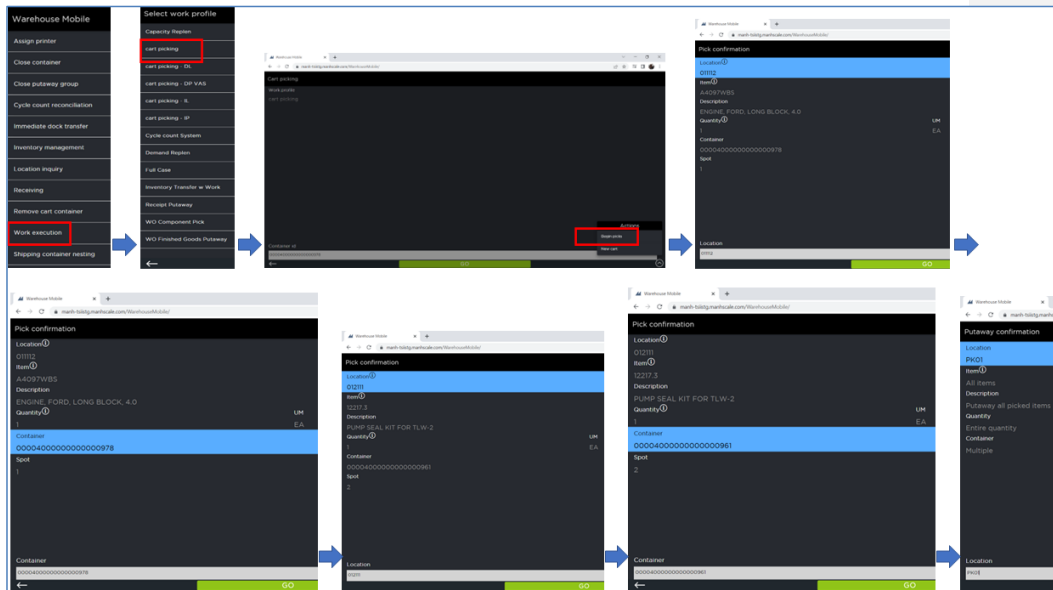


Figure: Cart pick work execution

23.2 Full Pallet Pick

Assumptions:

- Containers are created in wave.
- One Work unit is created per Pallet.
- At Wave release Container Content label and Vendor Label will be printed.

The user starts picking by signing into the **RF on warehouse mobile** and selecting a Work Profile of 3PL Pick. The user is then prompted to scan a work unit, and user will scan the ContainerID from the container content label. SCALE then presents the user with the pick which displays the location, item, lot and quantity to be picked **[EX24 - Display Item Alias in RF Pick confirmation screen]** displays the Item Alias if exists for an item. The picker confirms the pick by scanning the location and item **[EX39 - Work Confirmation Item validation]** for validation. User will also scan the ContainerID from the content label as part of validation. User will apply the container content label to the pallet.

While the user is picking, if there is not enough inventory in the location that the user is prompted to pick from, if the picker has permission to short, picker will adjust the quantity and then click on the short button. SCALE will ask for a short reason and user must select the reason from the drop

Commented [BK280]: 7.2 if a short pick is necessary - permission to short pick should only be allowed by user level permission.

Commented [NC281R280]: NC11222024: this functionality exists - just need to be implemented for all facilities

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down. If picker doesn't have permission to short, the picker presses the 'Pass' button to back out of the work unit (to leave it suspended) and contacts a supervisor. The supervisor intervenes and determines how to handle the out-of-stock situation.

Note: Short option is controlled using security and Knipper can decide which pickers will have access to short option.

After the picks are complete the user confirms the putaway to the PACK location.

Note: Full Pallet Pick is majorly applicable for 3PL.

23.3 PTL Pick (MHE Controlled)

Assumptions:

- Containers are created in Wave.
- One Work unit is created per shipping container.
- Shipping Label / Vendor Label, Container Content label are printed
- Work units messages are downloaded to Pick to Light system
- The picking is executed using Pick to Light system

Once picking is completed by the user, pick confirmation messages will be sent to SCALE. **[EX19 – Pick to Light Integration]** will process the messages and complete the work units in SCALE. The status will be advanced based on the status flow configuration.

23.4 PTL Case Pick (MHE Controlled)

Assumptions:

- Containers are created in Wave.
- One Work unit is created per shipping container.
- Shipping Label / Vendor Label, Container Content label are printed
- Work units messages are downloaded to Pick to Light system
- The picking is executed using Pick to Light system

Once picking is completed by the user, pick confirmation messages will be sent to SCALE. **[EX19 – Pick to Light Integration]** will process the messages and complete the work units in SCALE. The status will be advanced based on the status flow configuration.

23.5 LTL Pallet Pick

Assumptions:

- Containers are created in wave.

- Pallet building wave step will nest cases and repack boxes on to pallets
- One Work unit is created per pallet.
- At Wave release Container content labels and Vendor labels are printed.
- Picker will carry empty pallet with them.

The user starts picking by signing into the **RF on warehouse mobile** and selecting a Work Profile of 3PL Pick/3PL Pick DSCSA. The user is then prompted to scan a work unit. User will scan the ContainerID from the Vendor Label and click on the GO button. SCALE then presents the user with the pick which displays the location, item, lot and quantity to be picked **[EX24 - Display Item Alias in RF Pick confirmation screen]** displays the Item Alias if exists for an item. The picker confirms the pick by scanning the location for validation and Item **[EX39 - Work Confirmation Item validation]**. If the item is serial number tracked, then user needs to scan the serial number at the unit level. The user will place the cases picked on to an empty pallet they carried with them. If it is Loose item, then the picker will place that item into a Tote. User will apply the container Content label to individual cases or place the container content labels inside the tote. User will apply the vendor label on to the Pallet.

While the user is picking, if there is not enough inventory in the location that the user is prompted to pick from, the user presses the 'Skip' button to continue with the next instruction on that work unit. If after all the picks are complete for the work unit and they are returned to the screen with the short item, if the picker has permission to short, picker will adjust the quantity and then click on the short button. SCALE will ask for a short reason and user must select the reason from the drop down. If picker doesn't have permission to short, the picker presses the 'Pass' button to back out of the work unit (to leave it suspended) and place the pallet and label stock in a hospital area. The supervisor intervenes and determines how to handle the out-of-stock situation.

After the picks are complete the user confirms the Putaway to the pack location.

[EX38 – Serialization Integration with Rfxcel for Outbound] will enable company, GTIN and Serial Number validation for DSCSA. The extension will also capture serial numbers at multiple UM levels.

23.6 REPS Pick

Assumptions:

- Containers are created in Wave.
- One Work unit is created per Wave.
- Shipping Label / Vendor Label and Container Content label printed in the wave.

The user starts picking by signing into the RF Work option and selecting a Work Profile of REPS picking. The user is then prompted to scan a Work Unit. User will scan the barcode form Break Label to initiate work. SCALE presents the user with the pick which displays the location, item and quantity to be picked **[EX24 - Display Item Alias in RF Pick confirmation screen]** displays the Item Alias if exists for an item. The user scans the

location and item for validation [EX39 - Work Confirmation Item validation]. When complete, the user scans the Container ID from the Container Contents Label to verify and press the OK button. User will apply Shipping / Vendor Label on to the container and place the container on the line.

While the user is picking, if there is not enough inventory in the location that the user is prompted to pick from, the user presses the 'Skip' button to continue with the next instruction on that work unit. If after all the picks are complete for the work unit and they are returned to the screen with the short item, the user presses the 'Pass' button to back out of the work unit (to leave it suspended) and contacts a supervisor. The supervisor intervenes and determines how to handle the out-of-stock situation (whether to short that line using the 'Short Pick' button or to cancel the entire order, etc.).

Note: Security is set up to only allow specific users (supervisors) to have the Short Pick button on the RF screen

After the picks are complete, the user confirms the putaway to the Packing location.

While the user is picking, if there is not enough inventory in the location that the user is prompted to pick from, the user presses the 'Skip' button to continue with the next instruction on that work unit. If after all the picks are complete for the work unit and they are returned to the screen with the short item, if the picker has permission to short, picker will adjust the quantity and then click on the *short* button. SCALE will ask for a short reason and user must select the reason from the drop down. If picker doesn't have permission to short, the picker presses the 'Pass' button to back out of the work unit (to leave it suspended) and contacts a supervisor. The supervisor intervenes and determines how to handle the out-of-stock situation.

Note: Short option is controlled using security and Knipper determines this. If picker doesn't have short option, then once they click on the pass button, they need to take the cart to a designated hospital area and leave the cart there for the supervisor to research on the issue.

Once all the picks are completed user will confirm putaway to PACK location and will take the cart to packing station and unload the containers.

23.7 Cooler Pick

Assumptions:

- Containers are created in Wave.
- One Work unit is created per container.
- Shipping Label / Vendor Label and Container Content label is printed per container.

Knipper will utilize Group picking for Cooler picking. User will manually build carts prior to picking.

User starts picking by signing into the WHM work option and selecting the work profile of Cooler picking. Based on the containers that can fit on the cart, the user will start scanning the container ID and click on the Assign option. SCALE will assign the containers to slot beginning from Slot 1. Once all the container that are on the cart are scanned and assigned spots, the user will click on Begin Picks to initiate work execution.

SCALE then presents the user with the pick which displays the location, item and quantity to be picked **[EX24 - Display Item Alias in RF Pick confirmation screen]** displays the Item Alias if exists for an item. The picker confirms the pick by scanning the location and item for validation **[EX39 - Work Confirmation Item validation]**.

While the user is picking, if there is not enough inventory in the location that the user is prompted to pick from, the user presses the 'Skip' button to continue with the next instruction on that work unit. If after all the picks are complete for the work unit and they are returned to the screen with the short item, the user presses the 'Pass' button to back out of the work unit (to leave it suspended) and contacts a supervisor. The supervisor intervenes and determines how to handle the out-of-stock situation (whether to short that line using the 'Short Pick' button or to cancel the entire order, etc.).

Note: Security is set up to only allow specific users (supervisors) to have the Short Pick button on the WHM screen

After the pick is complete for the item, if the item is only part of one container, then SCALE displays the slot number on the pick screen itself and user needs to scan the ContainerID that they are picking into.

If the item that is being picked is present in multiple slots, then once the pick is confirmed SCALE will display the slot and ContainerID in the next screen. User needs to scan the ContainerID that they are picking into for validation.

For Cooler pick the Auto Putaway will be configured. So, user doesn't have to confirm the putaway once all the picks are completed.

24.0 PACKING

24.1 QC Confirmation

For Knipper, outbound QC is leveraged for 3PL orders This is needed to ensure the right item and lot are picked and packed. If Pallets are created as part of wave, QC should be enabled at Pallet level. Pallet level QC will be handled as an extension (**EX46 – Pallet level QC**) in SCALE.

To perform QC, the user will utilize the QC insight screen. The user will scan the ContainerID from the container content label to initiate QC. In this screen, SCALE displays item, Lot and quantity that is expected in the container. User then needs to scan (or manually enter) every single item present in the container (Grocery scanning). Once scanning is done of all the items, user will click on confirm button to complete the process. At this stage SCALE determines if the QC has passed or failed. If failed, SCALE will highlight the item and ask user to enter a reason code for either the shortage or overage to complete the entire QC process. If QC is passed, user can proceed with Close container operation. If QC is failed, then a supervisor is responsible to fix the issue with the container.

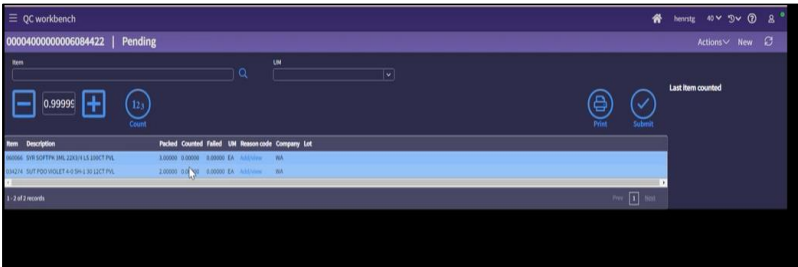


Figure: QC Workbench

If the items and quantities scanned match the system, the user receives a message that QC was successful. If not, the user receives a message that QC failed and need to provide reason codes for any missing quantities. These reason codes are configurable by Knipper to record things such as damaged, incorrect item, etc. The user needs to correct any failures and have a successful QC before continuing to Close Container.

Commented [NC282]: NC11222024: We need QC for each level of UOM and picker can't be QC'er

Commented [RS283R282]: This will be developed as part of a custom functionality for Pallet. However, base already supports QC for full case, unopened full pallets, loose shipper case containers. We will need to develop a custom functionality to support QC for containers nested on a Pallet.

Commented [NC284R282]: NC1252024: This is resolved.

Commented [NC285]: NC11222024: We need this functionality for RF gun

Commented [RS286R285]: The basic QC functionality exists in RF. However, we need to know the custom Pallet level QC functionality is needed in RF as well. If so, we will need to add that to the scope

Commented [NC287R285]: NC12052024: This is resolved.

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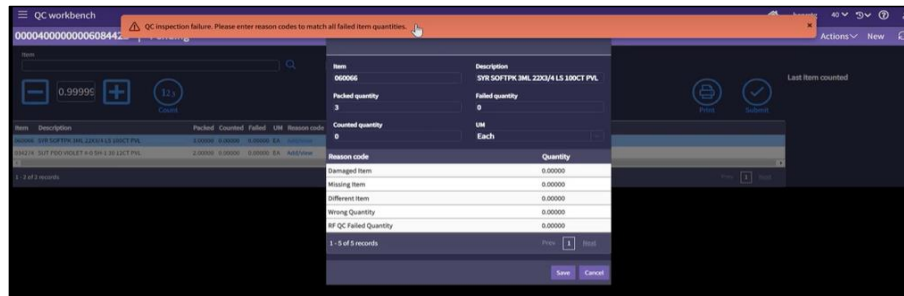


Figure: QC failure – reason code selection

Knipper uses Force QC pass several times in the workflow to use QC [workbench] as a visual QC platform.

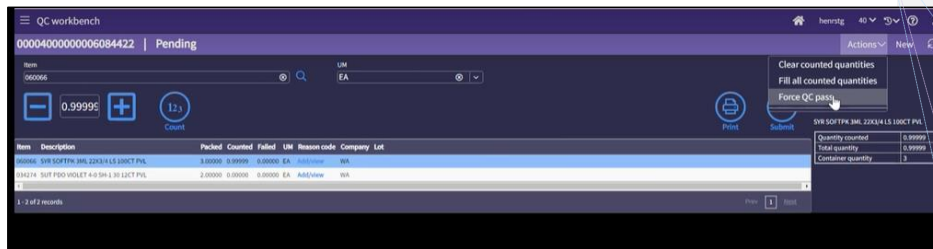
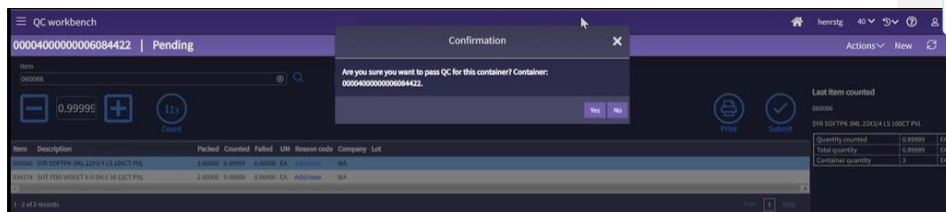


Figure: Force QC pass



Internal id	Activity date time	Process	Action	Message	Identifier 1	Identifier 2
2623025	8/24/2023 11:33:05 AM	QC Confirmation	Force QC Pass	User invoked the Force QC Pass action for the Sh...	Container id:00004000000006084422	Previous QC status:Pendi
2623026	8/24/2023 11:34:00 AM	QC Confirmation	Force QC Pass	User invoked the Force QC Pass action for the Sh...	Container id:00004000000006084439	Previous QC status:Pendi

Figure: Force QC pass process history

Commented [RS288]: Review further with Knipper

Commented [NC289R288]: NC1118224: Please confirm when this will happen

Commented [RS290R288]: We need an input from Knipper if they would like users to force pass the QC when there are issues. This will be rare exception but the standard process is to resolve any QC issues on the floor and confirm it in the system

Commented [NC291R288]: NC12052024: We want this implemented

Commented [RS292R288]: This is a base option available already of Knipper uses the base QC workbench. Please note that we have proposed an extension for Pallet level QC from RF screen. If this is needed there, we can discuss further when we start documenting that

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24.2 Close Container

Close container action/transaction identifies that the container is packed (sealed), and no other item can be put into it. This also advances the status of the container (and shipment) as per the status flow and determines the next action on the staging/dock area.

[EX12 – Scheduled Ship Date Validation Before Close Container] allows changing the scheduled ship date on the shipment before container manifesting.

For parcel and each picks, those containers need to be closed by the user to advance the status. To perform close container, user accesses **close container insight** screen or the Close container option on the Warehouse mobile menu and scans the Container ID. SCALE displays the expected weight of the container. User then clicks the *Close* button to complete the operation.

When close container is successful, SCALE will advance the status and print the parcel shipping label (LBL04) as well as the shipment packing list (DOC02). The shipment pack list is printed at close of last container.

For International parcel shipments, user needs to perform close container for every individual container and when the last container of the shipment is closed, user can select shipment level manifest and print shipping labels for all the containers for that shipment. For International shipments, commercial invoice is also printed (DOC04)

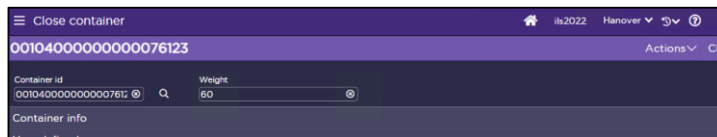


Figure: Close Container Screen – initiation

The screenshot shows the 'Close container' interface with the following details:

- Container ID:** 001040000000000076123
- Weight:** 60
- Container info:**
 - Container count:** 3
 - Total containers:** 3
 - Carrier:** InHouse
 - Carrier service:** (empty)
 - Container type:** MPL
 - Length:** 48
 - Width:** 40
 - Height:** 108
 - NMFC code:** (empty)
 - Tracking number:** (empty)

Figure: Close Container – Container details populated

24.3 Shipping Container Insight

If necessary, the **Shipping Container Insight** can be used to modify the contents of the Shipping Container. The user can update the quantity to pack to 0 for any items needing to be unpacked from a container. Once unpacked, the user can repack into new containers using the Packing screen.

If user doesn't have Container ID to Scan but still must perform the close container, then they can use **Shipping Container Insight** and filter based on the Shipment ID. From the Container insight, user can select the Container ID and then use the *Close Action* to close the container.

The screenshot shows the 'Shipping container insight' interface with the following details:

- Basic criteria:**
 - Container ID: (empty)
 - Shipment ID: (empty)
 - Item: (empty)
 - Current location: (empty)
 - Status: (empty)
 - Container type: (empty)
 - QC: (empty)
 - Warehouse: (empty)
 - Wave number: 162551
 - Internal shipment number: (empty)
 - Shipping load: (empty)
 - Show only top level: Off
 - Show only manifested: Off
- Summary:**
 - Loads: 0
 - Shipments: 1
 - Containers: 14
 - Locations: 2
- Table:**

Shipment ID	Container ID	Shipment ID	Tracking number	Container type	Shipping load number	Status
00000327001116673045	DSK5AR01	00000327001116673045	DSK5AR01	LR	0	Picking
00000327001116673052	DSK5AR01	00000327001116673052	DSK5AR01	LR	0	Picking
00000327001116673069	DSK5AR01	00000327001116673069	DSK5AR01	LR	0	Picking
00000327001116673076	DSK5AR01	00000327001116673076	DSK5AR01	LR	0	Picking
00000327001116673083	DSK5AR01	00000327001116673083	DSK5AR01	LR	0	Picking
00000327001116673090	DSK5AR01	00000327001116673090	DSK5AR01	LR	0	Picking
00000327001116673106	DSK5AR01	00000327001116673106	DSK5AR01	LR	0	Picking
00000327001116673113	DSK5AR01	00000327001116673113	DSK5AR01	LR	0	Picking
00000327001116673120	DSK5AR01	00000327001116673120	DSK5AR01	LR	0	Picking
00000327001116673127	DSK5AR01	00000327001116673127	DSK5AR01	LR	0	Picking
00000327001116673134	DSK5AR01	00000327001116673134	DSK5AR01	LR	0	Picking
00000327001116673153	DSK5AR01	00000327001116673153	DSK5AR01	LR	0	Picking Pending
- Context Menu:**
 - Edit
 - Delete
 - Edit accessories
 - Close
 - Confirm QC
 - Dock transfer
 - Manifest
 - Mark for QC
 - Nest
 - Print preview
 - Print default docs
 - Print selected docs
 - Remove from manifest
 - Remove from nesting
 - Transfer
 - Update packed quantity
 - VAS activity

Figure: Shipping Container Insight

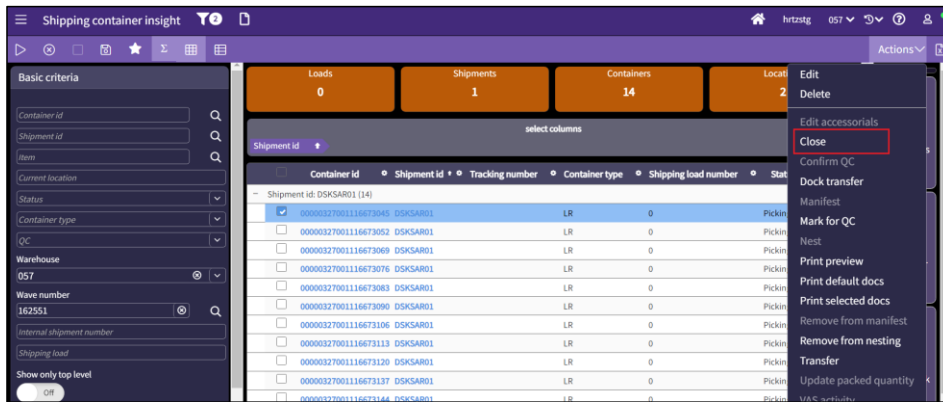


Figure: Shipping Container Insight with Close Container Option

Note: To edit container contents, a container must be in a location with a subclass of Packing.

24.4 Carrier changes

Carrier is assigned as part of load planning before the wave is run.

To change an LTL carrier before loading into truck, the user will use the **Shipment Insight** screen and utilize the *Transfer Shipment* option to change the carrier.

When *Transfer Shipment* option is selected, user will be presented with an option to provide the destination load. If the user knows the load number, then they can enter the shipping load number and SCALE will transfer the shipment to that load.

If the load is not known, then user can select to create a new load and select the carrier for the new load. SCALE will create new load and then transfer the shipment to new load.

Carrier change after waving happens at a minimal and is an exception. Knipper handles this using a SOP outside of SCALE.

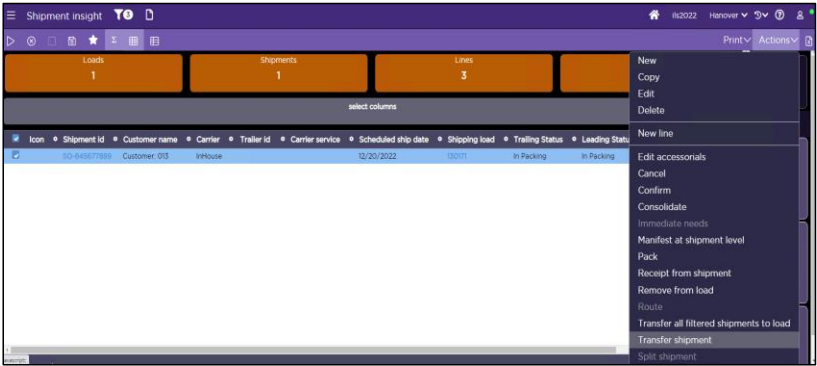


Figure: Shipment Insight – Transfer Shipment Option

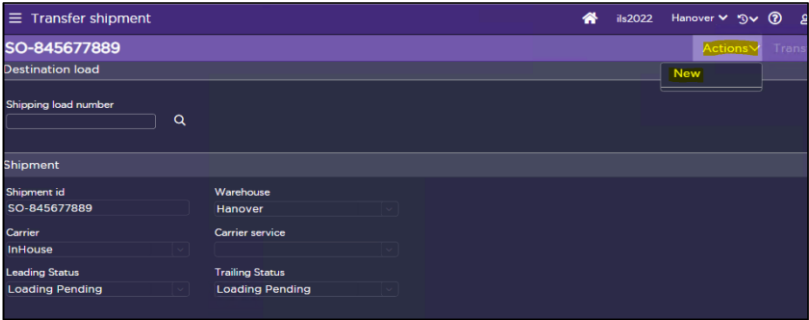


Figure: Shipment Insight – Transfer to new load

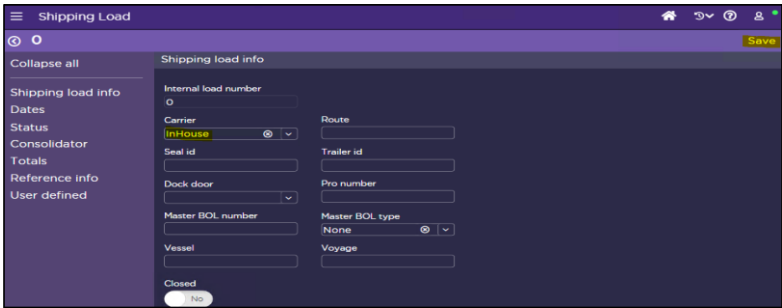


Figure: Shipment Insight – Enter Destination Load Number or create New Load

25.0 DOCK MANAGEMENT

Dock management represents the process of tracking shipments after they have arrived in the shipping dock area of the warehouse. These processes include the consolidation of items/containers at the packing area location, the placing of containers on the shipping dock at staging locations, and the loading of containers onto the truck, referred to in the system as dock door locations.

Dock management is used at Knipper LTL shipments. Knipper utilizes dock doors by manually assigning them to the shipping load.

25.1 Dock Door Assignment

Dock Door assignment at Knipper is done manually.

Shipping loads are created for LTL and parcel carrier using the load building wave steps. When the Parcel / LTL carrier shows at the warehouse, the shipping supervisor instructs the driver to go to a specific dock door. In SCALE the user will go to shipping load insight screen and search for the shipping load and will edit the shipping load to assign a dock door. Loading work will get generated and user can perform this work to load the pallet to truck.

Figure– Shipping Load Insight

25.2 Loading

The user starts the process by signing into the **Work execution/Loading** option on warehouse mobile. The user is then prompted to scan a Container ID. The user scans the container ID and SCALE then presents the user to confirm the pick from staging location. After that, user is prompted to scan the dock door to confirm the loading. This updates the container status to Ship Confirm Pending.

Commented [SG293]: Dock Management for Parcel?

Commented [NC294R293]: NC11182024: Please confirm

Commented [RS295R293]: This is currently being configured/tested in SCALE 2013 version. We have also provided some recommendations. Once Knipper goes live with the functionality, the same will be ported over to new version

Commented [NC296R293]: NC12052024: Per Ops/Vimal - this is not working and this will need a custom extension

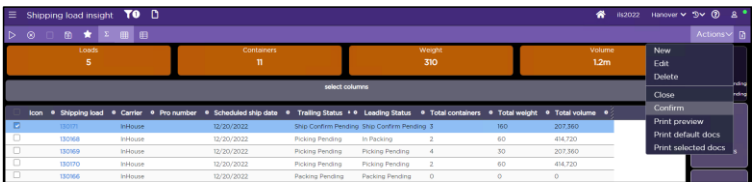
Commented [RS297R293]: We have recommended a base solution which needs manual scanning containers to stage/dock door which will avoid extension. Please note that even if we design an extension, it will be based on a lot of assumptions.

26.0 LOAD CONFIRMATION

26.1 Processing

Knipper Personnel use the **Shipping Load Insight** screen to monitor the status of shipments and loads. Once all shipments for a Load are in *Load Confirm Pending* status [**Ship Confirm All scheduled job confirms shipments and advance status to Load Confirm Pending**], the Load can be confirmed. To Confirm the Load, a user selects the Load and uses the Confirm Load action in the **Shipping Load Insight** options. This relieves the inventory from the Shipping Dock locations in SCALE. At this point, the shipments are available for upload to the host system the next time the interface job runs. In addition, the shipment can no longer be modified in SCALE.

Knipper does not split shipments. The Shipping load leading and trailing status should be ship confirm pending. If the status is not Ship confirm pending, then user moves any shipments that are not in ship confirm pending to another load. This is necessary as Knipper doesn't want to Split the shipment. This is based on the restriction on the host system.



icon	Shipping load	Carrier	Pro number	Scheduled ship date	Trailing Status	Leading Status	Total containers	Total weight	Total volume
☒	10001	inHouse	12/20/2022	Ship Confirm Pending	Ship Confirm Pending	5	160	207,560	
☐	10008	inHouse	12/20/2022	Picking Pending	In Packing	2	60	414,720	
☐	10009	inHouse	12/20/2022	Picking Pending	Picking Pending	4	30	207,360	
☐	10070	inHouse	12/20/2022	Picking Pending	Picking Pending	2	60	414,720	
☐	10068	inHouse	12/20/2022	Picking Pending	Picking Pending	0	0	0	

Figure – Load confirm using Shipping Load insight

Additionally, personnel may print documents before, or after the load is confirmed using the **Shipping Load Insight** options.

- Bill of Lading (**DOC03**) Knipper uses a customized BOL implemented using [**EX14 – Custom BOL Changes**]

For international shipments, the commercial invoice is printed. Knipper uses a custom commercial invoice template configured in SCALE.

The load confirmation process also validates if the ship events have been generated in RTS system for all DSCSA tracked items. If the ship events are not generated for any shipment in the load, load confirmation process will not proceed further and displays an error. This is handled through [**EX30 – Serialization Integration with Rfxcel for Outbound**]

Once a load has been confirmed, all statuses (load, shipments, details, containers) are moved to Closed, and inventory is relieved from the Shipping Dock (is officially out of the building). This generates the shipment upload interface files for the Host system.

27.0 PARCEL MANIFESTING PROCESS

All parcel manifest processing is executed from the **Manifest Insight** screen. Users can view any pertinent manifest information from this option. Any action for FedEx carrier is excluded from this screen. There is no end of day processing necessary for FedEx parcel carrier. Knipper will continue to utilize this screen for UPS, USPS and UPS Mail Innovations.

27.1 End of Day Processing

Manifests can be closed at the end of the day by clicking on a Manifest from the **Manifest Insight** and choosing the Close option. Rating Systems Value “*Allow Multiple Manifests per Day*” is set to Yes.

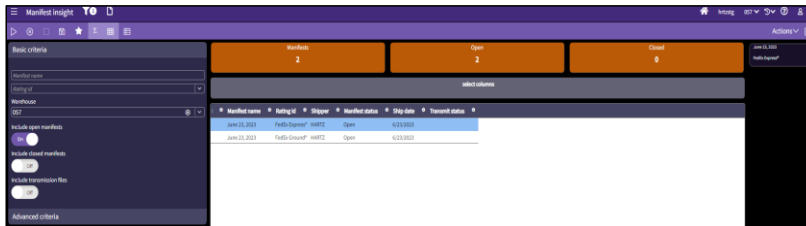


Figure: Manifest Insight Screen

27.2 Printing

Required documents can also be printed from the Manifest Insight. These documents can print out automatically when closing the manifest or be printed manually.

The documents include:

Container Manifest
UPS Summary Label

Once a Manifest has been closed, the Manifest is no longer open to receive additional containers; all additional containers manifested for that day are put on a new manifest, and Electronic Manifest is sent to Parcel Carrier. Containers on closed manifests may not be changed.

Commented [CC298]: Ups, UPS MI, USPS Priority

Commented [NC299R298]: NC11182024: MAH to confirm

Commented [RS300R298]: updated

Commented [NC301R298]: NC12052024: This is resolved

VI. CONVERSION NOTE

The below points should be part of the conversion process.

1. Review and remove any database objects not compliant with Azure SQL. The findings have been already shared with Knipper team. This should be one of the steps to be part of the conversion checklist.
2. Configuration related to Warehouse split should be documented and executed during the conversion.
3. Inventory conversion by Warehouse after the split.
4. Planning of conversion to consider the timing due to data migration. Knipper and Manhattan to finalize on number of years of data to be migrated to SCALE Active.

VII. SYSTEM EXTENSIONS

Interface Integration Extensions

Extension	Description (Defined above in the interface section)

Performance Management Extensions

Extension	Description	Module
N/A		

Warehouse Management Extensions

Extension	Description
EX12	Scheduled Ship Date Validation Before Close Container
	Pick to Light Wave Splitting
	Wave Steps: Cancel Replenishment for Rejected Orders DTP Location Assignment DTP Wave Splitting
EX13	Exit point: Release Wave - Before
EX14	Custom BOL
EX16	FedEx Express Reference Fields
EX17	Re-Cartonization of Containers based on Season
EX18	Whole Number Allocation for Work Orders
EX19	MHE Interface - Scale Pick to Light Integration PTLViewer Outbound msg for pick confirmation
EX19A	MHE Integration Changes to Support Multiple MHE Server
EX24	Display Item Alias in RF Pick confirmation screen
EX26	Display Expiry Date on cycle count screen
EX36	Corrugate Item Tracking

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KNIPPER SOLUTION DESIGN DOCUMENT

EX37	Serialization Integration with Rfxcel Inbound
EX37A	SERIALIZATION INTEGRATION INBOUND FOR LOT VALIDATION
EX38	Serialization Integration with Rfxcel Outbound
EX38A	Serialization Integration with Rfxcel Outbound
EX38B	Verification & Address Validation
EX38C	Downgrade UOM Picking & Serialization Changes
EX38D	Additional Changes to Outbound Serialization
EX39	GS1 Scanning for Work Confirmation Item Validation
EX40	Printing multiple work unit document from work insight
EX41	Mass updates - Change Carrier, Service, Scheduled ship date for multiple shipments at Wave level
EX42	Custom Return from Shipment
EX43	Flash title – Inventory Conversion from one Item to another
EX44	LPN Combine
EX27	HazMat documents/Accessorial
EX46	Pallet level QC after nesting for DSCSA
EX47	Cubiscan Integration
EX48	Auto refresh Wave Insight screen
EX49	TPM Order Status Insight Changes

Stored Procedures for Generic Data Bind API

SP	Description
N/A	To be documented if identified during tech design

Warehouse Management Gaps for future considerations

Extension	Description
N/A	

Documents

Document	Description
DOC01	Receiving Worksheet

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KNIPPER SOLUTION DESIGN DOCUMENT

DOC02	Pack List
DOC03	Bill of Lading
DOC04	Commercial Invoice

Labels

Label	Description
LBL01	Receipt Container Label
LBL02	Vendor Label
LBL03	Container Contents Label
LBL04	Shipping Label
LBL05	Pallet Label

***Note:** For each of the documents and labels (unless part of the Top 100 Retailers for outbound labels) listed above, Knipper owns the development and certification process.

Exit Points

Exit Point	Description
EP01	Cancel Shipment – Before
EP02	Close Container – Before
EP03	Close Container UI Validation
EP04	Container Manifesting – After
EP05	Load Confirmation – Before
EP06	Release Wave – Before
EP07	Release Wave - After
EP08	RF Group Assignment – After
EP09	Ship Label Custom Text – Prologistics
EP10	Shipment Allocation Request After

Commented [RS302]: There are multiple exit points in use which are developed as part of the current extensions. New exit points identified as part of new extension or re-design of current extensions will be added.

Commented [NC303R302]: NC11182024: I do not know what this means

Commented [RS304R302]: These are currently used extensible features to implement some customizations. I have updated the list with the ones being used in the current version.

Commented [NC305R302]: NC12052024: This is resolved

Notifications

Notification	Description
Several	TBD as needed

Labor Management Extensions

Extension	Description
N/A	

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VIII. OPEN ISSUES



JKNP_ParkingLot_11
122024.xlsx

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IX. RESOLVED ISSUES



JKNP_ParkingLot_11
122024.xlsx

X. FUTURE FUNCTIONALITY

APPENDIX A – Configuration Notes

No key configuration change identified to existing workflow.

APPENDIX B – Override Data Wave Steps

Existing override steps will be evaluated in the build phase for use.

APPENDIX C – Security permissions

Knipper can review existing user security records on the Security Permissions Window. When a user attempts to access a SCALE window, the system determines if they have any user-level security records. If an employee has a user-level record, the system will apply that record each time the employee attempts to access the window. User-level security limits a single employee's ability to access and/or perform actions on a specific window. Or the Security Permissions Window can be used to define the security rights of a specific security group. Processing and/or configurations are defined for the security group and what security checkpoints can be used for the group. These are all defined using the Security Permissions Window.

Mass security changes are also allowed using this window. The system allows you to select the processing or configuration windows and assigning security to them. All the selected forms (windows) will be assigned the selected security levels and actions.

This window allows you to grant security permissions by a specific processing function or by a specific configuration.

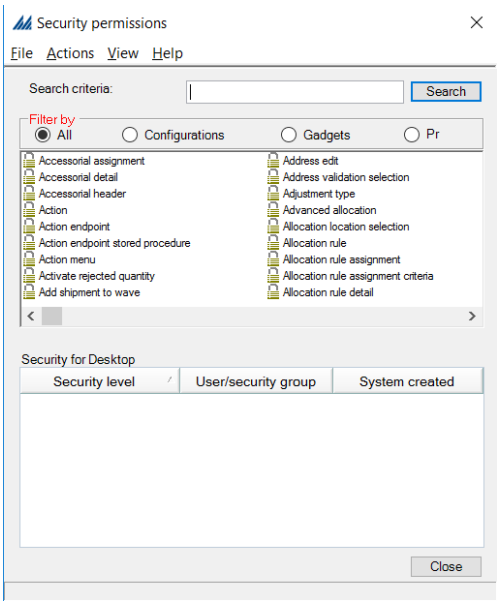


Figure: Security Permission Configuration window

APPENDIX D – Supplemental DC Ops Data

N/A

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REVISION HISTORY

Date:	Changed By:	Doc. Version:	Notes:
10/21/24	Ravishankar Suragihalli	1.0	Draft based on existing functional flow and discussions during kickoff.
11/12/2024	Ravishankar Suragihalli	1.1	Added Outbound flow.
12/05/2024	Ravishankar Suragihalli	1.2	Addressed the review comments
12/10/2024	Ravishankar Suragihalli	1.3	Addressed the review comments

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SOLUTION DESIGN DOCUMENT SIGN-OFF

As part of our project methodology, Manhattan Associates requires a written acknowledgement of the Solution design document at the end of the Design Phase.

By signing this letter Knipper indicates their understanding that the operations and extensions specified in this Solution Design Document (Knipper ***Solution Design Document v1.3***) will support the Knipper's Memphis warehouse operational procedures outlined during the functional design phase.

Please sign below to indicate your acknowledgement and email scanned document to sadinarayan@manh.com. Please contact at 678-597-7028 in the event you have any questions.

Sincerely,

Srivatsa Adinarayan Gotur
Project Manager

cc:
Ravishankar Suragihalli
Design Lead

Vimal Gopalakrishnan
Engagement Director

Signature

Printed Name / Title

Date